

# THE BASE- $\beta$ BRANCH OF $3\alpha + 2\beta = \pi$ : BOUNDARY WALKS, AND THE STRUCTURE OF A HYPOTHETICAL PRIME TILING

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**ABSTRACT.** This is a companion note to *A signed-direction invariant for triangle tilings, and the exclusion of primes  $\equiv 3 \pmod{4}$* . That paper excludes every prime  $N \equiv 3 \pmod{4}$  exceeding 3 except the base- $\beta$  candidates  $N = 3f^2 - e^2$ , which are exactly the primes  $\equiv 11 \pmod{12}$ . Here we record what is known about the internal structure of a hypothetical tiling in that one open branch: the boundary walk equations and their solutions, the apex-mismatch configuration, and the alignment and far-region theorems. None of this is needed for the main theorem; it is collected here so that the main paper can state the exception cleanly and so that the structural facts are available to anyone attacking the branch.

Two results are worth isolating. First, at  $m = 1$  the walk equations are completely solved in a regime: if  $f^2 > 2e^2$  then each equal side of the target carries no  $b$ -edge, and if  $f^2 > 2ef + e^2$  then the base carries exactly  $e$  of them, so  $R_{\text{base}} \in \{e, 2e\}$ . Second, for  $e = 1$  this pins the base walk to a single family, reducing that entire subfamily — the candidates  $N = 3f^2 - 1$ , among them 11, 47, 107, 191, 431, 587, 971 below 1000 — to the question of whether one explicit walk is realizable.

We also record, as prominently as the positive results, which attacks on this branch are now known to be dead: the whole class of linear direction invariants and the metric bounds derived from them; the hope of closing  $e = 1$  by proving that every side carries at least two  $c$ -edges; Euler and angle-sum counting, which collapse to a single relation; the non-abelian Conway–Lagarias tiling group, which collapses to its rank-two abelianisation; the reach ladder, which costs one geometric step per member; and two rounds of purpose-built machinery, both of which reached a true theorem whose hypothesis is never satisfied. The common cause is worth stating: every invariant the sweep produces *counts*, and what the open step needs is one that *locates*.

One further correction belongs in the abstract rather than the body. The complete-corner-wall hypothesis under which the main paper states its conditional theorem is, at  $e = 1$ , equivalent to its own conclusion: a complete west block requires the base word to begin  $a^f$ , while the reduction theorem for that subfamily makes it a permutation of  $(a^f, b, c)$  ending in  $a$ , so the hypothesis holds there exactly when no tiling exists. It is therefore not a weaker assumption at  $e = 1$ , and no exclusion of an  $e = 1$  member may cite it. At  $e \geq 2$  it is a genuine positional strengthening and remains open.

## 1. SCOPE AND CONVENTIONS

Throughout, the tile is the primitive  $3\alpha + 2\beta = \pi$  triangle

$$(a, b, c) = (ef, f^2 - e^2, f^2), \quad \gcd(e, f) = 1, \quad 1 \leq e < f,$$

with  $a$  opposite  $\alpha$ ,  $b$  opposite  $\beta$ ,  $c$  opposite  $\gamma$ , and  $2\gamma = \pi + \alpha$ . The base- $\beta$  target is the isosceles triangle with base angles  $\beta$ ,

$$\text{equal sides } S = f^3m, \quad \text{base } L = em(3f^2 - e^2), \quad N = m^2(3f^2 - e^2),$$

and  $N$  is prime exactly when  $m = 1$ . We write  $D = 4f^2 - e^2$ , so  $\sin \alpha = e\sqrt{D}/(2f^2)$  and the tiling’s coordinates lie in  $\mathbb{Q}(\sqrt{D})$ . Numbering of the form “Proposition 4.3” refers to the main paper [4]; results are stated here only when they are not there.

We use freely from [4]: the dissection basics and vertex figures, the corner figures (a base corner carries a single tile presenting  $\beta$ ; the apex carries exactly three tiles, each presenting  $\alpha$ ), the  $\gamma$ -trap (every side of the target carries at least one  $c$ -edge), the unsplittability of  $b$ , and the corner parallelogram.

## 2. BOUNDARY WALKS

A side of the target of length  $L$  is exactly partitioned by the tile edges lying on it, so it determines a *walk equation*

$$n_a e f + n_b (f^2 - e^2) + n_c f^2 = L, \quad n_a, n_b, n_c \geq 0,$$

together with  $n_c \geq 1$  from the  $\gamma$ -trap. The results of this section solve those equations and then constrain the order in which the edges occur.

**Theorem 1** (The base- $\beta$  family at  $e = 1$ ,  $f = 2$ ). [PROVED] *The isosceles triangle with base angles  $\beta$  and sides  $(8, 8, 11)$  admits no tiling by  $N = 11$  copies of the tile  $(2, 3, 4)$ . Consequently the base- $\beta$  branch is closed for  $N = 11$ , unconditionally.*

*Proof.* Write  $\omega = \alpha/2$ . The law of cosines gives  $\cos \alpha = (2f^2 - e^2)/(2f^2)$ , hence  $\sin \omega = e/(2f)$ ; as  $1 \leq e < f$  places this strictly in  $(0, \frac{1}{2})$ , *Niven's theorem* forces  $\alpha/\pi$  irrational — for every  $(e, f)$ , with no appeal to [3] (`tile_alpha_irrational`). Irrationality pins the vertex figures: a vertex  $x\alpha + y\beta + z\gamma = k\pi$  forces  $y + z = 2k$  and  $2x + z = 3y$ , so a  $\pi$ -vertex is  $\{3\alpha, 2\beta\}$  or  $\{\alpha, \beta, \gamma\}$ , each base corner carries *exactly one* tile (its  $\beta$ -vertex) and the apex *exactly three* (`vertex_pi`, `vertex_beta_corner`, `vertex_apex`); in particular a  $\pi$ -vertex carries at most one  $\gamma$ , since  $2\gamma = \pi + \alpha > \pi$ . Each side of  $ABC$  is partitioned into whole tile edges (a supporting line meets a convex tile in a face).

Two citation-free constraints now cut the base. (a) *Corner parallelogram*: the unique  $\beta$ -corner tile  $T_B$  has its  $b$ -edge a chord  $[P, Q]$ , matched (by the mod- $f$  edge arithmetic) by a single tile  $T'$  whose  $\alpha, \gamma$  are swapped, so  $T_B \cup T'$  is a parallelogram with sides  $a, c$ ; hence  $P, Q$  are  $(1, 1, 1)$ -vertices whose third tile carries  $\beta$ , and the second base edge from each corner lies in  $\{a, c\}$ . So a  $b$ -edge can occupy only positions  $3, \dots, k-2$  of a base of  $k$  edges. (b)  *$\gamma$ -trap*: no boundary junction carries  $\gamma$  on both sides, and only a  $c$ -edge leaves the  $\gamma$ -state, so every side carries at least one  $c$ -edge.

At  $e = 1$ ,  $m = 1$  the base  $Y = 3f^2 - 1$  has  $q_{\text{base}} \equiv 1 \pmod{f}$ , and the only compositions are  $B_0 = \{b, c, c\}$  ( $k = 3$ ),  $B_1 = (a^f, b, c)$  ( $k = f + 2$ ) and  $B_2 = (a^{2f}, b)$  ( $k = 2f + 1$ ).  $B_2$  dies by (b).  $B_0$  dies by (a) for *every*  $f$ : its single  $b$  would need a position in  $3, \dots, 1$ . At  $f = 2$ ,  $B_1$  also dies by (a):  $k = 4$  leaves positions  $3, \dots, 2$ . So no base survives, and  $N = 11$  admits no tiling.  $\square$

**Theorem 2** (Boundary walks of the base- $\beta$  target at  $m = 1$ ). [PROVED] *Let  $(a, b, c) = (ef, f^2 - e^2, f^2)$  be the primitive  $3\alpha + 2\beta$  tile ( $1 \leq e < f$ ,  $\gcd(e, f) = 1$ ) and let  $m = 1$ , so that the target is  $(f^3, f^3, e(3f^2 - e^2))$  and  $N = 3f^2 - e^2$ . Write a walk along a side as  $P \cdot a + Q \cdot b + R \cdot c$  equal to that side's length. Then both walk sets are cut out by the same linear form  $\langle e, f, 1 \rangle$ , the base at level  $2e$  and the equal side at level  $f$ :*

$$\begin{aligned} \text{base: } (P, Q, R) &= (je + fp, e + fj, R), & pe + jf + R &= 2e, & j &\in \mathbb{Z}_{\geq 0}, \\ \text{side: } (P, Q, R) &= (qe + fp, fq, R), & pe + qf + R &= f, & q &\in \mathbb{Z}_{\geq 0}, \end{aligned}$$

with  $p \in \mathbb{Z}$  and  $P, R \geq 0$ . In the thin regime  $f > 2e$  the levels force short lists:

- (i) (trichotomy) the base is exactly one of  $\{b^e, c^{2e}\}$ ,  $\{a^f, b^e, c^e\}$ ,  $\{a^{2f}, b^e\}$ ; in particular it carries exactly  $e$   $b$ -edges;
- (ii) (dichotomy) each equal side is  $\{a^e, b^f\}$  or  $\{a^{fp}, c^{f-pe}\}$  for some  $p \geq 0$ .

Feeding in the  $\gamma$ -trap ( $R \geq 1$  on every side, Remark 22) removes the thinness hypothesis altogether. For every  $(e, f)$  at  $m = 1$ :

- (iii) no equal side carries a  $b$ -edge — each is  $\{a^{fp}, c^{f-pe}\}$ ,  $p \geq 0$ ; so all boundary  $b$ -edges lie on the base;
- (iv) the base's  $b$ -count  $Q = e + fj$  obeys  $j(f - e) \leq e - 1$ .

In particular  $e = 1$  forces  $j = 0$  for every  $f$  — the base carries exactly one  $b$ -edge — and  $f > 2e$  forces  $j = 0$ , recovering (i) minus its  $R = 0$  member.

*Proof.* Base. Reducing  $P \cdot ef + Q(f^2 - e^2) + Rf^2 = e(3f^2 - e^2) \pmod f$  gives  $Qe^2 \equiv e^3$ , so  $f \mid Q - e$  by  $\gcd(e, f) = 1$ ; as  $0 \leq Q$  and  $0 < e < f$ , the least nonnegative residue is  $e$ , whence  $Q = e + fj$  with  $j \geq 0$ . Substituting and cancelling one factor  $f$  leaves the *halved equation*

$$Pe + j(f^2 - e^2) + Rf = 2ef. \quad (1)$$

Reducing (1)  $\pmod f$  gives  $e(P - je) \equiv 0$ , so  $f \mid P - je$ , say  $P = je + fp$ ,  $p \in \mathbb{Z}$ ; substituting into (1) and cancelling  $f$  gives  $pe + jf + R = 2e$ . Conversely every such triple solves the original equation. For  $f > 2e$ : from (1),  $j(f^2 - e^2) \leq 2ef$ ; if  $j \geq 2$  then  $f^2 - e^2 \leq ef$ , i.e.  $f^2 - ef - e^2 \leq 0$ , whereas  $(f - 2e)(f + e) = f^2 - ef - 2e^2 > 0$  gives  $f^2 - ef - e^2 > e^2 > 0$  — contradiction. If  $j = 1$  then  $f \mid P - e$  forces  $P \geq e$ , and (1) gives  $e^2 + (f^2 - e^2) \leq 2ef$ , i.e.  $f^2 \leq 2ef$ , i.e.  $f \leq 2e$  — contradiction. So  $j = 0$ , whence  $f \mid P$ ,  $P = fp$  with  $p \geq 0$ , and  $pe + R = 2e$  forces  $p \in \{0, 1, 2\}$  and  $R = e(2 - p)$ , giving the three walks.

Side. The same reduction on  $P \cdot ef + Q(f^2 - e^2) + Rf^2 = f^3$  gives  $Qe^2 \equiv 0$ , so  $f \mid Q$ ,  $Q = fq$ ; cancelling  $f$  leaves  $Pe + q(f^2 - e^2) + Rf = f^2$ , whose reduction  $\pmod f$  gives  $f \mid P - qe$ , say  $P = qe + fp$ , and cancelling  $f$  again gives  $pe + qf + R = f$ . For  $f > 2e$ :  $q \geq 2$  would give  $2(f^2 - e^2) \leq f^2$ , i.e.  $f^2 \leq 2e^2 < f^2$ , absurd. If  $q = 1$  then  $Pe + Rf = e^2$  and  $f \mid P - e$  force  $P = e$ ,  $R = 0$ ; if  $q = 0$  then  $f \mid P$ ,  $P = fp$  and  $pe + R = f$ .

(iii) and (iv) are one computation, and need no thinness. Suppose a side walk has  $q \geq 1$ . From  $P = qe + fp \geq 0$  we get  $fp \geq -qe > -qf$  (using  $e < f$  and  $q \geq 1$ ), so  $p > -q$ , i.e.  $p \geq 1 - q$ . The level equation then gives

$$R = f - pe - qf \leq f - (1 - q)e - qf = (1 - q)(f - e) \leq 0,$$

contradicting  $R \geq 1$ . Hence  $q = 0$ , which is (iii). Identically for the base with  $P = je + fp \geq 0$ : if  $j \geq 1$  then  $p \geq 1 - j$  and

$$R = 2e - pe - jf \leq 2e - (1 - j)e - jf = e - j(f - e),$$

so  $R \geq 1$  forces  $j(f - e) \leq e - 1$ , which is (iv) (and it holds trivially at  $j = 0$  since  $e \geq 1$ ). For  $e = 1$  this reads  $j(f - 1) \leq 0$ , so  $j = 0$ ; for  $f > 2e$  we get  $je < j(f - e) \leq e - 1$ , so  $j = 0$ .  $\square$

**Theorem 3** (Apex mismatch: the pierced corner). [PROVED] *Let a base- $\beta$  target be tiled at any scale  $m$  such that both equal sides end (at the apex) with a  $c$ -edge — automatic at  $m = 1$ , where the sides are  $b$ -free. Then, writing  $T_2$  for the middle of the three apex tiles:*

- (i) *the outer apex tiles carry  $c$  on the boundary and  $b$  inward, and  $T_2$  carries  $b$  and  $c$  inward, so exactly one inner apex ray carries  $T_2$ 's  $c$ -edge against a neighbour's  $b$ -edge: a  $T$ -junction at distance  $b$  from the apex, whose vertex figure on the neighbour's side is  $(1, 1, 1)$  (one  $\alpha$ , one  $\beta$ , plus the neighbour's  $\gamma$ );*
- (ii) *the remaining segment of  $T_2$ 's  $c$ -edge has length  $c - b = e^2$ , and  $n_a ef + n_b(f^2 - e^2) + n_c f^2 = e^2$  has no solution in nonnegative integers, for any  $1 \leq e < f$  with  $\gcd(e, f) = 1$ ;*
- (iii) *hence some far-side edge crosses the far endpoint of  $T_2$ 's  $c$ -edge straight through: every such tiling contains a **pierced  $\beta$ -corner** at distance  $f^2$  along an inner apex ray, and the angles on  $T_2$ 's side of the piercing line beyond the corner form  $\{\alpha, \gamma\}$  or  $\{3\alpha, \beta\}$ .*

*Proof.* (i) The apex is  $3\alpha$  (three tiles, each  $\alpha$  there); the two edges of an  $\alpha$ -corner are  $b$  and  $c$ . The boundary edges at the apex are the sides' last edges,  $c$  by hypothesis, so the outer tiles point  $b$  inward and  $T_2$  has  $\{b, c\}$  inward. Whichever inner ray pairs  $T_2$ 's  $c$  with an outer  $b$ , the  $b$ -edge ends at distance  $b$ : its far endpoint is the  $\gamma$ -corner of its tile ( $b$  joins  $\alpha$  to  $\gamma$ ), and the vertex figure on that side of the straight  $c$ -edge sums to  $\pi$  with a  $\gamma$  present, hence is  $(1, 1, 1)$  by the vertex classification. (ii) Any solution has  $n_a = n_c = 0$  since  $ef, f^2 > e^2$ ; then  $n_b(f^2 - e^2) = e^2$  gives  $n_b f^2 = (n_b + 1)e^2$ , and  $\gcd(e, f) = 1$  forces  $f^2 \mid n_b + 1$ , so  $n_b \geq f^2 - 1$  and  $e^2 = n_b(f^2 - e^2) \geq f^2 - 1 \geq e^2$ , with equality forcing  $f^2 - e^2 = 1$  — impossible, as  $f^2 - e^2 \geq 2e + 1 \geq 3$ . (iii) The far side of the leftover segment is covered by consecutive whole tile edges starting at the T-junction; by (ii) no partial sum lands exactly on the segment's far endpoint, so some edge strictly crosses it. That endpoint is the  $\beta$ -corner of  $T_2$  ( $c$  joins  $\alpha$  to  $\beta$ ). The angles on  $T_2$ 's side of the crossing line beyond the corner fill  $\pi - \beta$ ; writing them as  $x\alpha + y\beta + z\gamma$  and clearing  $\beta, \gamma$ , irrationality of  $\alpha/\pi$  forces  $y + z = 1$ ,  $2x + z = 3y + 3$ , whose only solutions are  $(3, 1, 0)$  and  $(1, 0, 1)$ .  $\square$

**Remark 4** (Validation on the 44-tiling). The theorem can be checked against the genuine 44-tiling ( $m = 2$ ; its equal sides do end with  $c$ ), where it holds exactly: the apex rays pair one  $b \mid b$  and one mismatch  $c \mid b$ ; the middle tile's  $c$ -endpoint  $V = (10, 2\sqrt{15})$  is pierced by a straight  $b$ -edge, and the sector on the middle tile's side is  $\beta + \{\alpha, \gamma\}$  — the first continuation type. Parts (ii) and the vertex figures in (i), (iii) are machine-checked (`apex_leftover_nonrepresentable`, `pierced_corner_types`, `BaseBetaWalks.lean`). The configuration is an interior constraint, uniform in  $(e, f)$  and unconditional at  $m = 1$ : every hypothetical thick-regime tiling,  $N = 59$  included, must contain it.

**Proposition 5** (Rung two: the pre-piercer chain).

[PROVED] *In*

*the configuration of Theorem 3, every full far-side edge strictly between the T-junction and the piercer lies inside  $[b, f^2]$ , hence has length  $\leq e^2$ ; among the tile edges only  $b = f^2 - e^2$  can satisfy this, and only in the super-thick regime  $f^2 \leq 2e^2$ . So the piercer starts at  $(t + 1)b$  with  $tb < e^2$ , where  $t = 0$  outside super-thick; and if the piercer is itself a  $b$ -edge, its corner at its starting junction is exactly  $\alpha$  (the  $(1, 1, 1)$  figure offers  $\{\alpha, \beta\}$ , while a  $b$ -edge's endpoints are  $\{\alpha, \gamma\}$ ). In particular for  $N = 59$  ( $(e, f) = (4, 5)$ , super-thick:  $25 \leq 32$ ) the piercer starts at distance 9 or 18, and for every non-super-thick member it starts at the T-junction itself.*

**Theorem 6** (Alignment of the mismatch ray).

[VERIFIED] *Let  $m = 1$ ,  $e \geq 2$ , and let  $r$*

*be the mismatch inner apex ray of Theorem 3, of inner length  $f \cdot b$  (Proposition 5,  $b = f^2 - e^2$ ). Along  $r$ , the near side is  $T_2$ 's single unsplittable  $c$ -edge covering  $[0, f^2]$ , and the far side begins with  $T_1$ 's  $b$ -edge. Then:*

- (i) *no common junction lies in  $(0, f^2)$ : for  $1 \leq k \leq f - 1$ ,  $k \cdot b$  is never  $n_a ef + n_b b + n_c f^2$  with  $n_c \geq 1$  (so every far-side junction is a genuine T-junction, the near  $c$ -edge passing straight through it);*
- (ii) *the far side is exactly  $b^f$ : the only representation of  $f \cdot b$  as  $n_a ef + n_b b + n_c f^2$  with  $n_b \geq 1$  is  $(0, f, 0)$ ;*
- (iii)  *$V = f^2$  is pierced:  $b \nmid f^2$ , so no far-side junction  $k \cdot b$  equals  $f^2$  and a far  $b$ -edge straddles  $T_2$ 's  $\beta$ -corner.*

*All three are machine-checked (`far_near_disjoint`, `far_is_bpow`, `b_not_dvd_fsq`); (ii) alone uses  $\gamma$ -vertex induction (via the walk arithmetic) to fix the first far edge, the rest is pure number theory.*

*Proof.* (i) If  $kb = n_a ef + n_b b + n_c f^2$  then  $(k - n_b)b = f(n_a e + n_c f)$ , so  $f \mid (k - n_b)e^2$  and  $\gcd(e, f) = 1$  give  $f \mid k - n_b$ ; writing  $k - n_b = fs$ , the equation reads  $sb = n_a e + n_c f \geq f > 0$  (as  $n_c \geq 1$ ), so  $s \geq 1$  and  $k \geq n_b + f \geq f$ , against  $k \leq f - 1$ . (ii)  $f \mid n_b$  by the same reduction, and

$n_b \leq f$  by size, so  $n_b \in \{0, f\}$ ;  $n_b \geq 1$  forces  $n_b = f$ , and then  $n_a e f + n_c f^2 = 0$  gives  $n_a = n_c = 0$ . (iii)  $b \mid f^2$  would give  $b \mid e^2$ , hence  $b \mid \gcd(f^2, e^2) = 1$ , so  $b = 1$ , impossible as  $b \geq 2e + 1 \geq 5$ . The far junctions are the multiples  $kb$ , none equal to  $f^2$ , so the  $b$ -edge containing  $f^2$  straddles it.  $\square$

**Theorem 7** (The mismatch ray is completely determined). [PROVED] *Let  $m = 1$ ,  $e \geq 2$ , and let the family be thick,  $f \leq 2e$ . Then along the mismatch inner apex ray, of inner length  $f b$ :*

$$\text{far side} = b^f \text{ (} f \text{ edges),} \quad \text{near side} = \{a^{f-e}, c^{f-e}\} \text{ (} 2(f-e) \text{ edges),}$$

*and the near side is the unique possibility. In particular the near side carries equally many  $a$ - and  $c$ -edges, and no  $b$ -edge.*

*Proof.* The far side is Theorem 6(ii). For the near side,  $T2$ 's  $c$ -edge covers  $[0, f^2]$  and the remainder  $f b - f^2 = f(b - f)$  is covered by whole edges, say  $n_a$  of length  $a$ ,  $n_b$  of length  $b$  and  $n_c$  of length  $c$ . Reducing modulo  $f$  kills  $n_b$  as in Theorem 6(i), and cancelling one factor  $f$  leaves

$$n_a e + n_c f + f = b.$$

Now  $(n_a, n_c) = (f - e, f - e - 1)$  solves it, since  $(f - e)e + (f - e - 1)f = f^2 - e^2 - f = b - f$ . For uniqueness,  $\gcd(e, f) = 1$  and the equation give  $f \mid n_a + e$ , so  $n_a + e = ft$ ;  $t = 0$  is impossible and  $t \geq 2$  would force  $n_a e \geq (2f - e)e$ , whence  $2ef \leq f^2 - f$ , i.e.  $f \geq 2e + 1$ , contradicting thickness. Hence  $t = 1$ ,  $n_a = f - e$ , and then  $n_c f = f(f - 1 - e)$  gives  $n_c = f - e - 1$ . Adding  $T2$ 's own  $c$ -edge produces the multiset  $\{a^{f-e}, c^{f-e}\}$ , whose total length is  $(f - e)(ef + f^2) = f(f - e)(e + f) = f b$  as required.  $\square$

**Theorem 8** (Unique representation of  $kb$ ). [VERIFIED]

*For  $1 \leq k \leq f - 1$  the only way to write  $kb$  as  $n_a e f + n_b b + n_c f^2$  in nonnegative integers is  $b^k$ . (Machine-checked, `kb_unique_rep`; it generalizes Theorem 6(ii), which needed  $n_b \geq 1$ .)*

**Theorem 9** (The angle  $(\pi - \alpha)/2$ ). [VERIFIED] *A vertex figure filling  $(\pi - \alpha)/2$  is  $\alpha + \beta$ , and nothing else. (Machine-checked, `halfpi_minus_alpha_unique`.)*

**Theorem 10** (The  $b$ -run orientation lemma). [PROVED]

*Let a run of  $b$ -edges be laid end to end along a ray, seeded at an apex tile. Then every  $b$ -edge of the run carries  $\alpha$  at its apex end and  $\gamma$  at its far end, and every junction interior to the run is an  $(\alpha, \beta, \gamma)$   $\pi$ -vertex. In particular the  $3\alpha + 2\beta$  figure never occurs along such a run.*

*Proof.* A  $b$ -edge joins its tile's  $\alpha$ - and  $\gamma$ -corners, so at a junction the incoming tile presents  $\alpha$  or  $\gamma$ , and likewise the outgoing one. The  $3\alpha + 2\beta$  figure contains no  $\gamma$ , and two  $\gamma$ 's cannot share a side of a straight line since  $2\gamma = \pi + \alpha > \pi$ . Hence if the incoming corner is  $\gamma$  the figure is  $(1, 1, 1)$ , whose single  $\alpha$  must be the outgoing corner, so the next edge again ends in  $\gamma$ . An apex tile has  $\alpha$  at the apex, hence  $\gamma$  at the far end of its  $b$ -edge, which seeds the induction; it proceeds with no branching.  $\square$

**Theorem 11** (The far region is a scaled tile). [PROVED]

*Suppose the mismatch apex ray is a complete wall, of inner length  $f b$ . Then it cuts off a triangle with angle  $\alpha$  at the apex,  $\beta$  at the base corner and hence  $\gamma$  at its third vertex, and with sides  $f c$ ,  $f a$ ,  $f b$  — that is, the tile scaled by  $f$ . It therefore has area  $f^2$  times the tile's and is tiled by the standard  $f$ -subdivision into  $f^2$  copies, whose ray side is exactly  $b^f$  with the junction chain of Theorem 10.*

**Corollary 12** (The far side can never give a contradiction). [PROVED]

*Under the hypothesis of Theorem 11 — that the mismatch apex ray is a complete wall — no obstruction to a base- $\beta$  tiling can be derived from the far side of that ray, at any  $(e, f)$ : that side admits a genuine tiling. The hypothesis may not be dropped; if the ray is not a complete wall the far region is not a scaled copy of the tile and the conclusion does not follow.*

**Remark 13** (Where the obstruction must instead live). Corollary 12 is a negative result and an important one: it explains why the pierced-corner analysis resisted three independent attempts. The configuration it pins down is not contradictory but *realizable*, so no amount of further chasing on that side can close the branch. The check is exact for every family — for  $(e, f) = (4, 5)$  the far region has sides  $(45, 100, 125) = 5 \cdot (9, 20, 25)$  and contains  $25 = f^2$  of the 59 tiles — and it is confirmed on the genuine 44-tiling, whose mismatch ray has *no* straddling tile and whose far region contains  $16 = (mf)^2$  tiles with sides  $4 \cdot (c, a, b)$ .

The live direction is the other ray. For every thick member with  $m = 1$ ,  $e \geq 2$ , the *matched* inner apex ray is *not* a wall. Indeed if it were, the target would split into  $f^2 + f^2 + (f^2 - e^2)$  tiles, the last a middle triangle  $M$  with apex  $\alpha$ , legs  $fb$  and base  $eb$ . Since  $kb$  has the unique representation  $b^k$  for  $1 \leq k \leq f - 1$  (Theorem 8), both edges of  $M$  at its far corner are  $b$ -edges; but the angle there is  $(\pi - \alpha)/2$ , whose only decomposition is  $\alpha + \beta$  (Theorem 9), and a  $b$ -edge endpoint is an  $\alpha$ - or  $\gamma$ -corner, so no tile present can supply the  $\beta$ . This is the precise mechanism behind the failure of the clean-ray corollary retracted in Remark 15, and it is confirmed on the 44-tiling, whose matched ray carries exactly three straddling tiles against the mismatch ray's zero.

Consequently the branch now reduces to a single question: *can the residual region — the target minus the far wall, containing  $N - f^2$  tiles and crossed by a necessarily straddled matched ray — be tiled?* Every tool developed here (walk arithmetic, vertex figures, unsplittability) is wall-based and goes silent once a tile straddles. What is needed is a bound on where a tile may straddle a ray whose two ends are pinned by  $b$ -edges.

**Remark 14** (Tiles do straddle: the no-straddle hypothesis is false). Remark 13 asks for a bound on where a tile may straddle a ray whose two ends are pinned by  $b$ -edges. The strongest form one might hope for is that no such straddle occurs at all:

(DL) if a segment carries a whole  $b$ -edge at each end, no tile straddles it.

(DL) would make the walk arithmetic of §5.5 applicable to interior segments, and it is the form the funnel arguments implicitly assume. It is false.

Testing it against the three tilings of base- $\beta$  targets exhibited in this note and the main paper — all with tile  $(2, 3, 4)$ , all verified exactly — for every  $b$ -edge we take the line through it and count tiles with interior on both sides whose centroid projects near the segment:

| tiling                                     | $b$ -edges | lines with a straddling tile | worst |
|--------------------------------------------|------------|------------------------------|-------|
| 44-tiling N44B of $(16, 16, 22)$ , $m = 2$ | 44         | 19                           | 3     |
| 44-tiling N44C of $(16, 16, 22)$ , $m = 2$ | 44         | 30                           | 4     |
| 99-tiling of $(24, 24, 33)$ , $m = 3$      | 99         | 64                           | 4     |

Straddling near  $b$ -edges is common rather than exceptional. Consequently the missing ingredient is not a prohibition but a *quantitative* bound: a description of the set in which a straddling tile's crossing points must lie. Since every vertex of a tiling lies in a finitely generated  $\mathbb{Z}$ -module over  $\mathbb{Q}(\sqrt{D})$ , and  $b$ -pinning fixes the segment's endpoints inside that module, such a bound is the natural object to seek; we have not found it. The computation is `code/rust/walls` (binary `straddle`), and it is recorded here because a negative result of this shape saves the reader the attempt, and because it explains why the interior argument of §5.5 has never been written out.

**Remark 15** (Two corrections this forces). Theorem 10 is uniform in  $(e, f)$  and in  $m$ , and is confirmed on the genuine 44-tiling, where all four far  $b$ -edges of the mismatch ray and both sides of the clean ray read  $\alpha \rightarrow \gamma$ . It strictly generalizes Proposition 5, which established only the first step. But it also refutes two statements of an earlier version, which we withdraw.

First, the assertion that the *clean* apex ray is  $b^f$  against  $b^f$ . Its justification — “by the symmetry of the isosceles target both inner rays have the same inner length  $f b$ ” — conflates two objects. The symmetry does apply to the geometric rays; it does not apply to the tiling’s maximal edge *segment* along a ray, which may terminate early at a vertex where the edges turn away. In fact the clean ray’s segment can never reach the base: if it did, both final  $b$ -edges would contribute  $\gamma$  at an interior point of the base, where the tiles above sum to  $\pi$ , contradicting  $2\gamma > \pi$ . On the 44-tiling the two segments have lengths 12 and 9 respectively, and the clean ray terminates at an interior  $\beta + 3\gamma$  vertex — which is precisely the  $M_4$  vertex the census forces.

Second, and more seriously, the same objection applies to the hypothesis under which Theorems 6 and 7 were stated: that the *mismatch* ray’s maximal segment attains the full length  $f b$ . That too was justified by symmetry, and it too now requires a genuine proof. The arithmetic of those theorems is unaffected — given a chain of total length  $f b$  containing a  $b$ -edge, the conclusions stand and remain machine-checked — but their geometric hypothesis must be stated explicitly. Establishing that the mismatch segment reaches the base is the single highest-value open step in this branch: it would restore Theorems 6 and 7 unconditionally for every thick member.

**Remark 16** (Scope and consequences). Recorded in [1]. Full discussion in [1].

**Remark 17** (Scope, and why  $m = 1$  is the rigid case). Recorded in [1]. Full discussion in [1].

**Theorem 18** (Walk structure at  $m = 1$ ). [VERIFIED] *Let  $m = 1$ ,  $\gcd(e, f) = 1$ ,  $1 \leq e < f$ , and consider a tiling of the base- $\beta$  target.*

- (i) *If  $f^2 > 2e^2$ , then each equal side carries no  $b$ -edge. Its edges are  $a$ ’s and  $c$ ’s with  $n_a = fk$  and  $n_c = f - ke \geq 1$ , and it ends with a  $c$ -edge at the apex.*
- (ii) *If  $f^2 > 2ef + e^2$  — equivalently  $f/e > 1 + \sqrt{2}$  — then the base carries exactly  $e$   $b$ -edges, and  $n_a = f\ell$ ,  $n_c = e(2 - \ell)$  with  $\ell \in \{0, 1\}$ . In particular*

$$R_{\text{base}} \in \{e, 2e\}.$$

*Proof.* A side of length  $L$  is exactly partitioned by the tile edges on it, giving  $n_a ef + n_b b + n_c f^2 = L$  with  $n_a, n_b, n_c \geq 0$ , and  $n_c \geq 1$  by Proposition 25.

(i) Here  $L = f^3$ . Rearranging with  $b = f^2 - e^2$  gives  $n_b e^2 = f(n_a e + n_b f + n_c f - f^2)$ , so  $f \mid n_b e^2$  and hence  $f \mid n_b$  by coprimality; write  $n_b = ft$ . If  $t \geq 2$  the  $b$ -edges alone cost at least  $2fb = 2f^3 - 2fe^2$ , which together with  $n_c f^2 \geq f^2$  exceeds  $f^3$  unless  $f^2 \leq 2e^2$ . If  $t = 1$ , cancelling  $f$  leaves  $n_a e + n_c f = e^2$ , so  $n_a \leq e$  and  $f \mid e(e - n_a)$ ; coprimality with  $0 \leq e - n_a \leq e < f$  forces  $n_a = e$  and then  $n_c = 0$ , contradicting  $n_c \geq 1$ . So  $t = 0$ . With  $n_b = 0$  the equation is  $n_a e + n_c f = f^2$ , whence  $f \mid n_a$ ; writing  $n_a = fk$  gives  $n_c = f - ke$ . Finally, the apex tile on that side presents  $\alpha$  (Proposition 24), whose incident edges are  $b$  and  $c$ ; as the side has no  $b$ -edge, it ends with a  $c$ -edge.

(ii) Here  $L = e(3f^2 - e^2)$ , and the same rearrangement gives  $(n_b - e)e^2 = f(n_a e + n_b f + n_c f - 3ef)$ , so  $f \mid n_b - e$ ; write  $n_b = e + ft$ . Negative  $t$  makes  $n_b < 0$ . For  $t \geq 1$  the  $b$ -edges alone cost at least

$$(e + f)b = (f + e)^2(f - e) = e(3f^2 - e^2) + f(f^2 - 2ef - e^2),$$

which already exceeds  $L$  under the hypothesis, before the mandatory  $c$ -edge is placed. So  $n_b = e$ ; subtracting and cancelling  $f$  leaves  $n_a e + n_c f = 2ef$ , whence  $f \mid n_a$ , and writing  $n_a = f\ell$  gives  $n_c = e(2 - \ell)$ . Then  $n_c \geq 1$  forces  $\ell \leq 1$ .

Machine-checked (`equal_side_no_b`, `equal_side_shape`, `base_b_count` in `lean/BaseBetaWalkArith.lean`), and cross-checked by enumerating all walk solutions for the 198 coprime pairs with  $f \leq 25$ : no discrepancies.  $\square$

**Remark 19.** Both hypotheses are sharp:  $n_b = 2f$  on an equal side is admissible exactly when  $f^2 \leq 2e^2$ , and  $n_b = e + f$  on the base exactly when  $f^2 \leq 2ef + e^2$ , by the displayed identity. Of the 42 base- $\beta$  prime candidates below 1000, 28 satisfy (i) — including the thick values 23, 131 and 167, since (i) needs only  $f/e > \sqrt{2}$  — and 17 satisfy (ii). The theorem structures those cases; it does not by itself exclude any of them.

**Theorem 20** (The  $e = 1$  subfamily reduces to one walk). [PROVED] *Let  $e = 1$ ,  $m = 1$  and  $f \geq 3$ , so the target is  $(f^3, f^3, 3f^2 - 1)$ , the tile is  $(f, f^2 - 1, f^2)$  and  $N = 3f^2 - 1$ . Both hypotheses of Theorem 18 hold ( $f^2 > 2$  and  $f^2 > 2f + 1$ ). In any tiling:*

- (i) *neither equal side carries a  $b$ -edge; each equal side begins and ends with a  $c$ -edge, and therefore carries at least two;*
- (ii) *the base carries exactly one  $b$ -edge and exactly one  $c$ -edge — so  $R_{\text{base}} = 1$  — and its first and last edges are  $a$ -edges.*

*Thus the base walk is a permutation of  $(a^f, b, c)$  beginning and ending with  $a$ , with the  $b$  in position  $3, \dots, f$ .*

*Proof.* (i) An equal side has length  $f^3$  and its edges lie in  $\{f, f^2 - 1, f^2\}$ . Reducing  $n_a f + n_b(f^2 - 1) + n_c f^2 = f^3$  modulo  $f$  gives  $f \mid n_b$ . If  $n_b \geq 2f$  then  $n_b(f^2 - 1) \geq 2f^3 - 2f > f^3$  for  $f \geq 2$ . If  $n_b = f$  then  $f(f^2 - 1) = f^3 - f$  leaves  $f$ , forcing  $n_c = 0$  and  $n_a = 1$ , hence no  $c$ -edge on that side — contradicting Proposition 25. So  $n_b = 0$ , and then  $n_a + n_c f = f^2$ , so  $f \mid n_a$ ; write  $n_a = fk$ ,  $n_c = f - k$ .

By Proposition 24 the apex tile on that side presents  $\alpha$ , whose incident edges are  $b$  and  $c$ ; since the side carries no  $b$ -edge, it ends with a  $c$ -edge. Its other end is handled in (ii): there we show the base begins and ends with  $a$ -edges, so the base-corner tile's  $c$ -edge lies on the equal side, which therefore begins with a  $c$ -edge. Were  $n_c = 1$  the single  $c$ -edge would be both the first and the last, forcing the side to consist of one edge and  $f^3 = f^2$ ; so  $n_c \geq 2$ .

(ii) The base has length  $3f^2 - 1$ . Reducing modulo  $f$  gives  $n_b \equiv 1$ . If  $n_b \geq f + 1$  then  $n_b(f^2 - 1) \geq f^3 + f^2 - f - 1 > 3f^2 - 1$  for  $f \geq 3$ , so  $n_b = 1$ . Subtracting,  $2f^2 = n_a f + n_c f^2$ , so  $n_a = f(2 - n_c)$  with  $n_c \in \{0, 1, 2\}$ , and  $n_c \geq 1$  by Proposition 25.

If  $n_c = 2$  then  $n_a = 0$  and the base is three edges, a permutation of  $(b, c, c)$ . By Proposition 28 the first two edges lie in  $\{a, c\}$  and so do the last two; with no  $a$ -edge available this forces all three to be  $c$ -edges, contradicting  $n_b = 1$ . Hence  $n_c = 1$  and  $n_a = f$ , so the base is a permutation of  $(a^f, b, c)$  with  $n = f + 2$  edges, and Proposition 28 places the  $b$  away from the first two and last two positions.

It remains to exclude  $E_1 = c$ . Suppose it. Then the unique  $c$ -edge is  $E_1$ , so every one of  $E_2, \dots, E_n$  is an  $a$ - or  $b$ -edge and places a  $\gamma$  at one of its endpoints; there are  $f + 1$  of them and  $f + 1$  interior junctions  $J_1, \dots, J_{n-1}$ . The corner tile  $T_A$  has its  $c$ -edge on the base, so its  $\alpha$  vertex is  $J_1$  and its  $\gamma$  vertex lies on the equal side; by Proposition 28 its partner  $T_3$  presents  $\gamma$  at  $J_1$ . So  $J_1$  already carries a  $\gamma$ , and since a  $\pi$ -vertex carries at most one (Proposition 23),  $E_2$  must place its  $\gamma$  at  $J_2$ ; inductively  $E_i$  places at  $J_i$ , and  $E_n$  places at  $J_n$ , a corner — contradicting Proposition 24. Hence  $E_1$  is an  $a$ -edge, and the same argument at  $B$  gives  $E_n$  an  $a$ -edge. This is what (i) used.  $\square$

**Remark 21.** Theorem 20 reduces the whole  $e = 1$ ,  $m = 1$  subfamily — which contains the base- $\beta$  prime candidates  $N = 3f^2 - 1$ , among them 11, 47, 107, 191, 431, 587 and 971 below 1000 — to the single question of whether that one base walk is realizable. It also corrects a plausible-looking plan: one might hope to close the subfamily by proving  $R \geq 2$  on every side. That is false for the base. The walk with  $R_{\text{base}} = 2$  is exactly the one the corner parallelogram kills, and the survivor has  $R_{\text{base}} = 1$ ; on the base, “ $R \geq 2$ ” is not a lemma reducing the problem but a restatement of it. On the two equal sides  $R \geq 2$  does hold, and is proved above.

**Remark 22** (The  $e \geq 2$  residue). The  $e \geq 2$  residue ( $N = 23, 59, 71, 83, 131, 167, \dots$ ; the members 23 and 71 are now settled by search, Table 1) resists all three natural invariant methods, and in two cases *provably*:

- **Linear/colouring invariants** — dead by Remark 20.
- **$b$ -edge census.** On a maximal segment of length  $L$  the two sides carry  $q_1 \equiv q_2 \pmod{f}$  (both  $\equiv -e^{-2}L$ ), and the entire census — in *every* modulus, and in its per-direction refinement — is a formal consequence of the single parity identity  $N(a+b+c) \equiv 2X+Y \pmod{2}$ , valid for all  $(e, f, m)$ . So no congruence of this type can bite.
- **Euler / vertex counting.** For any such tiling  $V - E + F = 2$  is *identically* the sum of the three corner identities  $\#\alpha = \#\beta = \#\gamma = N$ ; the vertex-type system collapses to  $Q_{013} = 1 + P_{320} + 2Q_{640} + Q_{431} + W_{320}$  and  $3P_{320} + P_{111} + 6Q_{640} + 4Q_{431} + 2Q_{222} + 3W_{320} + W_{111} = N - 3$ , which is feasible for every  $e \geq 2$ ,  $m = 1$  target. So Euler carries no information here. Here  $W_{320}$  and  $W_{111}$  count the interior vertices at which some tile is met in the *relative interior* of one of its edges: such a tile contributes a straight angle, so the remaining tile angles sum to  $\pi$  rather than  $2\pi$  and the figure is  $(3, 2, 0)$  or  $(1, 1, 1)$  (Proposition 23). These types are not optional — the dissections here are not edge-to-edge (Remark 18) — and an earlier version of this census omitted them. Their presence only enlarges the feasible set, so the conclusion is unaffected.

What *is* gained is citation-free structure, valid for all  $(e, f, m)$ : the  $\gamma$ -trap (Proposition 25) and the corner parallelogram (Proposition 28), both proved above from the corner figures and the unsplittability of  $b$ , and both used by the exact search as prunes. They were checked against the two genuine certificates before being proved: the first-two-edges rule of Proposition 28 holds on all six sides of the 44- and 99-tilings, and at each of the four base corners exactly one of the two sides begins with a  $c$ -edge.

This isolates the residue to one clean statement. Since  $q_{\text{base}} \equiv em \pmod{f}$  while  $q_{\text{equal}} \equiv 0$ , we have:

**if the base carries no  $b$ -edge, then  $f \mid em$ , hence  $f \mid m$ , hence no tiling with  $m = 1$  — for every  $e$  at once.**

The 44-tiling ( $m = 2$ ) indeed carries *zero*  $b$ -edges on its entire boundary, consistent with this. But Remark 21 now shows the converse direction is a dead end: the 99-tiling has  $f \nmid m$  and its base carries seven  $b$ -edges, so “the base never carries a  $b$ -edge” is *false* in general, and the displayed implication — whose hypothesis is equivalent to  $f \mid m$  — cannot be used to exclude  $m = 1$  without already knowing  $f \mid m$ , which is false. A fourth method also fails, and provably: *boundary counting*. Each side  $S$  of  $ABC$ , cut into  $k_S$  whole edges, forces at least  $2k_S - 1$  tiles to meet it (the  $k_S$  inner tiles, plus a distinct “middle” tile at each of the  $k_S - 1$  junctions, since a triangle has no two collinear edges); this is exactly tight on the 44-tiling (base  $k = 8$ ,  $15 = 2 \cdot 8 - 1$  tiles). But at  $m = 1$  the boundary is *minimal* — for  $e = 1$  the least feasible total is  $\sum k_S = 3f + 2$  — so counting sees only  $O(f)$  of the  $N = \Theta(f^2)$  tiles, and  $\sum (2k_S - 1) = 6f + 1$  exceeds  $N = 3f^2 - 1$  only at  $f = 2$ , the case already closed. The ratio  $(6f + 1)/(3f^2 - 1) \rightarrow 0$ :  $m = 1$  is the case *least* constrained by counting, not the most.

So no *invariant* route survives: linear invariants, the census, Euler and counting are all provably vacuous, and the one arithmetic statement that would settle it ( $f \mid m$ ) is now known to be false. What does survive is *structure*. Theorem 2 classifies the boundary walks for every  $(e, f)$  at  $m = 1$ , and in the thin regime  $f > 2e$  reduces the whole branch to two possible bases and  $b$ -free equal sides — the  $e \geq 2$  analogue of Remark 14, and the first statement in this branch that is uniform in  $e$ . Parts (iii)–(iv) need no thinness at all: at  $m = 1$  *no equal side carries a  $b$ -edge*, for every  $(e, f)$ , so all boundary  $b$ -edges lie on the base; and the base’s  $b$ -count  $Q = e + fj$  obeys  $j(f - e) \leq e - 1$ . It does not close the branch, but it does locate the difficulty exactly. The bound  $j \leq (e - 1)/(f - e)$  is uniform, and it degrades precisely as  $f$  falls towards  $e$ : at  $e = 1$  it

pins  $j = 0$  for every  $f$  (one  $b$ -edge on the base, the  $e = 1$  rigidity of Remark 14, now a corollary); at  $f > 2e$  it still pins  $j = 0$ ; but in the *thick* regime  $f \leq 2e$  it leaves  $\lfloor (e-1)/(f-e) \rfloor + 1$  values of  $j$  and the trichotomy dissolves. That is precisely where the open members sit —  $N = 59$  is  $(e, f) = (4, 5)$ , so  $j \leq 3$ ;  $N = 66$  is  $(3, 5)$ , so  $j \leq 1$ ; likewise  $N = 23, 83, 131, 167$ . We record the branch as open and settle its members one at a time by exhaustive search. Table 1 lists every member below 110; of the eleven, eight are settled and 59 and 66 are now excluded by exhaustive search (apex-down with the walk prune; 1 838 175 and 7 232 464 nodes), leaving 83 under search. ( $N = 83$ ,  $(e, f) = (5, 6)$ , is the smallest base- $\beta$  prime not yet settled.) The walk classification also feeds back into the search: the  $\gamma$ -trap and the corner parallelogram have the multiset corollaries  $R \geq 1$  on every side,  $Q_{\text{base}} \leq k - 4$  and  $Q_{\text{side}} \leq k - 2$  ( $k$  the number of edges of the walk), and the engine now prunes any partial boundary walk that fits no surviving multiset (prune P5). The rule was validated positionally on all six sides of the two genuine tilings before use — the 44-tiling’s base *aaaaccca* and the 99-tiling’s base *aabbbbbbbcc*, which is exactly tight for  $Q \leq k - 4$  — the strengthened engine re-finds the 44-tiling from scratch (852,093 nodes; the found tiling passes the independent verifier: congruence, exact area, containment, pairwise disjointness) — and the corner rule is applied only after checking that the tile’s  $b$ -edge is unsplitable ( $n_a a + n_c c = b$  has no solution), which is what forces the parallelogram. For  $N = 59$  this cuts the admissible bases from seven to four, for  $N = 66$  from four to three.

### 3. THE $e = 1$ FAMILY

**Theorem 23** (The  $e = 1$  base- $\beta$  family).

[CONJECTURE:

EFFECTIVE, THE STRIP-AND-COLUMN EXPOSITION BEING DEFERRED] *Let  $f \geq 2$ , and consider the base- $\beta$  instance  $(e, f) = (1, f)$  at  $m = 1$ : the tile  $(f, f^2 - 1, f^2)$  on the isosceles target with base angles  $\beta$ , base  $Y = 3f^2 - 1$  and equal sides  $X = f^3$ , with tile count  $N = 3f^2 - 1$ . No such tiling exists. In particular no prime  $p = 3f^2 - 1$  whose only base- $\beta$  representation is  $(1, f)$  is a number of congruent triangles into which a triangle can be cut; this covers  $p = 11, 47, 107, 191, 431$ .*

The proof is a finite chain of forcing lemmas; we state the architecture and prove the three arithmetic pivots, deferring the full strip-and-column exposition to the extended version of this note. The chain: the walk trichotomy (base walks  $(2f, 1, 0)$ ,  $(f, 1, 1)$ ,  $(0, 1, 2)$ , the last dead by Theorem 20); the corner and partner forcing; for each corner, either the quadratic growth completes ( $k = f$ ) or a first break occurs, on the base (then the walk letter is forced into  $\{a, c\}$ , and two base-breaks are incompatible with the single  $c$  of  $(f, 1, 1)$ ) or on a side. A side-break at level  $j$  forces a horizontal column of  $f$  tiles whose junction fillers are exactly the triangles on consecutive apexes (Lemma 25); the structure recurses one level per step and reaches the base, where the apexes must be walk junctions (no vertex of a tile can lie interior to a boundary edge), pinning the orientations (Lemma 26) and forcing the letters  $a^f$  on  $[0, f^2]$  (Lemma 28). In  $(f, 1, 1)$  the walk then ends in  $b$  or  $c$ , contradicting the corner forcing; in  $(2f, 1, 0)$  the letter after  $f^2$  is the  $b$ , while the terminal vertex at  $(f^2, 0)$  carries exactly  $\gamma + \alpha$  from the filler and column, leaving a wedge of exactly  $\beta$ , whose tile presents  $a$  or  $c$  — never  $b$ . Every case terminates in one of these contradictions.

**Remark 24** (At  $e = 1$  the completion branch is empty, so the column is unconditional). Remark 47 already establishes that complete blocks are unavailable at  $e = 1$ , and concludes from it that the hypothesis there *is* the case rather than a step towards it. What follows is the structural dividend of that observation, which was not taken.

The chain above carries, at each corner, the alternative “either the quadratic growth completes ( $k = f$ ) or a first break occurs”. At  $e = 1$  the first alternative never occurs, and — unlike the argument of Remark 47, which reads the contradiction off the east corner’s  $c$ -foot — this can be seen at the west corner from the letter budget alone, which is what closes the west-only retreat

proposed in Remark 30. A complete west block has  $f$   $a$ -feet on the base; feet are consecutive from the corner, so it requires the base word to begin with  $a^f$ . A complete east block at  $e = 1$  has a single  $c$ -foot, so it requires the base word to end with  $c$ . But Theorem 20 makes the base word a permutation of  $(a^f, b, c)$  beginning *and* ending with  $a$ : the first requirement asks for  $f+1$  letters  $a$  out of  $f$ , and the second contradicts the final  $a$  outright.

So both corners break. A base-break consumes the walk's  $c$ , and  $(f, 1, 1)$  has only one, so at most one of the two breaks is on the base; hence at least one corner breaks on a *side*, and the forced horizontal column of  $f$  tiles exists in every tiling of every  $e = 1$  member. The column is therefore not one branch of a case split at  $e = 1$  — it is unconditional, and the subfamily rests on the single deferred step that the column reaches the base.

This also settles the status of Hypothesis 29 at  $e = 1$ , since the hypothesis is precisely the assertion that the completion branch is the one that occurs: it holds there exactly when no tiling exists, so no exclusion of an  $e = 1$  member may cite it. Formalised in `lean/Erdos634/WallsCircular.lean` (`west_block_never_complete`, `both_corners_break`, `some_side_break`), axiom-clean.

**Lemma 25** (Filler identity).

[PROVED]

With  $J_i$  the  $i$ -th column junction and  $\text{ap}_i$  the  $i$ -th apex, the triangle  $(J_i, \text{ap}_i, \text{ap}_{i+1})$  has sides exactly  $c, b, a$ :  $\left(\frac{3f^2-1}{2f}\right)^2 + c^2 \sin^2 \beta = f^4$  and  $\left(\frac{f^2-1}{2f}\right)^2 + c^2 \sin^2 \beta = (f^2 - 1)^2$ , both identities in  $f$  (clear denominators and expand; machine-checked in `lean/W2Core.lean`, axiom-free). Hence the column and its fillers tile the full inter-level strip and lay a run of  $a$ -edges one level down, whose far sides decompose only as single  $a$ -edges.

**Lemma 26** (Offset congruence).

[PROVED] The terminal column's base apex lies at

$[j(3f^2 - 1) + \sum_i \varepsilon_i]/(2f) + kf$  with  $\varepsilon_i \in \{-(f^2 - 1), 3f^2 - 1\}$  per level. Modulo  $f$  — for every  $f$ , of either parity, since  $f^2 \equiv 0$  — these are  $+1$  and  $-1$ . Let  $p$  be the number of levels taking  $+1$  and  $q = j - p$  the rest, so  $\sum \varepsilon_i \equiv p - q$  and

$$\sum \varepsilon_i - j = -2q.$$

Integrality of a base vertex gives  $f \mid 2q$  when  $f$  is odd, and  $2f \mid 2q$  when  $f$  is even; either way  $f \mid q$ , and  $q \leq j < f$  forces  $q = 0$ . Hence  $\sum \varepsilon_i = j$ , every level is  $(\gamma, \beta)$ -oriented, and the apexes lie at consecutive multiples of  $f$ .

**Remark 27** (The evenness hypothesis is removable). Earlier versions of Lemma 26 worked modulo  $2f$ , where the two offsets are  $+1$  and  $-1$  only when  $f$  is even, since that congruence needs  $f^2 \equiv 0 \pmod{2f}$ ; for odd  $f$  the residues are  $f+1$  and  $f-1$  and the computation does not transfer. That is why Theorem 23 was previously stated for even  $f$  only, leaving the members  $N = 26, 74, 146, \dots$  claimed by Theorem 31 but uncovered.

Both defects disappear at modulus  $f$ . There the residues are  $\pm 1$  with no parity hypothesis, and parametrising by the numbers  $p, q$  of  $+1$  and  $-1$  levels makes  $\sum \varepsilon_i - j = -2q$  manifestly even, so the parity check that the odd case appeared to require never has to be made: it is absorbed into the parametrisation. The forcing is `lean/OffsetForcing.lean` (`forcing_odd`, `forcing_even`, `forcing`), general in  $f$  and axiom-clean, superseding the per-member even- $f$  checks of `lean/W2Core.lean`.

**Lemma 28** (Interior multiples). [CONJECTURE: EFFECTIVE, RESTING ON THE DEFERRED COLUMN] A base edge of length  $f^2 - 1$  or  $f^2$  placed with its left end below  $f^2$  contains a multiple of  $f$  in its open interior, which would be a column apex interior to a boundary edge — impossible. Hence the letters covering  $[0, f^2)$  are exactly  $f$  copies of  $a$ .

The corner cases  $(1, 2)$ ,  $(1, 3)$ ,  $(1, 4)$  are additionally settled by exhaustive certificates (the seeded refutations of the side-break configurations; 267, 1809 and 1728 nodes, all branches

dead), and the members 11, 26, 47 by the full engine searches recorded in the main paper. The arithmetic layers of the chain are machine-checked in `BAdjacency.lean`, `Rigidity.lean`, `W2Core.lean`, `MidTriangle.lean` and `SurplusLattice.lean` (all kernel-only, axiom-clean).

#### 4. THE GENERAL BASE- $\beta$ ARCHITECTURE

The  $e = 1$  machinery of Section 3 extends to every coprime pair. Four pillars are identities in  $(e, f)$ : the column reach  $b \sin \gamma = c \sin \beta$  (law of sines); the filler-mesh identity (the column's junction fillers are exactly the triangles on consecutive apexes, sides  $c, b, a$ ; `lean/GeneralPillars.lean`); the sector inequalities  $\gamma - (\alpha + \beta) = \alpha$  and  $\cos \gamma = -e/2f < 0$ ; and the interior-multiple bounds. Two master lemmas complete the frame (`lean/MasterLemmas.lean`): a base segment of length  $a$  between forced junctions decomposes only as a single  $a$  (two applications of  $\gcd(e, f) = 1$ ), and the two corner-block types —  $f$   $a$ -feet, or dually  $e$   $c$ -feet — have identical footprints  $ef^2$  (the relation  $ec = fa$ ), so in every walk of the trichotomy the two corner structures consume exactly the non- $b$  letters and the middle is exactly the  $e$   $b$ -letters. Walks containing an  $a$ -block die at the middle's  $\alpha + \beta$  corner with  $(b, b)$  rays; the walk  $(0, e, 2e)$  is forced to  $c^e b^e c^e$ , its two dual blocks complete, and the resulting pentagon is refuted (certificate at (2, 3): 85 nodes, all branches dead; the general argument runs through the dual-junction overhangs). The orientation congruences force  $(\gamma, \beta)$  at every level for both parities of  $f$ . The assembled exposition of the general recursion is in preparation; every arithmetic component cited above is kernel-checked, and the certified refutations cover (1, 3), (1, 4), (2, 3).

#### 5. THE BASE- $\beta$ BRANCH: NO INSTANCE AT $m = 1$

This section assembles the complete argument. Throughout,  $(e, f)$  is coprime with  $1 \leq e < f$ , the tile is  $(a, b, c) = (ef, f^2 - e^2, f^2)$  with angles  $(\alpha, \beta, \gamma)$ ,  $\gamma = 2\alpha + \beta$ ,  $3\alpha + 2\beta = \pi$ ,  $\alpha/\pi$  irrational; the target  $\Delta$  is isosceles with base angles  $\beta$ , base  $Y = e(3f^2 - e^2)$ , equal sides  $X = f^3$ , and  $N = 3f^2 - e^2$ .

The argument below is *conditional*, on one geometric statement about the base corners that we isolate here and do not prove. We state it first, so that nothing downstream is read as unconditional.

**Hypothesis 29** (Complete corner walls).

[CONJECTURE: OPEN AT  $e \geq 2$ ,

AND AT  $e = 1$  EQUIVALENT TO ITS OWN CONCLUSION] Let  $\mathcal{T}$  be a tiling of the base- $\beta$  target at  $m = 1$ . Then neither base-corner structure is starved or broken: each is the complete block —  $f$   $a$ -feet at the west corner,  $e$   $c$ -feet at the east — whose hypotenuse is a straight segment, of length  $f b$  resp.  $e b$ , partitioned into whole  $b$ -edges.

**Remark 30** (Hypothesis 29 is asymmetric; the  $p = 0$  reading is its west half). Two readings of the hypothesis are in circulation in this development and they are not equivalent.

The block form above is asymmetric:  $a$ -feet west,  $c$ -feet east. By Lemma 99 a corner tile laying  $c$  on the base lays  $a$  on the equal side, so the block form puts an  $a$ -edge at the foot of the east side and gives  $p_{\text{west}} = 0$ ,  $p_{\text{east}} \geq 1$ . That asymmetry is visible in both tilings we possess: the certified 44-tiling (`Tiling44.lean`, the tiling N44B) has base walk  $a^4 c^3 a$  with west side  $c^4$  and east side  $ca^2 c^2$  read from the base corner, and the 99-tiling of (24, 24, 33) has base  $a^2 b^7 c^2$  with west side  $c^6$  and east side  $a^2 b^2 c^2 b^2$ . Both are at  $m > 1$ . (The words are the exact sweep of `code/interface.census.py`; an earlier version of this remark misquoted the 44-tiling's words as  $a^4 c^2 ac / a^4 c^2$ .)

Against this, §5.7 reads the hypothesis as “exactly the assertion  $p = 0$ ”, which is symmetric in the two sides, and Theorem 20 proves that at  $e = 1$ ,  $m = 1$ ,  $f \geq 3$  the base walk begins *and* ends with an  $a$ -edge — so its single  $c$ -edge is interior and neither corner carries  $c$ -feet. The block form and that theorem cannot both describe an  $m = 1$  tiling at  $e = 1$ .

The discrepancy is localised, and Remark 47 localises it. Its derivation reads: “the complete *west* block is the tile scaled by  $f$ , so its far side — which at  $m = 1$  is the whole equal side — is subdivided into exactly  $f$   $c$ -edges, i.e.  $p = 0$ .” That argument uses the west block only, and the conclusion is drawn for the west side; it is then stated for “the sides”. The east block, being the tile scaled by  $e$ , has far side  $ea$  and therefore places  $e$   $a$ -edges at the foot of the east side.

That resolution does not survive, and the reason is that the decisive clash is at the *west* corner, not the east. The complete west block has  $f$   $a$ -feet on the base; feet are consecutive and begin at the corner, so the block form asserts that the base word begins with  $a^f$ . By Theorem 20 the base word at  $e = 1$ ,  $m = 1$ ,  $f \geq 3$  is a permutation of  $(a^f, b, c)$  whose *last* letter is also an  $a$ , and  $f + 1$  letters  $a$  do not exist. Hence at every  $e = 1$  member the west corner of every tiling is starved or broken. Since the west block is the sole premise of the derivation of  $p = 0$  quoted above, the west-only reading is not a weaker surviving form of the hypothesis: it has no premise at all at  $e = 1$ . The correct conclusion is that at  $e = 1$  the hypothesis holds exactly when no tiling exists, so any exclusion of an  $e = 1$  member citing it is circular. This is `lean/Erdos634/WallsCircular.lean`, axiom-clean; the counting step is `BaseWordBlock.no_f_plus_one_a`. The  $e = 1$  members remain excluded by the reduction of Theorem 23, which nowhere mentions the hypothesis; that reduction rests in turn on the deferred strip-and-column step, so the subfamily stands reduced rather than closed, and the hypothesis is of no help in closing it.

So the hypothesis is genuinely asymmetric, and for  $e \geq 2$  the identification with  $p = 0$  holds for the west side. Both tilings we possess have west side all  $c$ -edges; the 99-tiling’s east side begins with  $a$ -edges as the block form predicts, while the 44-tiling’s east side begins with a  $c$  — consistent with Remark 33, whose east corner blocks are broken in exactly that tiling, so it is no counterexample to the block form’s prediction for *complete* corners. Every statement of the form “Hypothesis 29 is exactly  $p = 0$ ” should be read as applying to the west side; on the east the hypothesis gives  $p \geq 1$ .

That the block reading fails at  $e = 1$  is not new here: Remark 32 already records that Theorem 20, forcing the base walk to begin and end with  $a$ -edges, “is incompatible with the complete-block reading of the base, so *every* hypothetical  $e = 1$  tiling falls into the starved-or-broken clause”. What the present remark adds is only where the two readings part company — Remark 47 argues from the west block and states its conclusion for both sides — together with the confirmation from the two certified tilings.

The practical consequence is that the hypothesis should be split into its west and east halves before being used further, and that a result quoted as “establishing Hypothesis 29” should say which half.

Having split them, it is worth recording how unequal the two halves are, since the split might be read as doubling the work and it does not. The base accounts as

$$e(3f^2 - e^2) = \underbrace{fa}_{\text{west foot}} + \underbrace{eb}_{\text{middle}} + \underbrace{ec}_{\text{east foot}},$$

an identity in  $(e, f)$ . The west block is the tile scaled by  $f$  and carries  $f^2$  tiles; the east block is the tile scaled by  $e$  and carries only  $e^2$ . Three consequences.

First, *at  $e = 1$  the east block is a single tile*, and the east half of the hypothesis has no content beyond Lemma 99: a base corner tile laying  $c$  lays  $a$  on the side. So on the whole thin family the east half is free, and what remains there is the base-form question of Lemma 98, which the two halves share.

Second, the east block is smaller than the west at every member —  $e^2$  against  $f^2$ , a factor between 4.8 and 49 on the members tabulated in §5.10 — so any method that settles the west settles the east *a fortiori* in size, if not automatically in content.

Third, and most usefully, the side the east block presents to the equal side is its  $a$ -side, of length  $ea$ ; and Proposition 141 makes the  $a$ -side of an inflation *rigid*, namely  $a^e$ . The  $p = 0$  versus  $p = 1$  ambiguity that is the whole difficulty of the west half lives on the  $c$ -side, which for the east block lies along the base rather than along an equal side. The east half is therefore not a second copy of the crux: it is a smaller problem, and the feature that makes the west hard is absent from it. This does not affect §5.9, whose results are statements about  $p$  on a given equal side and stand on their own proofs.

**Theorem 31** (The base- $\beta$  branch, conditional on Hypothesis 29). [CONJECTURE] *Assume Hypothesis 29. Then  $\Delta$  admits no tiling by  $N$  congruent copies of the tile. Consequently no prime  $N \equiv 11 \pmod{12}$  is a number of congruent triangles into which a triangle can be cut, and with the results of the main paper the folklore conjecture would follow: no prime  $N \equiv 3 \pmod{4}$ ,  $N > 3$ , is such a number.*

**Remark 32** (Why the hypothesis is needed, and where it bites). Hypothesis 29 is exactly the step Remark 15 already identifies as “the single highest-value open step in this branch”; isolating it here makes the dependence explicit rather than leaving it distributed over the funnels. Two places consume it.

First, the funnels of §5.5 dispose of a walk by completing both corner blocks; if a corner is *starved or broken* the text routes the case to “the walk arithmetic ... exactly as in the  $e = 1$  case”, and the  $e = 1$  case is Theorem 23, whose proof defers the strip-and-column exposition. At  $e = 1$  this is not a corner case but the whole problem: Theorem 20 forces the base walk to begin and end with  $a$ -edges, which is incompatible with the complete-block reading of the base, so *every* hypothetical  $e = 1$  tiling falls into the starved-or-broken clause.

Second, Lemma 41 is stated “after both dual blocks complete”, and its stubs are region-boundary segments only because the opposite side is a wall of whole  $b$ -edges. In a dissection that is not edge-to-edge — which the main paper explicitly permits — a far-side edge may begin inside a stub and overhang it, so no such segment need be created. The arithmetic of the lemma is unaffected and is machine-checked general in  $(e, f)$  (Lemma 42, `lean/PentagonLemma.lean`); what is conditional is that a stub arises at all.

The unconditional consequences of this section are therefore the arithmetic ones, together with the members settled by certified exhaustive search independently of any of it:  $N = 11, 23, 26, 47, 59, 66, 71, 107$ . The member  $(1, 2)$ , i.e.  $N = 11$ , is moreover closed unconditionally by Theorem 1, whose argument is boundary-only and needs no corner-block hypothesis.

**Remark 33** (A corner block can break: the hypothesis needs  $m = 1$  essentially). It is worth recording what the corner blocks are, and that Hypothesis 29 is genuinely a statement about  $m = 1$  and not a local fact.

First the identification. At a base corner the complete block with  $f$   $a$ -feet has footprint  $fa = ef^2$  and, by the law of cosines against the base angle  $\beta$ , its far side has length exactly  $f^3 = fc$ ; so *the block is the tile scaled by  $f$* , and it contains  $f^2$  tiles. Dually the  $c$ -feet block is the tile scaled by  $e$ , with footprint  $ec = ef^2$ , far side  $e^2f = ea$ , and  $e^2$  tiles. At  $m = 1$  the  $f$ -block’s far side is the whole equal side, so its hypotenuse is the cevian from the apex. The two counts and the middle add up,

$$N = f^2 + e^2 + 2(f^2 - e^2),$$

which is  $3f^2 - e^2$ , as it must be.

Now the point. The local ingredients of the strip argument — the corner decomposition  $\{\beta\}$  with flanks  $\{a, c\}$ , the single- $b$  partner, the parallelogram mate, the kite kills, the surplus lattice — are all statements about the tile and about angles. None of them mentions  $m$ . If they sufficed to force a complete block they would force one at every  $m$ . They do not: among

the three genuine tilings of base- $\beta$  targets that we possess, all with tile  $(2, 3, 4)$  and all verified exactly, the 44-tiling N44B of  $(16, 16, 22)$  has base walk  $a^4c^3a$ , and at its right-hand corner *neither* candidate block is complete — the  $f$ -block (4 straddling tiles) nor the  $e$ -block (1 straddler) — because the last base letter there is a single  $a$  where a block would need the footprint  $ef^2 = 4$  to be cleanly composed. Of the other two, the 99-tiling of  $(24, 24, 33)$  has both unit-scale corner blocks complete (zero wall straddlers, exact sweep, `code/interface_census.py`); N44C has its east block complete but its west  $f$ -block broken — six tiles straddle that wall. Among the three genuine tilings exactly one realises both complete blocks.

So a broken corner is not merely permitted in principle; it occurs in a tiling we have exhibited. Any proof of Hypothesis 29 must therefore use  $m = 1$  in an essential way — the natural candidate being the base  $b$ -count, which is  $e$  at  $m = 1$  but is 0 for both 44-tilings and 7 for the 99-tiling — and no argument phrased purely in the local toolkit above can succeed. We record this because it is the sharpest constraint we have on how the gap can be closed, and because it disqualifies the shortest-looking route.

**5.1. The angle calculus.** Every corner argument below is an instance of one finite computation, which we isolate once.

Since  $\alpha/\pi$  is irrational (Sec. 1), so is  $\alpha/\beta$ : from  $\alpha/\beta = p/q$  and  $3\alpha + 2\beta = \pi$  one gets  $\beta = q\pi/(3p + 2q)$ , making  $\beta$  and hence  $\alpha$  rational multiples of  $\pi$ . Consequently an angle has *at most one* representation  $x\alpha + y\beta$  with  $x, y \geq 0$ : two would give  $(x - x')\alpha = (y' - y)\beta$ . Encoding

$$\alpha \mapsto (1, 0), \quad \beta \mapsto (0, 1), \quad \gamma = 2\alpha + \beta \mapsto (2, 1), \quad \pi = 3\alpha + 2\beta \mapsto (3, 2),$$

a vertex figure at an angle  $(X, Y)$  with  $n_\alpha, n_\beta, n_\gamma$  tile corners is exactly a solution of  $n_\alpha + 2n_\gamma = X$ ,  $n_\beta + n_\gamma = Y$  in nonnegative integers. All of the following are that system, solved; each is machine-checked in `lean/AngleArithmetic.lean` (core toolchain, no `sorry`).

**Lemma 34** (Angle calculus).

[PROVED] *For every member  $(e, f)$ :*

- (1) (No right angle.)  $\pi/2$  has no representation: it would force  $2x = 3$ . Hence no piece of any dissection of a base- $\beta$  region has a right angle. In particular the target cannot be split along its altitude, and no perpendicular cut is available anywhere in the branch.
- (2) (Apex.) At the apex,  $(X, Y) = (3, 0)$ , so  $n_\beta = n_\gamma = 0$  and  $n_\alpha = 3$ : the apex carries exactly three  $\alpha$ -corners.
- (3) (Base corner.) At  $(0, 1)$  the only figure is a single  $\beta$ -corner, with flanks in  $\{a, c\}$ .
- (4) ( $\gamma$ -trap.) At a straight angle,  $(3, 2)$ , the only figures are  $\{\alpha, \alpha, \alpha, \beta, \beta\}$  and  $\{\gamma, \alpha, \beta\}$ ; in particular at most one  $\gamma$  meets any interior boundary point.
- (5) (Wedge  $\alpha + \beta$ .) At  $(1, 1)$  the only figure is one  $\alpha$  and one  $\beta$ .

**Remark 35** (Where  $\beta$  sits).  $3\alpha > \beta$  iff  $\beta < \pi/3$  iff  $\cos \beta > \frac{1}{2}$  iff  $e(3f^2 - e^2) > f^3$ . For  $e = 1$  this is  $f^3 - 3f^2 + 1 < 0$ , which holds only at  $f = 2$ ; in general it is  $f/e < 2.879\dots$ , the real root of  $t^3 - 3t^2 + 1$ . So  $(1, 2)$  — the one member whose instance spectrum is completely determined (Theorem 183) — is the unique member of its family with  $\beta < \pi/3$ , and should not be treated as typical of  $e = 1$ .

**5.2. The  $\gamma$ -grading of directions.** Before the forcing arguments, one structural fact about the whole branch, which does not seem to have been noticed and which costs two lines.

**Proposition 36** ( $\gamma$ -grading).

[PROVED] *Let  $\mathcal{T}$  be any tiling of any target by a base- $\beta$  tile. Then every edge direction occurring in  $\mathcal{T}$  is an integer multiple of  $\gamma$  modulo  $\pi$ . Writing a direction as  $x\alpha + y\beta$ , the integer is  $L = 2x - 3y$ ; it is well defined because  $3\alpha + 2\beta = \pi$ .*

*Proof.* From  $\gamma = 2\alpha + \beta$  and  $3\alpha + 2\beta = \pi$ :

$$2\gamma = 4\alpha + 2\beta = 4\alpha + (\pi - 3\alpha) = \alpha + \pi \equiv \alpha, \quad -3\gamma - \beta = -6\alpha - 4\beta = -2(3\alpha + 2\beta) = -2\pi \equiv 0,$$

so  $\alpha \equiv 2\gamma$  and  $\beta \equiv -3\gamma$  modulo  $\pi$ . Hence  $\langle \alpha, \beta \rangle = \langle 2\gamma, 3\gamma \rangle = \langle \gamma \rangle$ , as  $\gcd(2, 3) = 1$ .

It remains to see that all directions lie in  $\theta_0 + \langle \alpha, \beta \rangle$  for one reference  $\theta_0$ . Read around a tile, the direction turns by the exterior angle,  $-(\text{interior})$  modulo  $\pi$ , so a tile's three directions are  $\theta, \theta - \alpha, \theta - \alpha - \beta$ : a translate of a fixed three-element set  $S$ . If two tiles share a boundary segment of positive length they share a direction, so their offsets differ by an element of  $S - S \subseteq \langle \alpha, \beta \rangle$ . The tile adjacency graph is connected, the target being connected and the tiles finitely many closed sets with disjoint interiors covering it; propagating along a chain from a reference tile gives the claim.  $\square$

Two consequences are used below. Round a tile the label shifts are  $-1, -2, +3$  at the  $\gamma$ -,  $\alpha$ - and  $\beta$ -vertices, so a tile's three edge labels are  $\{L - 2, L, L + 1\}$ , or  $\{L - 1, L, L + 2\}$  in the mirrored chirality: *span exactly three*, with the  $b$ -edge carrying  $L$  in either case. And the target's own sides are graded 0 (base) and  $\mp 3$  (the equal sides).

**Remark 37.** The point of Proposition 36 is that  $\alpha/\pi$  is irrational, so the group generated by  $\alpha$  and  $\beta$  is dense modulo  $2\pi$ ; one expects no finite structure on directions at all, and indeed  $\langle \gamma \rangle$  is dense in  $\mathbb{R}/\pi\mathbb{Z}$ . What the grading says is not that the direction set is discrete but that it is *cyclic*: the directions are the multiples of one angle, so they carry a canonical integer label, and a given tiling uses only a narrow window of those labels. This is worth isolating: the invariant no-go of the main paper is proved for functionals on a dense direction set, and does not automatically apply to functionals graded by  $L$ . The arithmetic is machine-checked in `lean/LabelCalculus.lean`, and the grading was tested against every tiling in this work — the 44-tilings, the 99-tiling, the 28-tiling on a cevian target and the scalene four-component 77-tiling — with no direction failing.

**Proposition 38** (The direction group of a branch). [PROVED] *Let  $T$  be a triangle whose angles satisfy a single primitive relation, i.e. the lattice  $\Lambda = \{(x, y) \in \mathbb{Z}^2 : x\alpha + y\beta \equiv 0 \pmod{\pi}\}$  is generated by one vector  $(p, q)$ . Then in any tiling by congruent copies of  $T$  all edge directions lie in one coset of  $\langle \alpha, \beta \rangle \subset \mathbb{R}/\pi\mathbb{Z}$ , and*

$$\langle \alpha, \beta \rangle \cong \mathbb{Z}^2 / \langle (p, q) \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/d, \quad d = \gcd(p, q).$$

*Writing  $(p, q) = d(p', q')$ , the free coordinate is the label  $L(x, y) = q'x - p'y$ , which is surjective onto  $\mathbb{Z}$  with kernel exactly  $\langle (p', q') \rangle$ .*

*Proof.* The coset statement is the propagation argument of Proposition 36, which used no property of the angles. For the quotient,  $\gcd(p', q') = 1$  gives  $u, v$  with  $up' + vq' = 1$ ; then  $L(vn, -un) = n$ , so  $L$  is onto, and  $L(x, y) = 0$  means  $q'x = p'y$ , whence  $p' \mid x$  and  $(x, y)$  is a multiple of  $(p', q')$ . The remaining generator contributes  $\mathbb{Z}/d$ .  $\square$

**Remark 39** (A torsion dichotomy across the classification). Solving for  $\Lambda$  in each branch of the Laczkovich classification:

| branch             | relation                  | $(p, q)$ | direction group                  |
|--------------------|---------------------------|----------|----------------------------------|
| base- $\beta$      | $3\alpha + 2\beta = \pi$  | $(3, 2)$ | $\mathbb{Z}$                     |
| $\gamma = 2\alpha$ | $3\alpha + \beta = \pi$   | $(3, 1)$ | $\mathbb{Z}$                     |
| $\gamma = 2\pi/3$  | $\alpha + \beta = \pi/3$  | $(3, 3)$ | $\mathbb{Z} \oplus \mathbb{Z}/3$ |
| $\gamma = \pi/3$   | $\alpha + \beta = 2\pi/3$ | $(3, 3)$ | $\mathbb{Z} \oplus \mathbb{Z}/3$ |
| right angle        | $\alpha + \beta = \pi/2$  | $(2, 2)$ | $\mathbb{Z} \oplus \mathbb{Z}/2$ |

The two branches in which  $\alpha/\pi$  is irrational — base- $\beta$  and  $\gamma = 2\alpha$  — are exactly those with  $d = 1$ , that is, with torsion-free direction group. Every special-angle branch carries torsion. We record the coincidence without asserting a mechanism.

The phenomenon that tiles related by rotation through a single irrational angle generate a cyclic orientation group is not itself new; it is familiar from substitution tilings, where the pinwheel tiling's orientations are the multiples of  $2\arctan(1/2)$  and are dense in the circle. What is recorded here is the computation of  $\Lambda$  for each branch, the resulting torsion dichotomy, and, in the base- $\beta$  case, the identification of the free generator as  $\gamma$  itself. The arithmetic is machine-checked in `lean/DirectionGroup.lean`.

**5.3. The forcing toolkit.** *Corners.* A vertex of angle  $\theta$  between two boundary rays admits only the tile-angle decompositions of  $\theta$ , and the extreme tiles must present flanks compatible with the rays' first edges; for  $\theta = \beta$  the unique decomposition is  $\{\beta\}$  and the flanks are  $\{a, c\}$ , one per ray (the two orientations). *Partners.* The corner tile's  $b$ -edge is a boundary-anchored chord; both sides partition into whole edges of total  $b$ , and  $xa + yb + zc = b$  forces  $(x, y, z) = (0, 1, 0)$  (reduce mod  $f$ ; sizes), so the far side is a single  $b$ -edge; of the two congruent mates the reflected one repeats the corner angle and overflows ( $2\gamma = \pi + \alpha$ ), leaving the parallelogram partner. *Strips.* Inductively, the  $k$ -th chord ( $k < f$  resp.  $k < e$  for the dual) carries  $k$  aligned  $b$ -edges; the surplus lattice (Sec. 4) forbids  $0 < |dQ| < f$ , so the far side matches  $b$ -for- $b$ ; kites die bottom-up ( $\gamma$ -repeat at the base, then  $2\gamma > \pi$  at each junction), partners are forced, and each junction wedge is exactly  $\beta$  resp. the dual angles. All statements are symmetric under the  $a$ -feet/ $c$ -feet duality; footprints agree by  $ec = fa$ .

**5.4. Runs, columns, fillers, recursion.** Along any covered run of an anchored line both sides partition into whole edges of equal total, and the surplus lies in  $\Lambda = \mathbb{Z}(f, 0, -e) \oplus \mathbb{Z}(f-e, -f, f-e)$ . A break tile's  $c$ -edge admits no whole-edge completion of  $c - a = f(f - e)$  (mod  $f$  and gcd), so the run closes minimally as the generalized column:  $e$   $c$ -edges against  $f$   $a$ -edges ( $ec = fa$ ). The column's junction fillers are exactly the triangles on consecutive apexes with sides  $c, b, a$  (the filler identity, an identity in  $(e, f)$ ), so column and fillers tile the inter-level strip completely and lay an  $a$ -run one level down; an interior  $a$ -edge's far side decomposes only as a single  $a$  (Master Lemma 1:  $\gcd(f^2 - e^2, ef) = 1$ ), so the run forces the next level's column. The recursion reaches the base; a tile vertex on the base is a walk junction (no one-sided  $T$ -junction), and the accumulated orientation offsets satisfy, modulo  $2f$ , the congruence that forces every level to the  $(\gamma, \beta)$  orientation — for both parities of  $f$  — placing the terminal apexes at consecutive multiples of  $a$ . Between consecutive apexes the walk letters decompose only as single  $a$ 's (Master Lemma 1 again), so the letters spanning the apex range are forced.

**5.5. The funnels.** By Master Lemma 2 the two corner structures —  $f$   $a$ -feet or  $e$   $c$ -feet — have equal footprints  $ef^2$ , so in each walk of the trichotomy they consume exactly the non- $b$  letters and the middle is the  $e$   $b$ -letters:  $(2f, e, 0) \mapsto (a, a)$ ,  $(f, e, e) \mapsto (a, c)$ ,  $(0, e, 2e) \mapsto (c, c)$ . (The step just used — that a stub is a genuine region-boundary segment, so that whole-edge partition applies — assumes no tile straddles the ray. That assumption is false in general; see Remark 14, and Remark 32 for how it is absorbed into Hypothesis 29.) If a corner structure is starved or broken, the recursion of the previous subsection forces the  $a$ -prefix regardless (the apex lattice), and the  $\beta$ -wedge at a column terminal (filler  $\gamma$  + column  $\alpha$  leave exactly  $\beta$ , whose tile presents  $a$  or  $c$ ) forbids a  $b$ -letter there; the walk arithmetic then kills exactly as in the  $e = 1$  case.

**Lemma 40** (Terminal wedge). [PROVED] *Let a base-reaching column terminate at base vertex  $V = (x_0 + fa, 0)$ ,  $x_0$  its anchor offset. Then the angle at  $V$  decomposes as  $\gamma$  (the last filler) +  $\alpha$  (the last column tile) +  $\beta$ , and the base edge east of  $V$  is an  $a$ - or a  $c$ -letter.*

*Proof.* The filler  $(J_{f-1}, \text{ap}_{f-1}, \text{ap}_f)$  has its  $a$ -edge on the base ending at  $V$  with the  $a$ - $b$  corner there, i.e.  $\gamma$ ; the column tile ends at  $V$  with its apex angle  $\alpha$ . The remaining budget is  $\pi - \gamma - \alpha = \beta$ , whose unique decomposition is a single  $\beta$ -tile; its flanks are  $a$  and  $c$ , one of which lies along the base east of  $V$ . A  $b$ -letter there would have to be that flank, and  $b \notin \{a, c\}$ .  $\square$

**Lemma 41** (Pentagon).

[PROVED] *In the walk  $(0, e, 2e)$ , after both*

*dual blocks complete, the middle region — base  $b^e$ , dual hypotenuses  $b^e$ , side segments of length  $fb$ , apex  $3\alpha$ , containing  $3(f^2 - e^2)$  tiles — admits no tiling.*

*Proof.* Consider the west base corner  $W_0 = (ef^2, 0)$ , of interior angle  $\pi - \alpha$  (the dual block's  $\alpha$  sits west of it). Its decompositions are  $\{\gamma, \beta\}$  and  $\{2\alpha, 2\beta\}$ ; in each, the tile adjacent to the dual hypotenuse presents along it a flank from  $\{a, c\}$  —  $\beta$ -tiles carry  $(a, c)$ ,  $\gamma$ -tiles  $(a, b)$  with the  $b$  unavailable (it would repeat the hypotenuse edge inside the wedge),  $\alpha$ -tiles  $(b, c)$  likewise contribute  $c$ . The hypotenuse is partitioned into whole  $b$ -edges of the dual strips, so a flank of length  $a$  or  $c$  never terminates at a hypotenuse junction: each placement leaves a stub of length  $s_1 \in \{c - b, |a - b|\} = \{e^2, |ef - f^2 + e^2|\}$  against the next  $b$ -edge. Such a stub is a region boundary segment and must be partitioned into whole tile edges. Neither can be: the first lies in  $(0, \min(a, b))$ , which contains no element of the numerical semigroup  $\langle a, b, c \rangle$ , and the second is a gap of that semigroup by Lemma 42. So no partition exists and the region dies. The certified refutation at  $(2, 3)$  (85 nodes, every branch dead, all kills of run or wedge type) instantiates the argument; the stub arithmetic above is member-independent.  $\square$

For the middles: any walk containing an  $a$ -block dies at the middle's  $a$ -side corner (Lemma 40 and the  $(b, b)$  row of the vertex table), of angle  $\alpha + \beta$  with both rays presenting  $b$  — the  $T_{\text{mid}}$  kill of the main paper. The walk  $(0, e, 2e)$  is forced to  $c^e b^e c^e$  (the dual break flanks are  $\{a, c\}$  and no  $a$  exists); both duals complete and the middle is the pentagon: base  $b^e$ , two dual hypotenuses  $b^e$ , two side segments of length  $fb$ , apex  $3\alpha$ . Its west wedge is  $\pi - \alpha$ ; every decomposition places against the dual hypotenuse a flank from  $\{a, c\}$ , never matching  $b$ : each placement leaves a stub, and the stubs walk in steps of  $e^2$  (through-vertex overhangs) or  $|a - b|$ ; each such stub is a gap of the numerical semigroup  $\langle a, b, c \rangle$  — the first,  $e^2 \bmod b$ , because it lies in  $(0, \min(a, b))$ , and the second,  $|a - b|$ , by Lemma 42 — so no whole-edge partition exists and the region dies. No iteration occurs: both stubs are gaps at the first step, for every member. (Certificate at  $(2, 3)$ : 85 nodes, all dead, kills exclusively of run/wedge type.)

**Lemma 42** (The second stub is a gap).

[PROVED] *Let  $\gcd(e, f) = 1$  with  $1 \leq e < f$ , and*

*let  $(a, b, c) = (ef, f^2 - e^2, f^2)$ . Then  $|a - b|$  is a gap of the numerical semigroup  $\langle a, b, c \rangle$ : it is nonzero and admits no representation  $xa + yb + zc$  with  $x, y, z \geq 0$ .*

*Proof.* First  $\gcd(a, b) = 1$ : a common divisor of  $e$  and  $b$  divides  $b + e^2 = f^2$ , hence divides  $\gcd(e, f^2) = 1$ ; a common divisor of  $f$  and  $b$  divides  $f^2 - b = e^2$ , hence divides  $\gcd(f, e^2) = 1$ ; so  $\gcd(ef, b) = 1$ . Next both generators exceed 1:  $a = ef \geq 2$ , and  $b = (f - e)(f + e) \geq 3$  since  $f - e \geq 1$  and  $f + e \geq 3$ . Hence  $a \neq b$  — equality with  $\gcd(a, b) = 1$  would force  $a = b = 1$  — so  $t := |a - b| > 0$ .

Now  $t < c$ : if  $a > b$  then  $t = ef + e^2 - f^2 < f^2$  because  $ef < f^2$  and  $e^2 < f^2$ ; if  $b > a$  then  $t = f^2 - e^2 - ef < f^2$ . So  $z = 0$  in any representation. Also  $t < \max(a, b)$  trivially, so the larger of  $a, b$  has coefficient 0 as well. What remains is  $t = k \cdot \min(a, b)$  for some  $k \geq 1$ , i.e.  $\min(a, b)$  divides  $|a - b|$  and therefore divides  $\max(a, b)$  too; with  $\gcd(a, b) = 1$  this forces  $\min(a, b) = 1$ , contradicting  $a \geq 2, b \geq 3$ .  $\square$

**Remark 43.** The lemma is stronger than the pentagon argument needs and removes its only iterative step. The earlier treatment sent the stub sequence walking by  $e^2$  modulo  $b$  until it entered  $(0, \min(a, b))$ , a gap whose reachability had to be argued geometrically; the statement above shows the second stub is *already* a gap, for every member, so no walk occurs. This is not merely a shorter route to the same place: an exhaustive check over every member with  $f \leq 200$  (12231 pairs, `code/rust/stubgap`) finds that *both* initial stubs are gaps at the first step in every single case, so the walk the earlier argument invoked never executes even one step. The mechanism can be deleted from the lemma rather than justified. It is formalized in

`lean/PentagonLemma.lean` (`pentagon_lemma`, general in  $(e, f)$ , standard axioms only), together with  $\gcd(a, b) = 1$ , and cross-checked by exhaustive semigroup search over all coprime pairs with  $f \leq 400$  (`code/rust/stubgap`). Note that  $(0, \min(a, b))$  and  $\{|a - b|\}$  are genuinely different gaps: at  $(1, 3)$  the tile is  $(3, 8, 9)$ ,  $\min(a, b) = 3$ , and  $|a - b| = 5$  lies outside  $(0, 3)$  while still being a gap of  $\langle 3, 8, 9 \rangle$ .

**Lemma 44** (Two  $c$ -edges are forced;  $p \leq f - 2$ ). [CONJECTURE] *At  $e = 1$ ,  $m = 1$ , the first and last edges of each equal side are  $c$ -edges: the first is the base-corner tile's side flank (its base flank is an  $a$ -edge by Theorem 20), and the last belongs to the apex figure  $\{3\alpha\}$ , whose side edge is  $b$  or  $c$ , with  $b$  excluded by Lemma 45. Hence  $f - p \geq 2$ , i.e.  $p \leq f - 2$ . In particular  $p = 0$  for  $f = 2$ : the member  $(1, 2)$  satisfies Hypothesis 29 outright, by boundary analysis alone, and the closure of  $N = 11$  in Theorem 1 is recovered through Remark 47. For  $f = 3$  the only surviving value is  $p = 1$ , whose side word is forced to  $c a a a c$ ; by the run rigidity its chirality word is  $(-4)^j(-2)^{3-j}$  for some  $0 \leq j \leq 3$ , so the entire boundary configuration space at  $(1, 3)$  consists of four explicit placements.*

**Lemma 45** (The equal sides carry no  $b$ -edge). [PROVED] *Let  $m = 1$  and let  $(e, f)$  be any member. Then every side walk  $P'a + Q'b + R'c = f^3$  obeying the  $\gamma$ -trap  $R' \geq 1$  has  $Q' = 0$ .*

*Proof.* Reducing the walk modulo  $f$  kills  $P'a$  and  $R'c$  and leaves  $f \mid Q'e^2$ ; as  $\gcd(e, f) = 1$ ,  $f \mid Q'$ . Write  $Q' = qf$  and suppose  $q \geq 1$ . Dividing the walk by  $f$ ,

$$P'e + qb + R'f = f^2, \quad b = f^2 - e^2,$$

so in particular  $qb + R'f \leq f^2$ . Reducing *this* modulo  $e$  gives  $f(qf + R') \equiv f^2$ , and  $f$  is invertible mod  $e$ , so  $R' \equiv f(1 - q) \pmod{e}$ .

*Case  $q = 1$ .* Then  $R' \equiv 0 \pmod{e}$ , and  $R' \geq 1$  forces  $R' \geq e$ , so  $R'f \geq ef$ . But  $qb + R'f \leq f^2$  gives  $R'f \leq f^2 - b = e^2$ , whence  $ef \leq e^2$  and  $f \leq e$ , contradicting  $e < f$ .

*Case  $q \geq 2$ .* From  $qb + R'f \leq f^2$  with  $q \geq 2$ ,  $R' \geq 1$  we get  $2(f^2 - e^2) + f \leq f^2$ , i.e.  $f^2 + f \leq 2e^2$ ; hence  $f < e\sqrt{2} < 2e$ , so  $u := f - e$  satisfies  $1 \leq u \leq e - 1$ . Put  $t := q - 1 \geq 1$ . Since  $f \equiv u \pmod{e}$ , the congruence above reads  $R' \equiv -ut \pmod{e}$ . Write  $ut = v + we$  with  $0 \leq v \leq e - 1$  and  $w \geq 0$ . Substituting  $f = e + u$  and  $b = u(2e + u)$ , the constraint is

$$(t + 1)u(2e + u) + R'(e + u) \leq (e + u)^2.$$

If  $v = 0$  then  $e \mid R'$  and  $R' \geq e$ , so the display gives  $(t + 1)u(2e + u) \leq (e + u)^2 - e(e + u) = u(e + u)$ , i.e.  $(t + 1)(2e + u) \leq e + u$ , impossible as  $t + 1 \geq 2$ . If  $v \geq 1$  then  $R' \geq e - v$ , so  $(t + 1)u(2e + u) \leq (e + u)(u + v)$ . But  $(t + 1)u = ut + u = v + we + u$ , so

$$(t + 1)u(2e + u) - (e + u)(u + v) = (v + u)[(2e + u) - (e + u)] + we(2e + u) = e[(v + u) + w(2e + u)] > 0,$$

a contradiction. Hence  $q = 0$ , i.e.  $Q' = 0$ .  $\square$

**Remark 46.** The special case  $e^2 < f$  — which already covers every  $e = 1$  member — has the one-line proof “ $Q' \neq 0 \Rightarrow Q' \geq f \Rightarrow Q'b + R'c \geq f^3 - fe^2 + f^2 > f^3$ ”, and it is that case which is machine-checked in `lean/SideNoB.lean` as `side_no_b`, with `side_no_b_e_one` the  $e = 1$  specialisation. The unconditional statement is `side_no_b_uncond` in the same file, formalised by the identity  $(\star) \, tb + R'f + P'e = e^2$  used in the proof above; both depend only on the standard axioms. Cross-checked by exhaustive search over every member with  $f \leq 200$ .

**Remark 47** (At  $e = 1$ , Hypothesis 29 is exactly “the sides carry no  $a$ -edge”). Lemma 45 is one of the few statements in this development genuinely special to  $m = 1$  — at  $m = 2$  it is false, by the identity  $e(ef) + f(f^2 - e^2) + f \cdot f^2 = 2f^3$  — and by Remark 33 any proof of Hypothesis 29 must use  $m = 1$  essentially. It is therefore worth following where it leads at  $e = 1$ , where it reduces the hypothesis to a single integer.

With  $Q' = 0$  the side equation  $P'f + R'f^2 = f^3$  gives  $P' + R'f = f^2$ , so  $f \mid P'$ ; writing  $P' = fp$ ,

$$(P', R') = (fp, f - p), \quad 0 \leq p \leq f - 1,$$

the  $\gamma$ -trap being  $R' \geq 1$ . The complete west block is the tile scaled by  $f$ , so its far side — which at  $m = 1$  is the whole equal side — is subdivided into exactly  $f$   $c$ -edges, i.e.  $p = 0$ . Hence, at  $e = 1$ ,

Hypothesis 29 at the west corner  $\iff p = 0 \iff$  the equal side carries no  $a$ -edge.

Two corner facts pin the ends of that walk. At the base corner the unique figure is a single  $\beta$ -tile with flanks  $\{a, c\}$ , and Theorem 20 puts an  $a$ -edge at the base end, so the side's first edge is a  $c$ . At the apex the figure is exactly three  $\alpha$ -corners (Lemma 34(2)), and the two edges at a tile's  $\alpha$ -vertex are  $b$  and  $c$ , so with  $Q' = 0$  the side's last edge is a  $c$  as well. Every equal side therefore reads  $c(\cdots)c$  with  $fp$  interior  $a$ -edges.

So the open step at  $e = 1$  is the single statement “ $p = 0$ ”, a direct analogue of Lemma 45 with  $a$  in place of  $b$ , and it is not arithmetic: the walk equation admits every  $p \leq f - 1$ . Finally, note that  $p = 0$  closes the subfamily outright, without any further funnel analysis: complete blocks put  $f$   $a$ -feet at one base corner and  $e = 1$   $c$ -foot at the other, so the base reads  $a^f b c$  and ends in a  $c$ -edge, whereas Theorem 20 forces  $a$ -edges at both ends. The contradiction is immediate. Hypothesis 29 is thus not a step towards the  $e = 1$  case; at  $e = 1$  it *is* the case.

**Remark 48** (Correction to Remark 47:  $p = 0$  is not the open step, and fails for one side). Both supports of the previous remark's identification have to be withdrawn, and the target replaced.

The *derivation* of  $p = 0$  there is “the complete west block is the tile scaled by  $f$ , so its far side — the whole equal side at  $m = 1$  — is subdivided into exactly  $f$   $c$ -edges”. A complete west block puts  $f$  consecutive  $a$ -feet at the corner, so it needs the base word to begin with  $a^f$ ; with exactly  $f$  letters  $a$  and a final  $a$  forced by Theorem 20, that is  $f + 1$  of them. The premise never holds, so the derivation is vacuous.

The *sufficiency* of  $p = 0$  — “note that  $p = 0$  closes the subfamily outright” — is justified there by an argument every step of which is about complete blocks: “complete blocks put  $f$   $a$ -feet at one base corner and  $e = 1$   $c$ -foot at the other, so the base reads  $a^f b c$ ”. That silently substitutes complete blocks for  $p = 0$ , which requires the converse implication  $p = 0 \Rightarrow$  (walls). The converse appears only as part of the statement of Conjecture 60, proved at  $f = 2, 3, 4$  alone.

Moreover the symmetric reading is not merely unsupported but false. By Remark 24 both corners break; a base-break consumes the walk's single  $c$ , so at most one break is on the base, and some corner breaks on a side. A corner block breaks on a side by failing to read a  $c$  where the block demands one, and the block's partial far side lies on the adjacent equal side. Hence

at  $e = 1, m = 1, f \geq 3$ : at least one equal side carries an  $a$ -edge,

so no tiling has  $p = 0$  on both sides, and a single symmetric  $p$  satisfies  $p \geq 1$ . Formalised as `Erdos634.WallsCircular.some_side_carries_a` and `symmetric_p_zero_false`.

What survives as the open step is the *cascade*, not the parameter: whether the chain from such an  $a$ -edge terminates in a collision, which is Conjecture 60. Its premise, “suppose an equal side carries an  $a$ -edge”, is now discharged rather than hypothetical — it holds in every tiling of every member. In the language of the crossing statement,  $\ell$ , the initial  $c$ -block of the side, is exactly the level at which the corner block breaks, and the open inequality  $\ell \geq b_p - 1$  is a race between where the block gives out and where the base  $b$  sits. It was never the claim that the block does not give out.

**Lemma 49** (First-run orientation). [CONJECTURE] *In any tiling of the  $(1, f)$  target at  $m = 1$ , every tile of the first  $a$ -run of an equal side is mirrored: its  $\gamma$  sits at the upper junction and its*

label is  $-2$ . In particular at  $(1,3)$ , where Lemma 44 forces the side word  $c a a a c$ , the boundary configuration is unique: both  $c$ -tiles direct with label  $-1$ , all three  $a$ -tiles mirrored with label  $-2$ .

*Proof.* The corner tile's  $b$ -edge is a chord with both endpoints on the target boundary: its upper end is the first side junction  $J$ , its lower end the base point  $(f,0)$ , and the ray beyond either end leaves the target (past  $J$  it crosses the side, the two directions being non-parallel; past  $(f,0)$  it descends below the base). So no covering edge overshoots, the far side partitions into whole edges of total  $b$ , and by the partner relation  $xa + yb + zc = b \Rightarrow (x,y,z) = (0,1,0)$  (machine-checked, general in  $(e,f)$ : `PentagonLemma.partner_unique`) the far side is a single  $b$ -edge: a partner tile  $P$  shares the whole edge. The ends of a  $b$ -edge host  $\alpha$  and  $\gamma$ , so  $P$  contributes  $\alpha$  or  $\gamma$  at  $J$ ; the corner tile contributes  $\alpha$  there. If the first  $a$ -tile were direct it would contribute  $\gamma$  at  $J$ , and the straight-angle figure at  $J$  would contain one  $\alpha$ , one  $\gamma$ , and a further corner from  $\{\alpha, \gamma\}$ ; neither admissible figure does (machine-checked: `OrderForcing.first_run_kill`). So the first  $a$ -tile is mirrored, and the run rigidity (`gamma_far_absorbing`: mirrored is absorbing along the run) makes the whole run mirrored.  $\square$

**Lemma 50** (Top junction;  $p \leq f - 3$  for  $f \geq 4$ ). [CONJECTURE] *Let an  $a$ -run end at the last side junction  $J$  (final  $c$ -block of length 1) with the run's last tile mirrored. The junction figure is  $\{\gamma, \alpha, \beta\}$ , its interior  $\alpha$ -filler has its edges along the two junction rays, and the edge along the top tile's  $a$ -ray is  $c = f^2$  (mirrored filler) or  $b = f^2 - 1$  (direct filler), with endpoint  $J + (f^2, 0)$  resp.  $J + (f^2 - 1, 0)$ . Testing these against the right side of the target reduces, exactly, to the signs of*

$$f^3 - 3f^2 + 1 \quad (\text{mirrored filler outside} \iff > 0 \iff f \geq 3), \quad f^3 - 3f^2 - f + 1 \quad (\text{direct filler outside} \iff > 0 \iff f \leq 2)$$

Hence for  $f \geq 4$  no such junction exists; for  $f = 3$  the filler is forced direct with label 0. Consequently:  $p = f - 2$  gives  $c$ -count 2, forcing the single-run word  $c a^{f(f-2)} c$ , which is all-mirrored by Lemma 49 and ends at such a junction — dead for  $f \geq 4$ . So

$$p \leq f - 3 \quad (f \geq 4),$$

and at  $(1,4)$  in particular  $p \leq 1$ . The first polynomial is the thin/thick discriminant of Remark 35: the member  $(1,2)$  is on its other side, which is again why  $f = 2$  is the degenerate case. The sign reductions are verified exactly (rational cross products over  $\mathbb{Q}(\sqrt{D})$ ) and the two positivity facts are machine-checked (`OrderForcing.mirrored_filler_outside`, `direct_filler_outside`), as are the single-run combinatorics and the count bound (`two_c_single_run`, `p_le_f_sub_three`).

**Lemma 51** ( $c$ -chord dichotomy). [VERIFIED] *Let a chord of length  $c$  be blocked at both ends, so that its far side partitions into whole tile edges of total  $c$ . Then  $xa + yb + zc = c$  forces  $(x,y,z) = (0,0,1)$ , or, exactly when  $e = 1$ ,  $(x,y,z) = (f,0,0)$ : the far side is a single  $c$ -edge or (at  $e = 1$  only) a run of  $f$   $a$ -edges. Machine-checked general in  $(e,f)$  (`CChord.c_chord_dichotomy`); the modular step is the identity  $(y-1)f^2 = e(ye - xf)$ .*

**Remark 52** (Tile-interior blocking). Two ends of a chord can be blocked by the target boundary; they can also be blocked by the interior of an already-forced tile, since a covering edge extended past the chord's end would enter it. Forced tiles thereby create new fully-blocked chords, and the forcing propagates.

**Theorem 53** (Hypothesis 29 holds at  $(1,3)$ ). [CONJECTURE] *No tiling of the  $(1,3)$  target at  $m = 1$  carries an  $a$ -edge on an equal side. Hence, by Remark 47, no tiling of the  $(1,3)$  target at  $m = 1$  exists at all.*

*Proof.* Suppose an equal side carries an  $a$ -edge. By Lemma 44,  $p = 1$  and the side word is  $c a a a c$ ; by Lemma 49 the boundary configuration is unique, and the corner tile  $T_1$ , its  $b$ -partner

$P$ , and the three mirrored  $a$ -tiles are all pinned. Write the base corner at the origin,  $J$  for the first side junction, and  $Q = J + (f, 0)$  for the end of  $P$ 's horizontal  $a$ -edge, so that  $P$ 's  $c$ -edge runs from  $(f, 0)$  to  $Q$  parallel to the side. That edge is blocked at  $(f, 0)$  by the base and at  $Q$  by the interior of the second side tile (Remark 52; the directions differ, so the crossing is transversal), and Lemma 51 applies: its far side is a single  $c$ -edge or a run of three  $a$ -edges.

The vertex figure at  $(f, 0)$  already contains  $T_1$ 's  $\gamma$  and  $P$ 's  $\alpha$ , leaving exactly one  $\beta$  (the representation of  $\pi - \gamma - \alpha$  is unique). The tile carrying it presents, from its  $\beta$ -corner with flanks  $\{a, c\}$ , the base's second edge.

*Single- $c$  branch.* The  $c$ -tile  $C$  has  $\beta$  at  $(f, 0)$  and its  $a$ -edge on the base: the base word is  $aabca$ , so the third base edge is the  $b$ -edge  $[2f, 2f + b]$ . Label consistency forces  $C$  direct, and  $C = T_1 + (f, 0)$ : the corner block grows by one cell. Now  $C$ 's  $b$ -edge is blocked at  $(2f, 0)$  by the base and at  $Q$  by the same tile interior, so its far side is a single  $b$ -edge (Lemma 45's companion fact, **partner\_unique**), carried by a tile  $Z$  whose corners at the two ends are  $\alpha$  and  $\gamma$  in one of two orientations. The reflected orientation puts  $\gamma$  at  $(2f, 0)$ , where  $C$ 's  $\gamma$  already sits: the straight-angle figure admits at most one  $\gamma$ . So  $Z = P + (f, 0)$ , with  $\alpha$  at  $(2f, 0)$  and  $\gamma$  at  $Q$ , and the figure at  $Q$  closes as  $\{\beta, \alpha, \gamma\}$ . At  $(2f, 0)$  the figure now contains  $C$ 's  $\gamma$  and  $Z$ 's  $\alpha$ , leaving exactly one  $\beta$ , whose tile presents the base edge from  $(2f, 0)$  out of flanks  $\{a, c\}$ . But that edge is the  $b$ -edge of the forced base word. Contradiction.

*Three- $a$  branch.* The first  $a$ -tile  $A_1$  has its corner at  $(f, 0)$  among the  $a$ -edge's end angles  $\{\beta, \gamma\}$ ; the single open slot is  $\beta$ , so  $A_1$  has  $\beta$  at  $(f, 0)$  and its other flank, the  $c$ -edge, on the base: the base word is  $acbaa$ , with the  $b$ -edge at  $[f + c, f + c + b] = [12, 20]$ . Label consistency forces  $A_1$  mirrored.  $A_1$ 's  $b$ -edge runs from the first trisection point of  $P$ 's  $c$ -edge down to  $(f + c, 0) = (12, 0)$ ; it is blocked at  $(12, 0)$  by the base and at the trisection point by  $P$ 's interior, so its far side is again a single  $b$ -edge, carried by  $Z'$ . The reflected orientation repeats  $\gamma$  at the trisection point, where  $A_1$ 's  $\gamma$  already sits — impossible. So  $Z'$  is the parallelogram mate with  $\gamma$  at  $(12, 0)$ . The figure at  $(12, 0)$  then contains  $A_1$ 's  $\alpha$  and  $Z'$ 's  $\gamma$ , leaving exactly one  $\beta$  — but the base edge from  $(12, 0)$  is the  $b$ -edge of the forced word, whose ends host  $\alpha$  or  $\gamma$ . Contradiction.

Both branches of Lemma 51 are impossible, so no such tiling exists;  $p = 0$ .  $\square$

**Remark 54.** Three remarks. First, the terminal mechanism is the same in both branches: the cascade advances one cell from the corner and the base word, fixed at the fork, collides with it one base point later. Second, the proof is search-free; it re-derives the exclusion of the  $(1, 3)$  instance ( $N = 26$ ) established by the certified exhaustion (2025 nodes), and the two agree. Third, the arithmetic components are machine-checked (**partner\_unique**, **c\_chord\_dichotomy**, **first\_run\_kill**, **straight\_junction\_cases**, the slot classifications **beta\_slot\_unique**, **alpha\_slot\_unique**, **wedge\_gamma\_cases**, and the polynomial signs of Lemma 50); the geometric steps are exact coordinate computations over  $\mathbb{Q}(\sqrt{35})$ . Hypothesis 29 is thereby proved for the members  $(1, 2)$  (Lemma 44) and  $(1, 3)$ ;  $f = 4$  is settled by Theorem 61, so on the thin family the hypothesis is open for  $f \geq 5$ , where  $p \leq f - 3$ , and it is open for  $e \geq 2$  generally — see Conjecture 60 for the thin case and §5.9 for what is known at  $e \geq 2$ .

**Remark 55** (The cascade at  $(1, 4)$ , and the general mechanism). Running the same cascade at  $(1, 4)$  ( $p \leq 1$  by Lemma 50) kills five of the *six* admissible base words (an earlier count of five missed  $acbaa$ , which dies at once: its  $b$ -edge sits at  $[f + c, f + c + b]$ , exactly the four- $a$  branch's first  $\beta$ -slot). For every side word with initial  $c$ -block 1, the corner partner and the  $c$ -chord dichotomy force, at each base point  $(kf, 0)$  reached, a  $\beta$ -slot whose tile presents the next base edge from flanks  $\{a, c\}$ : the words with  $b$  in position 3 die at  $(2f, 0)$ ; the word  $aabca$  dies at  $(3f, 0)$ ; the word  $aacbaa$  dies at  $(f + c, 0) = (24, 0)$  through the  $a$ -run partner, exactly as at  $(1, 3)$ . Two configurations survive the present tools, and the obstruction is located precisely in each: for the base word  $acbaa$  the deviating  $a$ -run's new  $c$ -chord  $[(20, 0), (20, 0) + cu]$  has its

upper endpoint beyond the second side tile's  $c$ -edge ( $25.875 > 21.875$  exactly), so tile-interior blocking fails and the dichotomy does not apply; and for the side word  $cc a^4 c$  the first run does not abut the corner tile, so Lemma 49 gives no chirality forcing. Hypothesis 29 at  $(1, 4)$  is thereby reduced to these two configurations; the first is now eliminated, and the second is narrowed, as follows. *The double- $c$  side word dies.* With the second  $c$ -tile forced direct and the run all-direct, the  $\beta$ -slot tile  $B$  at the run's first junction has exactly two label-consistent forms, and both are impossible. The junction ray toward the second  $c$ -tile's  $b$ -edge extends, exactly, to the base point  $(2f, 0)$ : the points  $2cu$ ,  $Q$  and  $(2f, 0)$  are collinear with  $|2cu \rightarrow (2f, 0)| = 2b$ . If  $B$  is direct, its  $a$ -edge covers the first 4 units of that line's far side, and the remaining stretch to  $Q$  has length  $11 \notin \langle 4, 15, 16 \rangle$ ; some covering edge must therefore cross  $Q$ , a through-edge forces the west  $\alpha$ -slot tile  $X = T_1 + (f, 0)$  (the mirrored  $X$  dies by base overshoot), so the whole segment  $[Q, (2f, 0)]$  is an edge and the far side must partition  $26 \notin \langle 4, 15, 16 \rangle$  with no crossing at the  $\gamma$ -end. If  $B$  is mirrored, its  $c$ -edge itself crosses  $Q$  by  $e^2 = 1$ , forcing  $X$  the same way, and the remaining stretch is  $14 \notin \langle 4, 15, 16 \rangle$  with no crossing at the  $\alpha$ -end. Both die; the side word  $cc a^4 c$  admits no completion for any base word. The crossing exclusions are instances of the general principle that a through-edge occupies the whole half-turn on its side (**through\_edge\_exclusive**); the three gaps are machine-checked (**gaps\_4\_15\_16**). Each remaining configuration is narrowed as follows. In the double- $c$  configuration  $cc a^4 c$ , the second  $c$ -tile is forced direct: mirrored would place its  $\alpha$  at the first junction beside the corner tile's  $\alpha$  and the partner's  $\gamma$ , and  $\pi - 2\alpha - \gamma$  has no representation (**first\_run\_kill**); moreover the final  $c$ -block has length 1, so by Lemma 50 the run must be entirely direct, and the configuration now hinges on the single unforced  $\beta$  at the run's first junction. In the base word  $acabaa$ , the  $\beta$ -slot at  $(20, 0)$  forces the tile  $T_1 + (20, 0)$ , whose new  $c$ -chord is covered on its far side, in the mirrored-filler sub-case, by the mate's  $a$ -edge, two further  $a$ -edges, and the filler's own  $a$ -edge, whose endpoint is exactly the escape point  $E = Q + (c, 0) = (20, 0) + cu$ ; the direct-filler sub-case leaves  $E$  unblocked and remains open.

**Proposition 56** (The double- $c$  kill at any initial block length). [PROVED] *Let  $e = 1$ ,  $m = 1$ ,  $f \geq 3$ , and let an equal side carry an  $a$ -edge and have initial  $c$ -block of length  $j \geq 2$ . Write  $u$  for the unit vector along that side and place the base corner at the origin. Then:*

- (i)  $j \leq f - 1$ , and the chord from  $jcu$  to the base point  $(jf, 0)$  has length exactly  $jb$ , divided into  $j$  segments of length  $b$  by  $R_i = icu + (j - i)(f, 0)$  (so  $R_1$  is the point called  $Q$  in the  $(1, 3)$  argument; the  $Q_j$  of Lemma 70 are a different family, spaced  $c$  apart along the interior wall);
- (ii) that chord admits exactly one partition into tile edges, namely  $j$   $b$ -edges;
- (iii) the  $\beta$ -slot tile at  $jcu$  presents, from its  $\beta$ -corner, a flank of  $\{a, c\}$ , and each choice leaves a run outside  $\langle a, b, c \rangle$ ;
- (iv) the  $\alpha$ -slot tile at  $R_{j-1}$  cannot lay a  $c$ -edge along the chord.

*Proof.* (i) The side begins and ends with a  $c$ -edge (Theorem 20(i)), so an initial block of length  $\geq 2$  leaves a further  $c$  at the far end and  $n_c \geq 3$ ; with  $n_c = f - k$  and  $k \geq 1$  (the side carries an  $a$ -edge) this gives  $j \leq n_c = f - k \leq f - 1$ . For the chord,  $\cos \beta$  is  $(3f^2 - 1)/(2f^3)$  by the law of cosines on  $(f, f^2 - 1, f^2)$ , which is the isosceles ratio of the target, so with  $cu = (f^2 \cos \beta, f^2 \sin \beta)$ ,

$$|jcu - (jf, 0)|^2 = j^2(f^2 - 1)^2 = (jb)^2,$$

and  $R_i$  is the affine combination dividing the segment in ratio  $i : (j - i)$ , so consecutive  $R_i$  are  $b$  apart (**chord\_jb**, **chord\_jb\_segment**).

(ii) Reducing  $xa + yb + zc = jb$  modulo  $f$  forces  $y \equiv j$ , while  $yb \leq jb$  forces  $y \leq j$ ; as  $0 \leq j \leq f - 1$  this gives  $y = j$ , hence  $xa + zc = 0$  and  $x = z = 0$  (**partition\_jb\_gen**, which is general in  $(e, f)$ ; **partition\_30** is its case  $f = 4$ ,  $j = 2$ ).

(iii) The  $\beta$ -slot exists: at the side junction  $jcu$  where the block ends, Lemma 70 gives  $C_j$ 's  $\alpha$  and  $M_j$ 's  $\gamma$ , which force the next side tile's corner to  $\beta$ , and that lemma is unconditional — it uses no base-letter hypothesis and no line closure, so it applies at every block length. Now  $\beta$  is opposite  $b$ , so its incident edges are  $a$  and  $c$ : the tile in the  $\beta$ -slot lays an  $a$ -edge (direct) or a  $c$ -edge (mirrored), never a  $b$ -edge. Both runs left on the chord are gaps, by the uniform statement that  $jb - t \notin \langle a, b, c \rangle$  whenever  $1 \leq j \leq f - 1$  and  $f \mid t, t > 0$  (`gap_jb_minus_multiple`): indeed  $a = f$  and  $c = f^2$  are both multiples of  $f$ . Concretely, if the tile is direct its  $a$ -edge does not reach  $R_{j-1}$  and the remainder  $b - a$  is a gap, so no edge ends at  $R_{j-1}$  and some edge crosses it; if it is mirrored its  $c$ -edge overshoots  $R_{j-1}$  by  $c - b = e^2 = 1$  and crosses it outright. Either way a through-edge occupies the whole half-turn on its side (`through_edge_exclusive`), the segment to  $(jf, 0)$  is a single edge, and the far side must partition  $jb - a$  resp.  $jb - c$  — gaps.

(iv) The last segment  $[R_{j-1}, (jf, 0)]$  of the chord has length exactly  $b$  by (i), and  $c - b = e^2 > 0$ , so a  $c$ -edge laid from  $R_{j-1}$  along the chord passes  $(jf, 0)$  by  $e^2$ . The chord descends:  $jcu$  has ordinate  $jf^2 \sin \beta > 0$  and  $(jf, 0)$  lies on the base, so the continuation beyond  $(jf, 0)$  has negative ordinate and leaves the target. This is the directional overshoot test of Remark 69: a  $c$  laid downward onto a base point exceeds the boundary and dies, whereas the same edge laid upward from the base would merely pass an interior point. Hence the  $\alpha$ -slot flank along the chord is a whole  $b$ -edge, and the reflected placement is excluded. Nothing here depends on  $f$ , nor on  $e$  beyond  $e^2 > 0$ .  $\square$

**Remark 57.** Proposition 56 is the  $f = 4$  argument of Remark 55 with every constant replaced by its general form: the three runs 11, 26, 14 against  $\langle 4, 15, 16 \rangle$  are  $b - a, 2b - a, 2b - c$ , i.e. the cases  $(j, t) = (1, a), (2, a), (2, c)$ , and the  $2b$  chord is  $j = 2$ . The bound  $j \leq f - 1$  of (i) is sharp for (iii): at  $j = f$  both  $jb - a = f(f^2 - 2)$  and  $jb - c = f(f^2 - f - 1)$  are multiples of  $f$  and lie in the semigroup.

Part (iv) supplies what the  $f = 4$  argument records as “the mirrored  $X$  dies by base overshoot”: there  $X = T_1 + (f, 0)$  is the triangle  $(f, 0), (2f, 0), R_1$ , and the exclusion is the case  $j = 2$ . The general reason is that the last segment of the chord has length exactly  $b$  while the competing edge has length  $c = b + e^2$ , and the chord's far end is a base point, so the excess leaves the target. Consequently escape route 2 of Conjecture 60 is closed at every initial block length, for every  $f \geq 3$ . Route 1 — a deviating  $a$ -run whose new  $c$ -chord ends beyond the forced region — is what now stands between this and Hypothesis 29 at  $e = 1$ , and the next remark says exactly how it behaves in  $f$ .

**Remark 58** (Route 1 escapes by exactly  $a$ , uniformly in  $f$ ). The escape recorded at  $(1, 4)$  as “25.875 > 21.875 exactly” is not a numerical accident of  $f = 4$ , and it does not improve with  $f$ . Write  $A = cu$  for the upper end of the side's first  $c$ -edge and  $B = (c + a)u$  for the upper end of the run's first  $a$ -edge. The second side tile carries its  $a$ -edge  $AB$  on the side and, being mirrored (Lemma 49), presents  $\beta$  at  $A$ , so its  $c$ -edge runs from  $A$ ; its third vertex is

$$V = cu + (c, 0), \quad |V - A| = c, \quad |V - B| = b,$$

so that  $c$ -edge is *horizontal*, at height  $c \sin \beta$ . The deviating  $a$ -run's new  $c$ -chord starts at the base junction  $(f + c, 0)$  and has upper endpoint  $E = (f + c, 0) + cu$ , at the same height. Hence

$$x(E) - x(V) = f$$

identically: the chord's endpoint overshoots the blocking edge by exactly one  $a$ -length, for every  $f$ . At  $f = 4$  this is  $25.875 - 21.875 = 4$ .

Two consequences. The escape condition  $x(E) > x(V)$  reads  $f > 0$ , so it holds in every member: there is no analogue here of the sign conditions  $f^3 - 3f^2 + 1$  and  $f^3 - 3f^2 - f + 1$  by which Lemma 50 disposes of its case, and no threshold in  $f$  beyond which route 1 closes itself.

Any proof must therefore recover the blocking by a different mechanism rather than by showing the overshoot absent.

Second, the configuration on that horizontal line is completely rigid, and says what the mechanism has to be about. Writing  $R_1 = cu + (f, 0)$  for the point called  $Q$  in the  $(1, 3)$  argument (the midpoint of the  $2b$  chord of Proposition 56), all four relevant points lie at the common height  $c \sin \beta$ , in the order

$$A \xrightarrow{a} R_1 \xrightarrow{c-a} V \xrightarrow{a} E, \quad |AV| = |R_1E| = c,$$

so the line carries *two overlapping  $c$ -segments offset by exactly  $a$* : the second side tile's  $c$ -edge  $[A, V]$ , which is a genuine tile edge, and  $[R_1, E]$ , whose right end is the chord's unblocked endpoint. In particular the gap  $[V, E]$  between the blocking edge's end and the chord's endpoint has length exactly  $a$  — the length of an  $a$ -edge. What route 1 needs is therefore a statement about what covers  $[V, E]$ : if that segment is forced to be a whole  $a$ -edge then  $E$  is a junction and the vertex figure there is constrained, which is the substitute for the tile-interior blocking that fails. We do not prove that here, and record only that the geometry leaves no other target: every length in the display is exact and independent of  $f$ .

**Remark 59** (The blocking hypothesis cannot be weakened to “both ends are vertices”). Recorded in [1]. Full discussion in [1].

**Conjecture 60** (Advance-and-collide). [CONJECTURE, PROVED AT  $f = 2, 3, 4$ ] Let  $e = 1$ ,  $m = 1$ ,  $f \geq 3$ , and suppose an equal side carries an  $a$ -edge. Then the corner cascade — the  $b$ -partner of the corner tile, the  $c$ -chord dichotomy on successive both-ends-blocked chords, and the  $\beta$ -slot at each base point reached — terminates in every branch with a collision: a  $\beta$ -slot, whose tile presents the next base edge from flanks  $\{a, c\}$ , at a base position where the word of Theorem 20 carries its  $b$ -edge, or a repeated  $\gamma$ . Consequently  $p = 0$ : Hypothesis 29 holds for every  $(1, f)$ .

The conjecture is proved here for  $f = 2$  (Lemma 44),  $f = 3$  (Theorem 53) and  $f = 4$  (Theorem 61, which exhausts all six admissible base words). The two ways a branch escapes the collision in the general argument are exactly: a deviating  $a$ -run whose new  $c$ -chord ends beyond the forced region (so that tile-interior blocking fails), and an initial side  $c$ -block of length at least 2 (so that the first-run orientation is not forced). Any proof of the general case must close these two gaps and no others.

The second gap is now general except for its enumeration. Its ingredients carry no restriction on  $f$ : `first_run_kill` and `through_edge_exclusive` are statements in  $(n_\alpha, n_\beta, n_\gamma)$  alone; Lemma 50 reduces to the signs of  $f^3 - 3f^2 + 1$  and  $f^3 - 3f^2 - f + 1$ ; the chord carrying the three runs has length exactly  $2b$  with  $Q$  its midpoint for every  $f$ , since with the base corner at the origin and  $u$  the unit vector along the equal side,

$$|2cu - (2f, 0)|^2 = 4(f^2 - 1)^2 = (2b)^2, \quad |2cu - Q|^2 = |Q - (2f, 0)|^2 = (f^2 - 1)^2 = b^2$$

(`chord_two_b`, `chord_two_b_half`); and the three runs  $b - a = f^2 - f - 1$ ,  $2b - a = 2f^2 - f - 2$ ,  $2b - c = f^2 - 2$  are gaps of  $\langle f, f^2 - 1, f^2 \rangle$  for every  $f \geq 3$  (`double_c_kill`), the values 11, 26, 14 against  $\langle 4, 15, 16 \rangle$  being the case  $f = 4$ . At  $f = 2$  the last two runs lie in the semigroup, which is where the hypothesis  $f \geq 3$  of Theorem 20 is used. What is not yet general is the enumeration: a side with initial  $c$ -block at least 2 begins and ends with a  $c$ -edge, so  $n_c \geq 3$  and the double- $c$  walks are  $(fk, 0, f - k)$  with  $1 \leq k \leq f - 3$ ; at  $f = 4$  that is the single word  $cca^4c$  treated in Remark 55, while for  $f \geq 5$  there are  $f - 3$  walks and the argument as written covers an initial block of exactly 2 followed by an  $a$ -run.

**Theorem 61** (Hypothesis 29 holds at  $(1, 4)$ ). [CONJECTURE] *No tiling of the  $(1, 4)$  target at  $m = 1$  carries an  $a$ -edge on an equal side; hence, by Remark 47, no such tiling exists, re-deriving the exclusion of the  $(1, 4)$  instance ( $N = 47$ ) established by certified exhaustion (12 440 nodes).*

*Proof.* Suppose a side (named the left) carries an  $a$ -edge; then  $p = 1$  there (Lemma 50). Both corner partners are forced with  $\gamma$  at their first junctions, unconditionally: the reflected orientation repeats the corner tile's  $\gamma$  at the base end and overflows.

*Left kills.* For left side words with initial  $c$ -block 1, Lemma 51 runs on  $P$ 's  $c$ -chord as at  $(1, 3)$ : base words with second letter  $a$  die at  $(2f, 0)$  or  $(3f, 0)$  ( $\beta$ -slot against the forced  $b$ -edge);  $aacbaa$  dies at  $(f + c, 0)$  through the  $a$ -run partner;  $acbaaa$  dies at  $(f + c, 0) = (20, 0)$ , where its  $b$ -edge occupies the four- $a$  branch's first  $\beta$ -slot. Five of the six words die, leaving  $acabaa$ . The double- $c$  left word dies for every base word by the line-total kill of Remark 55.

*Right kill of  $acabaa$ .* The right cascade is the left cascade of the reversed word  $abacaa$ . The right corner tile, its partner  $P'$ , and the  $\beta$ -slot at  $(Y - f, 0) = (43, 0)$  force  $C' = T'_1 - (f, 0)$ , whose  $b$ -edge is collinear with the second right tile's  $b$ -line and meets the base at  $(Y - 2f, 0) = (39, 0)$  — an endpoint of the base word's  $b$ -edge [24, 39]. If the right side is all  $c$ 's, the second right tile is forced direct (**first\_run\_kill** with  $P'$ 's  $\gamma$ ), the doubled  $b$ -line is edged on its east side, and its west side partitions  $2b = 30$ , whose unique decomposition is  $b + b$  (**partition\_30**): the split falls at the mid-junction, producing the two  $b$ -partners. If the right side carries an  $a$ -run with initial block 1, the second right tile's horizontal  $c$ -edge blocks the mid-junction and the partner of  $C'$  is forced with  $\gamma$  there by the through-edge figure. Either way,  $\gamma$  at  $(39, 0)$  would double  $C'$ 's  $\gamma$ , so the partner's  $\alpha$  sits at  $(39, 0)$ ; the figure there closes with exactly one  $\beta$  adjacent to the base-left ray, and that tile must present the  $b$ -edge [24, 39] from flanks  $\{a, c\}$  — impossible. The double- $c$  right word dies by the mirrored line-total kill. Every combination of side and base words is exhausted;  $p = 0$ .  $\square$

**Remark 62** (The reversal principle). The mechanism is symmetric: the corner cascade runs from both base corners, and the right cascade of a base word is the left cascade of its reversal. A base word survives only if both it and its reversal survive the left cascade; at  $(1, 4)$  the reversal involution pairs the six words so that none does. This is the intended proof shape for Conjecture 60: show the left cascade kills every word with second letter  $a$  and every word whose  $b$ -edge occupies the four- $a$  branch's first slot, and that the survivors' reversals always lie in the killed class.

**Theorem 63** (The cascade closes every initial-block-1 configuration, for all  $f$ ). [CONJECTURE] *Let  $f \geq 3$ ,  $m = 1$ , and suppose an equal side of the  $(1, f)$  target carries an  $a$ -edge whose first run is preceded by exactly one  $c$ -edge, and that the opposite side either has no  $a$ -edge or also has initial  $c$ -block 1. Then no tiling exists. Consequently Hypothesis 29 fails to hold at  $(1, f)$  only if some hypothetical tiling has a side whose first  $a$ -run sits behind at least two  $c$ -edges; for  $f \leq 4$  no such configuration survives either (Theorems 53, 61), so Hypothesis 29 holds outright there.*

*Proof.* Three lemmas. (*L1: the single- $c$  chain.*) With initial block 1 the second side tile is the first  $a$ -tile, mirrored (Lemma 49), and its horizontal  $c$ -edge spans offsets  $[0, c]$  above the corner. The cascade produces cells  $C_k = T_1 + (kf, 0)$  and partners  $Z_k = P + (kf, 0)$  by induction: the  $k$ -th chord's upper end is  $cu + (kf, 0)$ , interior to that horizontal edge exactly when  $kf < f^2$ ; a base word whose  $b$  sits at position  $q \leq f$  is reached at step  $q - 2 \leq f - 2$ , so every needed chord is blocked (**cascade\_reaches**). At each base point the figure closes with exactly one  $\beta$  whose flanks are  $\{a, c\}$ ; reflected partners die by repeating  $\gamma$  at the base end. The letter  $b$  therefore collides: every word with its  $b$  before its  $c$  dies. (*L2: the  $f$ - $a$  branch.*) If the  $c$  immediately precedes the  $b$ , the  $f$ - $a$  branch of Lemma 51 runs at the  $c$ 's position, its first  $a$ -tile's  $b$ -partner is forced (the reflected orientation repeats  $\gamma$  at the chord's division point), its  $\gamma$  lands at the next base point, and the  $\beta$ -slot there collides with the  $b$ -edge. (*L3: reversal.*) Every admissible word has its  $b$  before its  $c$ , or its  $c$  immediately before its  $b$ , or its  $c$  before its  $b$  with a gap; in the third case the reversed word's  $b$ -position is  $f + 3 - q \in [3, f]$  and precedes its  $c$  (**reversal\_covers**), so the mirrored cascade from the opposite corner kills it — the opposite corner's chain is available

because the opposite side has no  $a$ -run (then the doubled  $b$ -line's far side partitions  $2b$  uniquely as  $b + b$ , `partition_2b`, forcing the partners) or has initial block 1 (then its horizontal  $c$ -edge blocks the mid-junction and the through-edge figure forces the partner). In every case the collision closes the configuration.  $\square$

**Lemma 64** (The  $jb$ -line partition). [VERIFIED] *For  $1 \leq j < f$  the only decomposition of  $jb$  as  $xa + yb + zc$  is  $x = z = 0$ ,  $y = j$ . Hence a line of length  $jb$  that is covered by whole edges on a side splits there into exactly  $j$  whole  $b$ -edges. (Whether a given line is so covered is a separate question: see Remark 14.)*

*Proof.* Reducing  $xf + y(f^2 - 1) + zf^2 = j(f^2 - 1)$  modulo  $f$  gives  $y \equiv j$ ; the size bound gives  $y \leq j$ , and  $j < f$  then forces  $y = j$ , hence  $x = z = 0$ . Machine-checked (`partition_jb`).  $\square$

**Remark 65** (The block cascade). In a residual configuration (initial  $c$ -block of length  $\ell \geq 2$ ), the line through  $jcu$  in the  $b$ -direction reaches the base at  $(jf, 0)$  with total length  $jb$ : each step is the vector  $cu - (f, 0)$ , of length  $b$ , by the translation structure alone. Three consequences, all side-word-independent, and all conditional on the relevant segment being uncrossed (Remark 14). (i) The  $(f, 0)$ -slot closes with exactly one  $\beta$  whose flanks are  $\{a, c\}$ , so the second base letter is  $a$  or  $c$ , with its tile forced:  $X_1 = T_1 + (f, 0)$  or the run-start  $A_1$ . (ii) When the second letter is  $a$ , the  $2b$ -line  $[2cu, Q_1, (2f, 0)]$  is fully west-edged by  $C_2$ 's and  $X_1$ 's  $b$ -edges, and at  $Q_1$  the west side closes exactly ( $C_2$ 's  $\gamma + P$ 's  $\beta + X_1$ 's  $\alpha = \pi$ ); Lemma 64 forces the far side to split at  $Q_1$  into both  $b$ -partners, the reflected lower partner repeats  $X_1$ 's  $\gamma$  at the base point  $(2f, 0)$  and dies, and the  $(2f, 0)$ -slot closes with exactly one  $\beta$ : *the third letter is never  $b$* . Every residual configuration whose base word has second letter  $a$  and  $b$  in position 3 is dead. (iii) At the  $2cu$  junction the figure  $C_2$ 's  $\alpha +$  upper mate's  $\gamma +$  corner  $= \pi$  forces  $C_3$  direct when  $\ell \geq 3$ : the block's chirality forcing propagates upward with the line, one junction per depth, and Lemma 64 supplies the partition at every depth  $j < f$ . The remaining freedom is the row-2 chord  $[Q_1, Q_1 + cu]$ , whose east side forks single- $c/f$ - $a$  exactly as the base did: the corner block grows as a two-dimensional array of words, one chord fork per row. Closing the residual class for all  $f$  means closing this array; that is the precise remaining content of Conjecture 60 at  $e = 1$ .

**Lemma 66** (The east fan at the fork is forced). [PROVED] *Let  $f \geq 3$ ,  $\ell \geq 2$ , and let the second base letter be  $a$ , so that the brick  $X_1$  and the mate  $M$  are in place. At the fork point  $Q_1 = cu + (f, 0)$  the tiles east of the chord form the figure  $\{\gamma, \alpha, \beta\}$  in that angular order from the base side: the mate  $M_1$  of  $X_1$ , then a  $\beta$ -tile, then the direct partner  $D_2$  of  $C_2$ . The  $\beta$ -tile has flanks  $\{a, c\}$ , one along the floor and one against  $D_2$ 's  $c$ -edge, so exactly two configurations occur:  $a$  on the floor (matching  $M_1$ 's  $a$ -edge), or  $c$  on the floor — the tile  $A_2$ .*

*Proof.* The chord through  $Q_1$  joins the side vertex  $2cu$  to the base vertex  $(2f, 0)$ , so both ends lie on the boundary and no edge extends past them; its east side is a union of whole edges of total length  $2b$ , which is  $b + b$  for  $f \geq 3$  (Lemma 64), split exactly at  $Q_1$ . Hence  $Q_1$  is a vertex of that cover and each east tile meets it at a vertex. The lower half is  $X_1$ 's  $b$ -edge; the single whole  $b$  covering it from the east belongs to a partner of  $X_1$ , and the reflected partner would repeat  $X_1$ 's  $\gamma$  at the base junction  $(2f, 0)$ , giving  $2\gamma > \pi$  there since  $\cos \gamma = -1/(2f) < 0$  (`straight_junction_gamma_bound`). So it is the direct partner  $M_1$ , which presents  $\gamma$  at  $Q_1$  — its edges there being  $b$  and  $a$  — and lays an  $a$ -edge east along the floor. A straight-junction figure containing a  $\gamma$  is  $\{\gamma, \alpha, \beta\}$  (`straight_junction_cases`), so exactly two further tiles meet  $Q_1$ . The one adjacent to the chord shares  $C_2$ 's whole upper  $b$ , so its angle at  $Q_1$  is an endpoint angle of a  $b$ -edge, namely  $\alpha$  or  $\gamma$ ;  $\gamma$  is taken, so it is  $\alpha$ , and the reflected alternative would again double  $C_2$ 's  $\gamma$  at  $Q_1$ . That tile is therefore  $D_2$ . The remaining tile carries  $\beta$ , whose flanks are  $\{a, c\}$ .  $\square$

**Theorem 67** (The row fork kill). [PROVED] *Let  $f \geq 3$ . In a residual configuration (initial  $c$ -block  $\ell \geq 2$ ) whose base word has second letter  $a$ , the row-2 fork resolves single- $c$ : the  $\beta$ -tile lays  $a$  along the floor.*

*Proof.* Lemma 66 leaves exactly two configurations, and Proposition 68 excludes the second.  $\square$

**Proposition 68** (The  $A_2$  branch dies). [PROVED] *In a residual configuration (initial  $c$ -block  $\ell \geq 2$ ) whose base word has second letter  $a$ , suppose the tile east of the row-2 fork is the mirrored  $L = -2$  tile  $A_2$ , laying a horizontal  $c$ -edge along the floor line east of the fork point. Then no tiling exists.*

*Proof.*  $A_2$ 's  $c$ -edge is an edge of the tiling, so no tile crosses the floor east of the fork point, and the south side of that edge is a union of whole edges with vertices on it. At a junction  $J$  of that cover the fan below the floor is the previous element's  $\beta$ , the row-1 brick's  $\alpha$ , and a remaining wedge of exactly  $\gamma$ . Since  $\gamma < \pi$ , no tile can meet  $J$  through the interior of one of its edges (that contributes exactly  $\pi$ ), so the wedge is filled by tiles having a vertex at  $J$ : the fill is  $\{\gamma\}$  or  $\{2\alpha, \beta\}$  (`straight_junction_cases`). The wedge is bounded by the floor and by the brick's  $b$ -edge running down to the base. A filler laying  $c$  along that ray leaves the target (it exceeds the ray by  $c - b = 1$  past the base); a filler laying  $a$  along it makes that  $a$  the first element of the  $b$ -edge's east cover, leaving  $f^2 - 1 - f$ , which admits no completion (`east_cover_gap`); a filler laying a whole  $b$  along it is the reflected partner of the brick and repeats  $\gamma$  at the base junction (`straight_junction_gamma_bound`). Hence every fill is the brick's mate, the cover is  $f$  whole  $a$ -edges, their feet force base junctions at  $(kf, 0)$  for  $k = 2, \dots, f + 1$ , and the first  $f + 1$  letters are all  $a$  — contradicting the  $b$  at position at most  $f$ . (The non-brick  $a$ -up placements are excluded by `alpha_vertex_gap` and `anti_brick_side` as before.)  $\square$

**Remark 69** (Why  $f \geq 3$ , and what the  $(1, 2)$  tilings show). The hypothesis  $f \geq 3$  in Lemma 66 is doing real work, and the  $(1, 2)$  member shows exactly what it buys. At  $f = 2$  one has  $2b = 6 = b + b = a + c = 3a$ : the chord's partition is not unique, the split need not fall at  $Q_1$ , and a tile may straddle there. The kernel-checked 99-tiling does precisely this — its tile with vertices  $(4, 0)$ ,  $(\frac{23}{4}, \frac{\sqrt{15}}{4})$ ,  $(5, \sqrt{15})$  lays a  $c$ -edge from the base along the chord which covers the row-1 brick's entire  $b$ -edge and overshoots  $Q_1$  by exactly  $c - b = 1$ . For  $f \geq 3$  that escape costs  $2b - c = f^2 - 2$ , which lies below  $b$  and is not divisible by  $f$ : a gap (`gap_2b_minus_c`). So the phenomenon is real, is realised, and is confined to  $f = 2$ .

Two methodological points are worth recording. First, the overshoot test is *directional*: a  $c$  laid downward from a floor junction exceeds the boundary and dies, while a  $c$  laid upward from the base overshoots an interior point and is legal. Second, an argument of the form “a segment pinned at both ends is covered by whole edges” is sound only where the ends are genuinely pinned — on the boundary, or against a tile edge that blocks continuation — and should say which. Both uses in Theorem 63 are of that kind: its chords are blocked above by the first  $a$ -tile's horizontal  $c$ -edge, and its doubled  $b$ -line runs boundary to boundary.

**Lemma 70** (The wall climb). [PROVED] *For every  $\ell \geq 2$  and  $2 \leq j \leq \ell$ , the block tile  $C_j$  is direct and its mate  $M_j$  is forced, with  $M_j$ 's  $c$ -edge  $[Q_{j-1}, Q_j]$  continuing the interior wall; the first run tile above the block is mirrored. This is unconditional: no base-letter hypothesis and no line closure is used.*

*Proof.* Induction on  $j$ .  $C_2$  is direct by the  $cu$ -junction figure ( $T_1$ 's  $\alpha + P$ 's  $\gamma + \beta$ ). Given  $C_j$  direct, the west fan at  $Q_{j-1}$  consists of  $\beta$  ( $P$ 's for  $j = 2$ , else  $M_{j-1}$ 's),  $C_j$ 's  $\gamma$ , and a wedge between  $C_j$ 's  $b$ -ray and the  $u$ -line of exactly  $\pi - \gamma - \beta = \alpha$ : it is filled by a single  $\alpha$ -tile ( $\alpha$  is the minimal angle). Its flank along  $C_j$ 's  $b$ -ray cannot be a  $c$ -edge — it would overshoot the boundary point  $jcu$  — so it is a whole  $b$ -edge shared with  $C_j$ , and the reflected partner is excluded by

the wedge's  $\alpha$ : the filler is the mate  $M_j$ , its  $c$ -edge running up the  $u$ -line. At the side junction  $jcu$ ,  $C_j$ 's  $\alpha + M_j$ 's  $\gamma$  force the next side tile's corner to  $\beta$ :  $C_{j+1}$  is direct while the block lasts ( $\alpha + \gamma + \alpha$  and  $\alpha + \gamma + \gamma$  are impossible straight figures), and the first run tile is mirrored when it ends.  $\square$

**Remark 71** (The pincer). The assembly frame for the residual class: each corner runs its own chain, and a chain needs only its *own* second letter to be  $a$ . If  $cp > bp$  the left chain meets the  $b$  first (second letter  $a$  since  $cp > bp \geq 3$ ); if  $cp < bp$  the reversed word has  $b$  before  $c$  and the right corner — whose side is all- $c$ , block-1, or again residual — runs the mirrored chain (reversed second letter  $a$  since  $bp \leq f$ ). Both chains advance through slots, lines and row forks; with the three leaks excluded the forks resolve to bricks, the lines close at every depth (Lemma 64), and the colliding  $b$  kills the configuration. Hypothesis `hyp:walls` at  $e = 1$  is therefore reduced to the leak exclusion.

**Theorem 72** (L2 at every reached slot). [PROVED] *Let  $f \geq 3$  and suppose the chain from a corner has closed slots  $1, \dots, k$ . If the base letter presented at slot  $k$  is  $c$  and the next letter is  $b$ , no tiling exists. This holds for every block length of that corner's side.*

*Proof.* The slot- $k$   $\beta$ -tile lays  $c$  on the base and its  $a$  up the  $u$ -ray from  $(kf, 0)$ , which is the  $Z_{k-1}$ -mate's  $c$ -edge (for  $k = 1$ :  $P$ 's  $c$ -edge) — an edge of the chain. The branch tile's  $b$  runs from the  $fu$ -division point of that edge to the base point  $(kf + c, 0)$ ; its upper end abuts the edge's interior, so no covering edge crosses it, and the lower end is on the base: the far side is a single whole  $b$  (`partner_unique`). The reflected partner repeats  $\gamma$  at the division point, an interior straight junction, and  $2\gamma > \pi$ ; the mate's  $\gamma$  lands at  $(kf + c, 0)$ , the figure there closes with exactly one  $\beta$  whose flanks are  $\{a, c\}$ , and it must present the  $b$ -edge: impossible.  $\square$

**Theorem 73** (The block-two chain runs to arbitrary depth). [PROVED] *Let  $f \geq 3$  and let a side carry its first  $a$ -run behind exactly two  $c$ -edges. Then the chain from that corner closes every slot it needs: the base letters are presented one by one, a  $b$  collides at its slot, and a  $c$  followed by  $b$  collides by Theorem 72. In particular a base word whose  $b$  precedes its  $c$  dies from this corner.*

*Proof.* By Lemma 70,  $T_{a1}$  is mirrored and lays its horizontal  $c$ -edge at height  $2cu_y$ : the ceiling. The column- $k$  line through  $((k+1)f, 0)$  meets the ceiling line after two pieces, at a point interior to  $T_{a1}$ 's  $c$  (offset  $kf < c$ ), so its west-edged portion is  $X_k$ 's  $b$  plus the row-2 brick's  $b$ , pinned above by the ceiling edge and below by the base: `partition_2b` splits the east side at the joint, the reflected partners die by repeating  $\gamma$  (at the base, resp. at the joint), and both mates are forced: the next slot closes. The row-2 bricks advance by the fork at each column: the fork line is pinned the same way, the fan at the fork point is  $\gamma | \alpha | \beta$  exactly as in Lemma 66 with the ceiling in place of the wall, and the  $A$ -branch dies by Proposition 68, whose south feet lie on the base and whose leak inventory is closed by `east_cover_gap` and the directional overshoot (both valid one strip above the base). Induction on the column.  $\square$

**Theorem 74** (Reach three behind a thick block, and the pincer window). [PROVED] *Let  $f \geq 3$ . From a corner whose side has initial block  $\ell \geq 3$  (including the all- $c$  side), slots 1, 2, 3 close: the lines for columns  $k \leq 2$  terminate on the wall (Lemma 70) with  $k + 1$  pieces and split by Lemma 64; the single row-2 brick needed is forced by Theorem 67. Consequently every residual configuration dies unless its base word escapes all four kills: direct- $b$  from the  $b$ -first corner at depth  $\leq 4$ , and L2 (Theorem 72) from the  $c$ -first corner when the  $c$  sits at depth  $\leq 4$  with the  $b$  adjacent. For  $f \leq 5$  no admissible  $(bp, cp)$  escapes (`pincer_window`).*

**Corollary 75.**

[PROVED]

VIA THE CERTIFIED EXHAUSTION w74; THE ARGUMENT-ONLY ROUTE INHERITS THEOREM 63'S EFFECTIVE LABEL]

*Hypothesis 29 holds at (1,5): together with blocks  $\leq 2$  (Theorems 63, 73) the window covers every configuration. This is the first member closed with no engine certificate under the chain's own hypotheses; the member is independently certified by the exhaustion w74 (132 519 nodes), so its exclusion does not rest on the chain.*

**Lemma 76** (The apex). [VERIFIED] *The target's apex angle is  $\pi - 2\beta = 3\alpha$ , and the only vertex figure filling it is three  $\alpha$ -corners (**apex\_figure**). Since the sides carry no  $b$ -edge (**side.no\_b\_uncond**), the two outer apex tiles lay  $c$ -edges on the sides — the top letters of both side words are  $c$  — and the three wedges' rays sit at labels  $-3, -1, +1, +3$ . The apex therefore admits exactly two configurations, distinguished by the middle tile's chirality: its  $b$  shares the  $-1$ -ray with the left tile's  $b$  (both  $\gamma$ -ends meeting below), or its  $c$  covers that  $b$  with a  $T$ -junction at depth  $b$ . The side conclusion is specific to  $m = 1$ : the kernel-checked tilings at  $m \geq 2$  carry  $b$ -edges on their sides, and their apex regions, while all realizing the forced three- $\alpha$  figure, diverge from one another immediately below it. Any forcing chain from the apex must therefore anchor on the  $m = 1$  side structure; no  $m$ -independent reach exists.*

**Remark 77** (The straddler: the unique leak at rows  $\geq 3$ ). At every strip junction of a row- $j$   $A$ -branch the wedge is exactly  $\alpha$ , and its filler admits exactly two label assignments: the mirrored  $L = -2$  mate (whole  $b$  shared with the run tile; the strip structure stays rigid) or the direct  $L = 0$  straddler —  $b$  horizontal under the next  $c$ -line,  $c$  laid along the run tile's  $b$ -ray, covering that whole  $b$  and overshooting the run tile's  $\alpha$  by exactly  $c - b = 1$ , poking below the floor line just past the horizontal  $c$ -edge's east end. At row 2 the analogous fill dies against the boundary ( $2b - c = f^2 - 2$  is a gap, **gap\_2b\_minus\_c**); at rows  $\geq 3$  the poke is interior and no local figure excludes it. The straddler does not cascade: it couples strips only upward, so no descent-to-the-base argument applies. Killing this one shape at rows  $\geq 3$  is equivalent to the full chain reach, hence to Conjecture 60 at  $e = 1$ .

**Lemma 78** (Descent identities and the ladder). [PROVED] *Let  $f \geq 3$ . (i) A line in the  $b$ -direction (label  $-1$ ) crosses the horizontal strip lines at intervals of exactly  $b$  along itself and advances exactly  $c$  horizontally per crossing:  $\sin \beta = b\sqrt{D}/(2f^3)$  with  $D = 4f^2 - 1$ , and  $f u_x + b |\cos 2\gamma| = c$  (**descent\_ident**, **sinb\_ident**). (ii) A straddler's  $c$ -edge therefore covers the run tile's  $b$  and pokes one unit past its  $\alpha$ -vertex  $T$ -junction; at every  $T$ -junction on the line the fan-adjacent tile lays a whole further edge along it, so the straddler forces a doubly edged ladder: east breaks in  $b + \langle a, b, c \rangle$ , west breaks in  $c + \langle a, b, c \rangle$ , staggered by  $c - b = 1$ . (iii) The ladder cannot terminate on the base: the termination abscissa is  $(f^3(m+1) - K(f^4 - 3f^2 + 1))/f^2$  with  $K = (j-1)f + i$ , and  $f^4 - 3f^2 + 1 \equiv 1 \pmod{f^2}$  forces  $f^2 \mid K$ , impossible since  $K \equiv i \not\equiv 0 \pmod{f}$  (**ladder\_no\_base**). (iv) The ladder can terminate only at an interior mutual vertex, where the cumulative east and west sums agree; the stagger makes this  $s_1 = s_2 + 1$  with  $s_1, s_2 \in \langle a, b, c \rangle$ , realized earliest by  $(s_2, s_1) = (b, c)$ : every straddler forces at least  $b + c = 2f^2 - 1$  of doubly edged ladder, descending two full strips and gliding exactly  $c$  east of the strip ends.*

**Lemma 79** (Termination is one condition, boundary or interior). [PROVED] *A ladder terminates only where both of its staggered covers end: at an interior mutual vertex by definition, and at a boundary point because the boundary ends both covers at once. So side exit is not a separate escape route; every termination satisfies the same condition. With  $c = b + 1$  that condition reads:  $T - b$  and  $T - c$  are consecutive members of  $S = \langle a, b, c \rangle$ . Consecutive members of  $S$  begin only at  $f^2$  (**consecutive\_gap**), so*

$$T \geq b + c = 2f^2 - 1.$$

*Proof.* Write  $m + 1 = T - b$  and  $m = T - c$ , both in  $S$ . Reducing  $xf + y(f^2 - 1) + zf^2$  modulo  $f$  shows an element of  $S$  congruent to  $-y$ ; if  $m + 1 < f^2$  then both representations have  $z = 0$

and  $y \leq 1$ , and the four cases give  $f \mid 1$ ,  $f \mid 2$ , or a direct size contradiction. Hence  $m + 1 \geq f^2$  and  $T = b + (m + 1) \geq b + c$ .  $\square$

**Theorem 80** (Straddlers need three strips, so rows 1, 2, 3 are clean). [CONJECTURE] *Let  $f \geq 3$ . The target is exactly  $f$  strips tall ( $f^3 \sin \beta$  over  $c \sin \beta$ ), and a ladder descends exactly  $b$  along itself per strip (Lemma 78). By Lemma 79 its length satisfies  $T \geq b + c > 2b$ , so it crosses strictly more than two strips; since it cannot terminate on the base (Lemma 78(iii)), a straddler whose ladder starts at the floor of row  $j$  requires  $j - 1 > 2$ . Hence no straddler occurs at rows 1, 2 or 3, the row-3 fork admits only the two branches of Lemma 66. The A-branch's exclusion at row 3 invokes Proposition 68 whose south-cover feet land on the row-2 floor, edged only as far as the staircase has reached: that step is incomplete as written, and the structural claim of reach 4 is open pending its repair. The (1, 6) conclusion itself is unaffected: it is certified by the independent exhaustion w107.*

*What the repair still owes is now smaller than it was. The proposition's Lean skeleton carried two geometric hypotheses about the descent — that the feet reach the base, and that each lands on an a-junction — and the second is close to the conclusion being sought. It is removable. The south cover is  $f$  whole a-edges, so its  $f + 1$  endpoints span a length  $f \cdot a = f^2$ ; they descend to  $f + 1$  distinct base points, which are base-word junctions since no vertex of a tile lies interior to a boundary edge; and they therefore carve a run of  $k \geq f$  whole letters of total length  $f^2$ . The three edge lengths then decide the run by themselves: a single  $c$  would exhaust  $f^2$  in one letter, leaving  $k = 1 < f$ , and a  $b$  would leave a remainder of 1, below the shortest letter, so every letter is an  $a$  and  $k = f$  exactly. That is  $f$  consecutive a-letters, which the base word cannot carry, its  $f$  letters  $a$  forming one block while its first and last slots are both  $a$ . So the descent needs only to land; what it lands on is forced arithmetically (A2BranchRow3. span\_all\_a, row\_three\_dies\_of\_span, axiom-clean).*

**Corollary 81.**

[CONJECTURE: EFFECTIVE, RESTING ON REACH 4, WHICH IS OPEN] *Hypothesis 29 holds at (1, 6), hence for every  $f \leq 6$ : with reach 4 the four kills cover depth  $\leq 5$  and no admissible  $(bp, cp)$  escapes (pincer\_window\_four). The (1, 5), (1, 6) and (1, 7) cases are additionally certified by independent exhaustions of the full  $m = 1$  targets (132,519, 1,251,382 and 13,729,561 nodes, instances w74, w107, w146). These are exhaustion certificates for the targets themselves and are labelled as computations; they are not a reading of Hypothesis 29 at those members, which at  $e = 1$  would require the arrangement check of Remark 32.*

**Remark 82** (The window is mirror-symmetric, and what that leaves at  $N = 191$ ). The target is isosceles, so reflecting it swaps the two base corners and reverses the base word: position  $i$  goes to  $f + 3 - i$ , and a configuration  $(bp, cp)$  to  $(f + 3 - bp, f + 3 - cp)$ . The admissible box  $3 \leq bp \leq f$ ,  $2 \leq cp \leq f + 1$  is carried to itself. The four kills are stated asymmetrically — direct- $b$  from the  $b$ -first corner, L2 from the  $c$ -first corner — but the reflection swaps those corners, and it permutes the kills in two transpositions, exchanging the first with the second and the third with the fourth (PincerLadder.kill\_mirror, omega, axiom-clean). So the escape set is closed under the reflection and its members come in mirror pairs.

That halves the residue. Enumerating the admissible box against the *proved* reach 3 ( $R = 4$ ; reach 4 is open at prop:a2branch row 3) gives

| $f$ | $N$ | words | escapes | up to mirror |
|-----|-----|-------|---------|--------------|
| 5   | 74  | 12    | 0       | 0            |
| 6   | 107 | 20    | 2       | 1            |
| 7   | 146 | 30    | 6       | 3            |
| 8   | 191 | 42    | 14      | 7            |

with representatives at  $f = 8$  of  $(4, 2), (5, 2), (5, 3), (5, 6), (5, 7), (5, 8), (5, 9)$ , and at  $f = 7$  of  $(4, 2), (5, 2), (5, 3)$ . So  $N = 191$ , the smallest unsettled prime, stands seven configurations from an argument-only exclusion at today's reach, and three at reach 4 — not forty-two. Generated by `code/analysis/base_word_residue.py`, which reproduces the ladder independently.

The count is a measurement, not a route: Corollary 81 derives  $f \leq 6$  from reach 4, which is open, so at  $e = 1$  that corollary is conditional on it; the members  $f \leq 7$  are settled regardless by the certified exhaustions quoted there. And by Remark 48 the phrase “Hypothesis 29 holds at  $(1, f)$ ” should be read at  $e = 1$  as “the member is excluded”, those being equivalent there — it is not evidence for the hypothesis as a structural fact.

**Remark 83** (The reach ladder, uniform in the reach). Theorem 74 and Corollary 81 instantiate one statement twice. If the four kills cover depth  $\leq R$ , then no admissible  $(bp, cp)$  escapes for any  $f \leq R + 1$ ; since reach  $r$  makes the kills cover depth  $\leq r + 1$ , this reads

$$\text{reach } r \implies f \leq r + 2.$$

Reach 3 gives  $R = 4$  and  $f \leq 5$ , reach 4 gives  $R = 5$  and  $f \leq 6$ , recovering both statements (`pincer_ladder`, with `ladder_reach_three` and `ladder_reach_four` as its instances). The arithmetic half of the ladder is therefore settled at every height at once: extending Corollary 81 past  $f = 6$  requires only the geometric reach step, and no further case analysis in  $(bp, cp)$  at any  $f$ .

The bound is sharp. At  $f = R + 2$  the configuration  $(bp, cp) = (4, 2)$  escapes all four kills (`pincer_sharp`), and enumerating the escapes at that  $f$  gives exactly  $(4, 2)$  and  $(f - 1, f + 1)$  and nothing else (`first_failure_escapes`).

Those are not two families. They are one configuration seen from the two base corners:  $(f - 1, f + 1)$  is the mirror  $(f + 3 - 4, f + 3 - 2)$  of  $(4, 2)$ , and by Remark 82 the kills are invariant under that reflection. An earlier version of this remark concluded that excluding  $(4, 2)$  alone “changes nothing”; on the contrary, a proof that  $(4, 2)$  cannot occur, stated generally in  $f$ , transports to its mirror by reflecting the tiling, and so advances the bound by the full step. After the gain two further orbits appear,  $(5, 2)$  and  $(5, 3)$  with their mirrors. What survives is that each  $f$  costs its own reach step; what does not is the count — the step is a single configuration.

That makes the route to  $N = 191$  a named target rather than a programme. It needs the ladder driven from  $R = 4$  to  $R = 7$ , and each step is the exclusion of the base word  $acaba^{f-3}$ , which is what  $(bp, cp) = (4, 2)$  spells.

**Remark 84** (The  $(4, 2)$  configuration has nowhere for the descent to land). That target is smaller than it looks. The word  $(bp, cp) = (4, 2)$  spells is

$$acaba^{f-2},$$

$f+2$  letters with exactly  $f$  of them  $a$ , and its  $a$ -runs have lengths 1, 1 and  $f-2$ . The longest is  $f-2 < f$ , so the word carries *no* block of  $f$  consecutive  $a$ -letters (`A2BranchRow3.word42_no_a_block`).

Now recall the span dichotomy of Theorem 80's repair: a base run of  $k \geq f$  whole letters of total length  $f^2$  must consist of  $f$  consecutive  $a$ s, those and a single  $c$  being the only ways to make  $f^2$  from  $\{f, f^2 - 1, f^2\}$ . This word has no such run; and a descent landing on the single  $c$  instead would put its  $f - 1$  interior feet interior to a boundary edge, which no tiling vertex may be. In this configuration the south cover therefore has *nowhere to land at all* (`word42_no_landing`, axiom-clean).

So  $(4, 2)$  dies the moment the descent is known to reach the base, and needs nothing else: no reach step, no fan analysis, no case split on the row. With Remark 82 supplying the mirror and Remark 83 the ladder, discharging the single hypothesis “the feet reach the base” advances the bound by one level. Three such advances,  $R = 4$  to  $R = 7$ , reach  $N = 191$ .

The three open steps of the prime hole's smallest member are thus one step applied three times, and that step is now carried in its weakest form — the descent must land, and nothing is asked about what it lands on.

**5.6. The  $(4, 2)$  word dies by an  $\alpha$ -fan: a quantitative law.** The refutation of the  $(4, 2)$  base word was mined from full traces of the seeded engine (placements and prune sites, node-identical to the untraced runs) at  $f = 4, \dots, 7$ , and its structure is uniform in  $f$ . At the base junction between the word's first  $a$  and its  $c$ , the west corner tile presents  $\gamma$  and the  $c$ -tile presents  $\alpha$ , leaving a wedge of exactly  $\pi - \gamma - \alpha = \beta$ . The growing part of the tree is a fan of  $\alpha$ -tiles pinned at that junction, each pivoting on the previous tile's  $b$ -edge; the tree's binary choices are chiralities that do not alter the angle, which is why every branch terminates identically. After  $k = \lfloor \beta/\alpha \rfloor$  fan steps the remainder  $\beta - k\alpha$  is a corner smaller than every tile angle, and the branch dies.

**Proposition 85** (The fan law). [HEURISTIC] *For the  $(4, 2)$ -seeded instance at  $(1, f)$ , the engine's refutation consists of two such fans (behind the two chiralities of the  $c$ -tile, the second four levels deeper), a set of local side cases whose number is independent of  $f$ , and one secondary chain that advances one block per unit of  $f$  along the trailing  $a$ -run of the base word. The fan depth is exactly  $\lfloor \beta/\alpha \rfloor$ , with  $\beta/\alpha = \pi/(2\alpha) - 3/2$ .*

The law is verified at seven consecutive members:

| $f$                            | 4   | 5   | 6    | 7    | 8     | 9     | 10     |
|--------------------------------|-----|-----|------|------|-------|-------|--------|
| $\lfloor \beta/\alpha \rfloor$ | 4   | 6   | 7    | 9    | 11    | 12    | 14     |
| measured fan depth             | 4   | 6   | 7    | 9    | 11    | 12    | 14     |
| nodes                          | 139 | 517 | 1025 | 4019 | 15953 | 31873 | 127183 |

the last two columns as pre-registered predictions scored out of sample (research protocol: the  $f = 10$  row was frozen, with the exact leaf count  $2^{15}$  and depth 20, before that run returned; all five registered quantities matched). The side-case counts are constant — the traces at  $f = 4$  and  $f = 5$  agree leaf for leaf on all 21 of them — and the secondary chain is the march along the base word, its side branches dying by the overrun identity  $c - b = 1$ , a leftover of length 1 lying below every generator of  $\langle f, f^2 - 1, f^2 \rangle$ . Instrumenting the semigroup prune to print its failing run settles the side-case arithmetic completely: at  $f = 4, \dots, 7$  all seven  $f$ -independent semigroup kills fire on the run  $f^2 - f - 1 = b - a$  and every chain kill on the run  $1 = c - b$ , and both lengths were already formalized as gaps (`east_cover_gap`; `FanKill.one_is_gap`, packaged as `two_gap_contract`). The  $P2$  layer of the refutation therefore consumes exactly two verified identities.

The step itself is also closed arithmetically (`lean/Erdos634/FanStep.lean`, axiom-clean): in the remaining wedge after  $k \geq 1$  steps, a tile at the pin presents  $\alpha$ ,  $\beta$  or  $\gamma$ , or passes an edge through it;  $\gamma$  and the through-edge never fit,  $\beta$  fits only at  $k = 0$  where it dies on the two gaps, and the  $\alpha$ -step advances by exactly  $\alpha$  with both chirality stubs equal to the same gap  $c - b = 1$  (`step_exhaustion`, `stub_is_gap`). The termination arithmetic is formalized in `lean/Erdos634/FanKill.lean` (axiom-clean):  $k\alpha = \beta$  is impossible for every  $k$ , since it would force  $\alpha = \pi/(3 + 2k)$  against the irrationality of  $\alpha/\pi$ , which holds at every member; the remainder at  $k = \lfloor \beta/\alpha \rfloor$  is trapped strictly in  $(0, \alpha)$ ; and nothing whose angle is at least the minimal tile angle fits in it. What remains outside arithmetic has shrunk to two instantiations of theorems the corpus already holds in full strength: `Dissection.wall_partition` (a straight interior wall segment is measure-partitioned by the tile edges along it) at the pin's flanking rays, and `Dissection.vertex_multiplicities` (the angle arithmetic at a vertex) at the pin itself — together with the completeness of the candidate enumeration, which is the engine's certified soundness. The assembled conditional statement is `FanStep.word42_junction_dies`:

the only exact fill of the  $\beta$ -wedge is a single  $\beta$ -tile (`beta_fill.unique`), which dies on the two-gap contract, and every non-exact fill stalls at the sub- $\alpha$  fan remainder, which no tile angle fits.

In fact the fan induction is not needed for the mathematics at all. Since the angle sums are a theorem of the dissection layer (`sum_localAngle.eq`, the discharged  $G2$ ), a *completed* tiling has tile angles summing to exactly  $\pi$  at every base junction; between a  $\gamma$ -flank and an  $\alpha$ -flank the interior multiset must sum to exactly  $\beta$ , and the multiplicity system forces it to be a single  $\beta$ -tile (`PinLemma.pin_forces_single_beta`, via `vertex_multiplicities`); a tile holding the junction inside one of its edges is excluded outright, contributing  $\pi$  against two positive flanks (`no_through_tile`); and the forced  $\beta$ -tile dies on the two gaps (`beta_tile_dies`). The engine's fans are what this impossibility looks like to a solver: partial states advancing by  $\alpha$  that can never close, their depth  $\lfloor \beta/\alpha \rfloor$  being the law measured above. The argument applies to the direct orientation of the  $c$ -tile in *every* word with  $c$  in the second position,  $(3, 2)$  included; what remains per word is the mirrored orientation, where the forced fill is a single  $\alpha$ -tile (`pin_forces_single_alpha`) and the configuration advances to the next junction — the head of the engine's second cascade. The remaining instantiations are the flank identification (the corner theory of Theorem 20) and the covering data (`wall_partition`) at the junction. An adversarial control is recorded: the  $(3, 2)$  word shares the junction but not the fate — its tree grows sevenfold per unit of  $f$  — because the fan governs only the wedge-west branches, and the rest of a word's tree depends on the word; the window's  $L2$  kill, not the fan, is what covers  $(3, 2)$ -type words. Under those, the  $(4, 2)$  word dies at every  $f$  simultaneously; since  $(4, 2)$  and its mirror are the only escapes of the pincer window at each level (Remark 83, Remark 82), this would advance the reach ladder uniformly. The signature is not special to  $(4, 2)$ . The  $(5, 2)$ -seeded instances show the same structure at  $f = 5, \dots, 8$ : a constant side-case count (18), three fans in place of two, and binary fan depth again exactly  $\lfloor \beta/\alpha \rfloor$  (6, 7, 9, 11), with leaf counts  $3 \cdot 2^{\lfloor \beta/\alpha \rfloor}$ . The  $(5, 3)$  instances replicate it a third time, with identical fan counts  $3 \cdot 2^{\lfloor \beta/\alpha \rfloor}$  and their own constants (9 area, 30 corner-type). The fan constant is thus shared across all three orbits probed; what varies per orbit is the finite head. We state plainly that the forcing is an instance of the crossing question and is open; the contribution of the law is that the question now has a measured, quantitative instance with its arithmetic core proved, replicated on a second orbit.

**Remark 86** (Where the landing hypothesis can fail, and why it is not local). It is worth saying exactly what the remaining hypothesis costs. On the base the  $f+1$  feet are forced to be junctions, since no tiling vertex lies interior to a boundary edge, and the span dichotomy then leaves only  $f$  consecutive *as*. At an *interior* floor that premise is unavailable: a vertex of an upper tile may sit interior to a lower tile's edge. Dropping it, the length arithmetic still decides the cover of a length- $f^2$  stretch, but with two cases (`A2BranchRow3.cover_dichotomy`):  $f$   $a$ -edges, or a single  $c$ -edge. In the first the run is reproduced one level down and the descent continues; in the second the  $f - 1$  interior feet become  $T$ -vertices inside that  $c$  and the descent is blocked.

So the feet reach the base *unless* at some interior level a single  $c$ -edge lies directly under the run. That is the whole residue, and it is one named configuration.

It is not locally refutable, and we record why so the ground is not walked again. At each interior foot the wedge from below is  $\pi$ , so the tiles above sum to  $\pi$  and the figure is  $\{\alpha, \beta, \gamma\}$  or  $\{\alpha, \alpha, \alpha, \beta, \beta\}$  — at most one  $\gamma$ . Each upper tile lays an  $a$ , whose flanks are  $\beta$  and  $\gamma$ , contributing exactly one of each; by the  $a$ -run rigidity of `OrderForcing` the orientation word is monotone,  $M^i D^{f-i}$ , and the  $\gamma$ s then fall on the feet  $\{0, \dots, i-1\} \cup \{i+1, \dots, f\}$  —  $f$  of them on  $f$  distinct feet, one apiece, with foot  $i$  free. Every constraint is met exactly. The endpoints give nothing

either: they are interior to the floor, so the tiles above them close to  $\pi$  by themselves and never meet the lower tile's  $\alpha$  and  $\beta$  in one equation.

Any proof must therefore be global, which is the boundary-versus-interior barrier already recorded for the strip iteration. What has changed is that the barrier is now a single configuration rather than a general claim about descents, and that its base case is settled outright.

The configuration is moreover not refutable by any argument blind to  $m$  and  $f$ , and we record the ground covered. At  $e = 1$  a run of  $f$   $a$ -edges has length  $f \cdot ef = f^2 = c$ , so it matches one  $c$ -edge exactly; for  $e \geq 2$  the match needs  $e \mid f$  and cannot occur. The interface census (`code/interface.census.py`) reports the two-sided edge words of every maximal straight interface in all eight certificates, and the equation  $c = aa$  — which at  $f = 2$  is the relation  $c = f \cdot a$  — occurs in every one of them: 3 times in N44C, 2 in N44B, 2 in Tiling44, 10 in Tiling99, 1 in PgramTiling22, 2 in CevianTiling28, 7 in CevianTiling63, 2 in Tiling28 and 5 in Tiling77. So the configuration is not merely locally consistent, it is abundant in the tilings we possess, all of which have  $f = 2$  and  $m > 1$ . Any proof that it cannot occur must consume  $m = 1$  and  $f \geq 3$  essentially, which is the same asymmetry Remark 33 identifies as the only admissible shape of an argument here.

That survival is predicted, not accidental. `ScaleRigidity.no_similarity_invariant_proof` states that a criterion depending only on similarity-invariant data of the tile and the target cannot exclude the  $m = 1$  case, since it would equally exclude  $m = 2$ , where Tiling44 exists. The  $c$ -under-run configuration asserts a ratio between edge lengths and is therefore scale-free, so the barrier applies to it directly. What the same theorem names as surviving is the scale-sensitive walk input: at  $m = 1$  every admissible equal-side walk is free of  $b$ -edges, whereas at  $m = 2$  the walk  $(P, Q, R) = (3, 2, 1)$  carries two. Any proof that the descent lands must consume that, or another condition separating the scales; a purely local or purely metric argument provably cannot.

Following that instruction reduces the residue further, from a configuration to two points. The mechanism of `side_no_b_m1` is a  $\gamma$ -trap: on a side of length  $f^3$  the walk gives  $f \mid Q$  modulo  $f$ , then  $Q \geq f$  forces  $R = 0$  on size, and  $R \geq 1$  kills that; the  $R \geq 1$  holds because every  $a$ - and every  $b$ -edge puts a  $\gamma$  at a junction of the side, no  $\gamma$  sits at a base corner or the apex, and no other node carries two, so  $\#a + \#b \leq \#\text{edges} - 1$ .

The same count runs on the interior stretch. Each of the  $f$  tiles above lays an  $a$ -edge, whose flanks are  $\beta$  and  $\gamma$ , so it contributes exactly one  $\gamma$  among the  $f + 1$  junctions, and a straight junction carries at most one. Hence *the configuration dies if both endpoints of the stretch are  $\gamma$ -free*,  $f$   $\gamma$ s being unable to fit in the  $f - 1$  interior junctions (`A2BranchRow3.c_under_run_dies_if_ends_gamma_free`, axiom-clean).

That criterion is true but *vacuous*, and we state so rather than leave it looking like a route. By the  $a$ -run rigidity the orientation word is monotone,  $M^i D^{f-i}$ , beta-far tiles first and gamma-far after; so tile 1 presents  $\gamma$  at its left end, the stretch's left endpoint, unless  $i = 0$ , and tile  $f$  presents  $\gamma$  at its right end unless  $i = f$ . Both endpoints  $\gamma$ -free would need  $i = 0$  and  $i = f$  together, impossible for  $f \geq 1$  (`ends_gamma_free_impossible`). Since `gamma_count_tight` shows the count is exactly tight —  $f$   $\gamma$ s do fit in  $f + 1$  junctions at one apiece — the endpoint condition was the only way to break it, and it cannot hold. The  $\gamma$ -count is a dead end.

What survives is the diagnosis. On a side the two ends are a base corner and the apex, both  $\gamma$ -free for boundary reasons; on an interior stretch an endpoint's figure from above may be  $\{\alpha, \beta, \gamma\}$ , which closes at  $\pi$  since  $\gamma + \alpha + \beta = 3\alpha + 2\beta$ . The  $\gamma$ -trap behind `side_no_b_m1` is a boundary phenomenon and does not survive the move inward. Any further attempt must couple the run to something other than its own junction count.

**Lemma 87** (Corner lines are column lines). [PROVED] *Let  $f \geq 3$ ,  $m = 1$ . Every  $b$ -direction line through two vertices of the corner lattice  $\mathbb{Z}(a, 0) + \mathbb{Z}cu$  meets the base at  $(kf, 0)$  and the left side at  $kcu$  for an integer  $k$ , and its horizontal distance to the side at row  $r$  is exactly  $(k - r)a$ . Its length is  $kb$  and both ends lie on the boundary, so for  $k < f$  Lemma 64 splits both of its sides into exactly  $k$  whole  $b$ -edges, breaking at every strip junction on both sides at once: the line is unstaggered. Since  $kc \leq f^3$  forces  $k \leq f$ , the only  $b$ -direction line of the left corner that escapes is  $k = f$  — the line from  $(f^2, 0)$  to  $fcu$ , which is the apex. A ladder is staggered by  $c - b = 1$ , so no straddler sits on a corner line with  $k < f$ .*

**Lemma 88** (The chain never needs the apex line). [PROVED] *The base  $b$ -letter sits at a position  $bp \in [3, f]$ . With the preceding letters forced to  $a$  it begins at  $((bp - 1)f, 0)$ , and the collision figure there consumes only the brick and mate of column  $bp - 2$ , hence only the lines  $L_k$  with  $k \leq bp - 1 \leq f - 1$ . Every line the chain requires therefore has  $k < f$ , where Lemma 87 applies (*chain\_needs\_small\_lines*); the apex line  $L_f$ , the one line escaping Lemma 64, is never required.*

**Remark 89** (The residue, as one statement, and why it must be  $m = 1$ ). Combining Lemmas 87 and 88: no straddler can sit on any line the chain needs, *provided no tile crosses that line*. Since  $L_k$  runs boundary to boundary and every needed  $k$  satisfies  $k < f$ , this crossing statement is the only remaining hypothesis at  $e = 1$ ; with it, every fork closes, the chain reaches the  $b$ , and Hypothesis 29 follows at  $e = 1$  for every  $f$ .

The statement is false without the hypothesis  $m = 1$ . Tested against the five kernel-checked tilings, all of which have  $m \geq 2$ , tiles cross boundary-to-boundary  $b$ -lines freely: 42, 15, 24, 10 and 98 crossing incidences respectively. This is consistent with Remark 33 — at  $m = 2$  the corner block can break — and it settles the shape of any proof: the crossing statement admits no  $m$ -independent argument, and must consume the  $m = 1$  hypothesis exactly where the corner block is rigid. That input is not supplied here.

**Lemma 90** (The vertex census). [VERIFIED] *In every tiling of the base- $\beta$  target, write  $n_1, n_2$  for the numbers of straight figures  $\{\gamma, \alpha, \beta\}$  and  $\{3\alpha, 2\beta\}$ , and  $v_1, \dots, v_4$  for the interior figures  $\{\beta, 3\gamma\}$ ,  $\{2\alpha, 2\beta, 2\gamma\}$ ,  $\{4\alpha, 3\beta, \gamma\}$ ,  $\{6\alpha, 4\beta\}$ . The base corners fill uniquely as  $\{\beta\}$  and the apex as  $\{3\alpha\}$ , and balancing the three corner types across the  $N$  tiles gives*

$$v_1 = 1 + n_2 + v_3 + 2v_4 \quad (\text{vertex\_census}).$$

*In particular  $v_1 \geq 1$ : a  $\{\beta, 3\gamma\}$  vertex is mandatory, and each  $\gamma$ -poor figure demands one more. The identity is verified exactly on all five kernel-checked tilings. It is also the entire content of corner counting: the three balance equations admit exactly one relation, and it does not involve the side structure — corner counting alone cannot see  $p$ , consistent with the census-based routes logged as dead.*

**Definition 91** (The frontier invariant). A *frontier* is a finite family of disjoint uncovered polygons inside the target together with, for each target side, the multiset of tile-edge types already laid on it. Say a frontier is *admissible* if all five of the following hold.

- (P1) Each polygon's area is a positive integer multiple of the tile's area, and the multiples sum to the number of tiles remaining.
- (P2) Every maximal straight run of a polygon's boundary joining two convex corners has integral length lying in  $\langle a, b, c \rangle$ .
- (P4) Every convex corner of every polygon has angle at least  $\alpha$ .
- (P5) The per-side edge-type multiset is dominated componentwise by an admissible walk profile.

(P6) An edge lying in a target side and meeting a corner of that side has the type forced at that corner.

Admissibility is decidable, and it is preserved by no tiling step: a tiling of the target is exactly a sequence of placements through admissible frontiers ending at the empty frontier.

**Lemma 92** (The angle threshold). [PROVED] *For the tile  $(a, b, c) = (f, f^2 - 1, f^2)$  one has  $\cos \alpha = (2f^2 - 1)/(2f^2)$  (`cos_alpha_closed`), alongside  $\cos \gamma = -1/(2f)$  and  $\cos \beta = (3f^2 - 1)/(2f^3)$ . Condition (P4) is therefore the statement that the uncovered region never acquires a corner sharper than the tile's own sharpest angle.*

**Remark 93** (The exhaustion invariant, and the shape of a uniform proof). Definition 91 is the invariant maintained by the exhaustive search whose verdicts certify the small members. Its content relative to the hand arguments of this paper is that (P2) generalises the semigroup gap lemmas used throughout, while (P4) is a constraint on the uncovered region rather than on completed vertices, and is not used anywhere in the arguments above. In the certified cases the searches reach a contradiction long before the tiling is complete. For  $f = 2, 3, 5, 6, 7$  the maximum refutation depth is 6, 13, 24, 29, 35 against  $N = 3f^2 - 1$  placements, so the fraction of the tiling ever built is 0.55, 0.50, 0.32, 0.27, 0.24 and falling, while the share of (P4) violations rises from 50% to 98%. The vanishing fraction is the robust part of this observation. The linear form is not: the ratio depth/ $f$  reads 3.00, 4.33, 4.80, 4.83, 5.00 and is still increasing at  $f = 7$ , so five instances do not separate a bound  $Cf$  from a slowly superlinear law. A proof of Hypothesis 29 for all  $f$  would follow from a uniform bound of the form: from a corner-anchored start with  $p \geq 1$ , no admissible frontier survives past depth  $Cf$ . That statement is open and the figures quoted are measurements, not evidence for the constant.

**Lemma 94** (The thick-member monochotomy). [PROVED] *Let  $e \geq 2$ ,  $e < f$ ,  $\gcd(e, f) = 1$ , tile  $(a, b, c) = (ef, f^2 - e^2, f^2)$ . The only decomposition of  $c$  as  $xa + yb + zc$  is the single  $c$ : writing  $xef + y(f^2 - e^2) = f^2$ , coprimality forces  $e \mid y - 1$ , and both  $y = 1$  (then  $xf = e < f$ ) and  $y \geq e + 1$  (then the sum exceeds  $f^2$  since  $f^2 > e^2 + e$ ) are impossible (`c_chord_unique_thick`). Consequently the  $f$ -a branch of Lemma 51 is an  $e = 1$  phenomenon: at thick members every  $c$ -chord fork of the corner chain is forced, and the chain machinery, where it applies, is stiffer than at  $e = 1$ .*

**Lemma 95** (Thick-side quantization). [CONJECTURE] *At any member  $(e, f)$ ,  $e \geq 1$ , a side's edge counts satisfy  $xe + zf = f^2$  with no  $b$  (`side_no_b_uncond`), so  $f \mid x$  by coprimality (`side_quantized`): the number of  $a$ -edges on a side is a multiple of  $f$ . For  $e \geq 2$  this quantizes the walls violation: any bound  $p < f$  forces  $p = 0$ , the side condition of Hypothesis 29.*

**Lemma 96** (The thick base trichotomy). [PROVED] *For a separated member  $(f^2 > 2ef + e^2)$  the base admits exactly three edge decompositions:  $(x, y, z) \in \{(0, e, 2e), (f, e, e), (2f, e, 0)\}$  (`base_trichotomy`); the walls form is the middle one,  $f$   $a$ -feet,  $e$   $b$ -edges,  $e$   $c$ -feet. Close pairs  $(f^2 \leq 2ef + e^2)$  admit in addition finitely many decompositions with  $y > e$ , computable per member.*

**Lemma 97** (The shadow at a  $c$ -corner). [PROVED] *Suppose a base corner tile lays  $c$  on the base (then it is mirrored with its  $a$ -edge first on the side), and the first side run has length at least 2. The run is  $\gamma$ -upper by absorption, so its second tile lays a horizontal  $c$ -edge at height  $a \sin \beta$  — exactly one tile-height (`strips_tall` family identities). Every tile in the strip below spans it and carries its  $c$  on the floor or the ceiling; a tile with  $b$  on the floor and  $c$  on the ceiling would have two parallel sides. Hence every base edge meeting the shadow  $(0, a \cos \beta + c)$  is a  $c$ -edge. Consequences: at such a corner either the first side run has length 1, or the second base edge is a  $c$  and no  $b$ -edge meets the shadow. At  $e = 1$  the two-sided*

shadow window is always smaller than the  $b$ -edge (*shadow\_footage.e1*: the comparison reduces to  $3f^2 > 1$ ), an independent exclusion of  $c$ -corners; at  $e \geq 2$  the comparison reverses, and the shadow is a constraint rather than a kill.

**Lemma 98** (The thick base dichotomy). [PROVED] *The  $\gamma$ -trap (Proposition 25, every side of a base- $\beta$  target carries a  $c$ -edge, general in  $(e, f)$ ) forces  $z \geq 1$ , deleting the third column of Lemma 96. At a separated thick member only two base decompositions survive:  $(0, e, 2e)$  and the walls form  $(f, e, e)$  (*base\_dichotomy\_thick*).*

**Lemma 99** (The  $c$ -corner is rigid). [CONJECTURE] *Suppose a base corner tile lays  $c$  on the base. Since the corner angle is  $\beta$ , with flanks  $\{a, c\}$ , that tile lays  $a$  on the side. Its  $b$ -edge is a chord joining the base point  $(c, 0)$  to the side point  $au$ , so both of its ends lie on the boundary and its far side carries a single whole  $b$  (*partner\_unique, lean/PentagonLemma.lean*). The corner tile presents  $\alpha$  at the base end of that chord and  $\gamma$  at the side end; the reflected partner would repeat  $\gamma$  at the side end, where the junction is straight and  $2\gamma > \pi$  because  $\cos \gamma = -e/2f < 0$ . Hence the direct partner is forced, contributing  $\gamma$  at the base end and  $\alpha$  at the side end, and each end closes with exactly one  $\beta$  whose flanks are  $\{a, c\}$ : the second base edge and the second side edge are each an  $a$  or a  $c$ .*

*In the base column  $(0, e, 2e)$  of Lemma 98 the base carries no  $a$ -edge, so both corner tiles lay  $c$  on the base and the hypothesis holds at both corners: both sides begin with an  $a$ -edge, and by Lemma 95 each side then carries at least  $f$  of them. The  $\gamma$ -trap bounds the side profiles to  $a$ -count  $fj$  with  $je \leq f - 1$ .*

*This is the unconditional entry point that Lemma 41 lacks: that lemma disposes of the walk  $(0, e, 2e)$  only after both dual blocks are complete, and its stubs are region-boundary segments only under that hypothesis, whereas the forcing here uses nothing but the boundary position of the corner chord and  $2\gamma > \pi$ . What remains for the column is to run the chain from the rigid  $c$ -corner far enough to create the stubs of Lemma 41 unconditionally; the stub arithmetic itself is already general and machine-checked (Lemma 42).*

**Theorem 100** (The column  $(0, e, 2e)$  admits no tiling). [CONJECTURE: PROVED AT  $e = 1$ ; OPEN AT  $e \geq 2$ , THE INDUCTION BEING GAPPED (REMARK 105)] *Let  $(e, f)$  be a separated member with  $e \geq 1$ ,  $\gcd(e, f) = 1$ ,  $f \geq 2$ , and suppose the base walk is  $(0, e, 2e)$ : no  $a$ -edge on the base. Then no tiling exists.*

*Proof.* Write  $\gamma = 2\alpha + \beta$ ; a wedge  $X\alpha + Y\beta$  filled by corner counts  $(x, y, z)$  satisfies  $x + 2z = X$ ,  $y + z = Y$ , so the fills of  $\beta$ , of  $\alpha$  and of  $\alpha + \beta$  are  $\{\beta\}$ ,  $\{\alpha\}$  and  $\{\alpha, \beta\}$  respectively (*fill\_beta, fill\_alpha, fill\_alpha\_beta*).

The corner angle is  $\beta$ , so a single tile  $T_0$  occupies the corner; its flanks are  $\{a, c\}$  and the base carries no  $a$ -edge, so  $T_0$  lays  $c$  on the base and  $a$  on the side, and its  $b$ -edge joins  $(c, 0)$  to  $au$ . At the side vertex  $au$  the junction is straight and  $T_0$  presents  $\gamma$ , leaving  $\alpha + \beta$ , so exactly two further tiles meet there, one with  $\alpha$  and one with  $\beta$ . Their order is forced. The edge  $T_0$  presents along the chord is an edge of a tile, so no tile crosses it, and both of its ends lie on the boundary, so no edge overruns them; hence the far side of that chord is a union of whole edges of total length  $b$ , which by *partner\_unique* is a single  $b$ . The tile carrying it meets  $au$  at an endpoint of a  $b$ -edge, so presents  $\alpha$  or  $\gamma$  there, and  $\gamma$  is already taken. It is therefore the  $\alpha$ -tile, and it is the partner  $P_0$ , forced with no hypothesis on corner blocks.

Inductively, suppose  $T_k$  (a translate of  $T_0$  by  $(kc, 0)$ ) and  $P_{k-1}$  are in place. At the base vertex  $((k+1)c, 0)$ ,  $T_k$  presents  $\alpha$  and  $P_k$  will present  $\gamma$ ; at the vertex  $V_k = (kc, 0) + au$ ,  $T_k$  presents  $\gamma$  and  $P_{k-1}$  presents  $\beta$ , leaving  $\alpha$ , whose unique filler has flanks  $\{b, c\}$ ; a  $c$  laid along  $T_k$ 's  $b$ -ray would run from  $V_k$  past  $((k+1)c, 0)$ , a point of the base, and so leave the target, so the filler lays a whole  $b$  there and is the partner  $P_k$ . At the base vertex  $(kc, 0)$ ,  $T_{k-1}$ 's  $\alpha$  and  $P_{k-1}$ 's  $\gamma$

leave exactly  $\beta$ , whose filler has flanks  $\{a, c\}$ : the next base edge is an  $a$  or a  $c$ , hence a  $c$ , and its tile is  $T_k$ . The induction therefore forces every base edge to be a  $c$ -edge.

The base has length  $e(3f^2 - e^2)$ , so this requires  $f^2 \mid e(3f^2 - e^2)$ , hence  $f^2 \mid e^3$ , hence  $f = 1$  by coprimality (`base_not_all_c`), contradicting  $f \geq 2$ .  $\square$

**Corollary 101.** [CONJECTURE: RESTS ON THEOREM 100 AT  $e \geq 2$ , WHICH IS OPEN] *At a separated thick member the base walk is the walls form  $(f, e, e)$ : Lemma 96 gives three columns, the  $\gamma$ -trap deletes  $(2f, e, 0)$  (Lemma 98), and Theorem 100 deletes  $(0, e, 2e)$ .*

Separation enters that chain only through Lemma 96, to rule out columns with  $y > e$ . That can be done under the weaker hypothesis  $f > 2e$ , so the trichotomy and the dichotomy both extend. Proposition 179 is not available for this purpose, because it assumes  $x, z \geq 1$  while Lemma 96 allows  $x = 0$  and  $z = 0$ ; the argument below is run with  $x, z \geq 0$  throughout.

**Corollary 102** (The base trichotomy and dichotomy without separation). [PROVED] *Let  $\gcd(e, f) = 1$  and  $f > 2e$ . Then the base admits exactly the three decompositions  $(0, e, 2e)$ ,  $(f, e, e)$ ,  $(2f, e, 0)$  of Lemma 96, and after the  $\gamma$ -trap exactly the two of Lemma 98, namely  $(0, e, 2e)$  and the walls form  $(f, e, e)$ .*

*Proof.* Reducing  $xa + yb + zc = e(3f^2 - e^2)$  modulo  $f$  kills  $a = ef$  and  $c = f^2$  and leaves  $-ye^2 \equiv -e^3$ , so  $f \mid e^2(y - e)$  and hence  $f \mid y - e$  by coprimality. Write  $y = e + kf$ ; since  $0 \leq y$  and  $e < f$ ,  $k \geq 0$ .

Suppose  $k \geq 1$ . The base equation is then  $xe + zf = 2ef - kb$  (Proposition 178(i), whose derivation uses no positivity), and  $f \mid x - ke$ , so  $x = ke - mf$  for an integer  $m$ ; from  $x \geq 0$ ,  $mf \leq ke$ . The identity

$$(ke - mf)e + k(f^2 - e^2) - 2ef = f(kf - (m + 2)e)$$

together with  $z \geq 0$ , i.e.  $xe + kb - 2ef \leq 0$ , gives  $kf \leq (m + 2)e$  after dividing by  $f > 0$ . Now  $2m < k$ : for  $m \leq 0$  this is clear from  $k \geq 1$ , and for  $m \geq 1$  it follows from  $2me < mf \leq ke$ . Finally  $f > 2e$  and  $k \geq 1$  give  $2ke < kf \leq (m + 2)e$ , hence  $2k < m + 2$ , and with  $2m \leq k - 1$  this yields  $4k < k + 3$ , i.e.  $k < 1$ , a contradiction.

So  $y = e$ . Then  $f \mid x$ , say  $x = jf$ , and the base equation becomes  $je + z = 2e$ , i.e.  $z = e(2 - j)$ , so  $z \geq 0$  gives  $j \in \{0, 1, 2\}$  and the three columns are exactly those listed. The  $\gamma$ -trap forces  $z \geq 1$ , deleting  $j = 2$ . We cite the  $\gamma$ -trap (Proposition 25) rather than Lemma 98: the latter's formal counterpart `base_dichotomy_thick` carries a separation hypothesis, whereas the  $\gamma$ -trap is proved for an arbitrary side of a base- $\beta$  target and its corner input (Proposition 24) puts no  $\gamma$  at a base corner or at the apex, so it applies here unchanged. (`no_extra_column_of_f_gt_two_e`).  $\square$

**Remark 103** ( $f > 2e$  is sharp, and  $(1, 2)$  is the unique exception). The inequality cannot be weakened to  $f \geq 2e$ . Coprimality makes  $f = 2e$  possible only at  $e = 1$ ,  $f = 2$ , and that member does carry a fourth decomposition:  $1 \cdot a + 3 \cdot b + 0 \cdot c = 2 + 9 = 11 = e(3f^2 - e^2)$ , i.e.  $(x, y, z) = (1, 3, 0)$  with  $y = 3 > e$ . It escapes the argument above exactly at  $z = 0$ . Checked by structured enumeration of the base equation on all 165038 members with  $f > 2e$ ,  $e \leq 300$ : no decomposition outside the three, and the same enumerator does find  $(1, 3, 0)$  at  $(1, 2)$ .

**Remark 104** (The band this opens, and what still blocks the walls form there). Separation is  $f > (1 + \sqrt{2})e$ , so Corollary 102 is strictly weaker in hypothesis and covers the band  $2e < f \leq (1 + \sqrt{2})e$  that Lemma 96 left to a per-member computation: 5064 of the 17296 close pairs with  $e \leq 200$  lie in it. For those members the base dichotomy now holds for the same reason it holds at a separated member.

To reach the walls form one further needs Theorem 100. Its separation hypothesis is indeed inert — no step of its proof invokes  $f^2 > 2ef + e^2$ , and its arithmetic inputs (`partner_unique`,

$c > b$ , `base_not_all_c`) were checked on all 36896 coprime members with  $e \leq 200$ ,  $e < f < 4e+2$ , of which 17296 are close pairs, with no failure. But deleting the hypothesis does not deliver the conclusion, because the proof of Theorem 100 has a gap at  $e \geq 2$ , recorded next.

**Remark 105** (A gap in the induction of Theorem 100). Recorded in [1]. Full discussion in [1].

**Remark 106** (The gap is exactly one crossing, and there is no cheaper repair). Recorded in [1]. Full discussion in [1].

**Remark 107** (The overrun configuration at  $V_k$ , determined). The crossing that has to be excluded is more rigid than it looks. By Remark 59's accounting, a tile whose edge passes through  $V_k$  contributes exactly  $\pi$  there, so an overrun forces the tiles with a *vertex* at  $V_k$  to form a  $\pi$ -figure. The forced ones are  $T_k$ , presenting  $\gamma$ , and  $P_{k-1}$ , presenting  $\beta$ , and the only  $\pi$ -figure containing  $\{\gamma, \beta\}$  is  $\{\gamma, \alpha, \beta\}$ . So an overrun admits *exactly one* further tile at  $V_k$ , and it presents  $\alpha$ .

That tile has room. The chord leaves the base at angle  $\alpha$ , so its extension at  $V_k$  lies at  $\pi - \alpha$ , while  $P_{k-1}$ 's  $c$ -edge lies along  $\pi$ ; the wedge between them is exactly  $\alpha$ . The configuration is therefore angularly consistent and cannot be refuted by counting angles, which is why it has survived.

Its two edges at  $V_k$  are  $b$  and  $c$ , one along the chord's extension and one along  $P_{k-1}$ 's  $c$ -edge. Along the latter, of length  $c$ :

- laying  $c$  matches exactly;
- laying  $b$  falls short of  $V_{k-1}$  by  $c - b = e^2$ , and  $e^2$  lies in  $\langle a, b, c \rangle$  for no member (`gap_e_squared`), so that residue admits no whole-edge closure either.

The overrun therefore forces either a second crossing further along the same horizontal, or the  $\alpha$ -tile to share  $P_{k-1}$ 's  $c$ -edge exactly and lay its  $b$  along the chord's extension. We record this because it reduces the open question to a single configuration, not because it settles it: the second branch remains open, and a descent on the first is blocked by the fact that the  $e^2$ -residue may be closed by an edge overrunning its *other*, interior end rather than  $V_{k-1}$ .

Where that surviving branch lands is worth recording, because it is not an accident. The chord's direction vector is  $au - (c, 0)$ , which is  $T_0$ 's own  $b$ -edge, so

$$W = V_k + (au - (c, 0)) = ((k-1)c, 0) + 2au.$$

At  $k = 1$  this is  $W = 2au$ , a point of the equal side, and  $R$  then lays an  $a$ -edge on that side from  $au$  to  $2au$ . Far from being absurd, that is exactly the configuration Lemma 99 predicts for this column. So the overrun is geometrically coherent, not merely angularly so, and it will not be refuted locally.

**Remark 108** (Theorem 100 at  $e \geq 2$ , repaired conditionally). The previous remark also supplies the honest repair. Lemma 99 already observes that in the column  $(0, e, 2e)$  the base carries no  $a$ -edge, so both corner tiles lay  $c$  on the base, both equal sides begin with an  $a$ -edge, and by Lemma 95 each carries at least  $f$  of them. In the notation of §5.7 that is  $p \geq 1$  on both sides. Hence

**Proposition 109.** [CONJECTURE] *Assume Hypothesis 29 at the member  $(e, f)$ , i.e.  $p = 0$ . Then no tiling has base walk  $(0, e, 2e)$ , for every  $e \geq 1$ .*

*Proof.* By Lemma 99 the column  $(0, e, 2e)$  forces  $p \geq 1$  on each equal side, contradicting  $p = 0$ .  $\square$

This is two lines and uses no induction, no chord pinning, and no crossing question. It is weaker than Theorem 100 as printed, being conditional; but Hypothesis 29 is assumed throughout this

programme anyway, so for every downstream use the conditional form is as good. Corollary 101 and Corollary 116 may accordingly be read as holding under Hypothesis 29 at every member, and unconditionally at  $e = 1$ , where Remark 105's three-letter argument applies.

The moral is worth stating. Theorem 100's induction was attempting to prove, unconditionally and by a long chord cascade, a statement that follows from Hypothesis 29 immediately. Its gap is therefore not an accident of that particular proof: excluding the overrun at  $V_1$  means exactly bounding the side's run of  $a$ -edges, which is Hypothesis 29 again. The two are one problem, and the cascade was the long way round.

**Remark 110** (Counting cannot decide the crossing question). Recorded in [1]. Full discussion in [1].

**Remark 111** (Anchoring one end is not enough, and the data says which edges get crossed). It is tempting to hope for a lemma of the shape “a tile edge with one end on  $\partial ABC$  is not crossed”. No such lemma exists. Across the four kernel-checked tilings, the split edges with exactly one endpoint on the boundary number 19, 8, 6 and 5 respectively, so crossings of that shape are routine.

Sorting them by edge type is more informative. Writing crossed/total for edges with exactly one end on the boundary:

|                  | $a$  | $b$  | $c$   |
|------------------|------|------|-------|
| 99-tiling        | 0/27 | 4/12 | 15/22 |
| 77-tiling        | 0/9  | 1/5  | 7/13  |
| 44-tiling        | 0/10 | 1/13 | 5/9   |
| parallelogram 52 | 0/8  | 3/14 | 2/13  |

Two things follow. First,  $b$ -edges with one anchored end *are* crossed, in every one of the four. The chord at  $V_k$  in Theorem 100 is exactly such an edge, so the crossing there cannot be excluded by any statement about anchoring alone: the exclusion has to consume the cascade's forced tiles, and that is precisely what has resisted.

Second,  $a$ -edges with one anchored end are crossed in none of them. That one is not a coincidence, and the next lemma explains it with a sharp constant.

**Lemma 112** (Blocked-end quantization). [PROVED] *Let  $[X, Y]$  be an edge of a tile of the tiling whose extension beyond  $X$  is blocked — either because  $X \in \partial ABC$ , or because the continuation enters the interior of another tile (Remark 52). Then every vertex of the tiling lying in the relative interior of  $[X, Y]$  is at distance from  $X$  belonging to the numerical semigroup  $\langle a, b, c \rangle$ . Consequently  $[X, Y]$  carries no interior vertex whenever*

$$\langle a, b, c \rangle \cap (0, |XY|) = \emptyset,$$

*and in particular whenever  $|XY| = \min(a, b, c)$ .*

*Proof.* Let  $T$  be the tile having  $[X, Y]$  as an edge. On  $T$ 's side of the segment nothing can contribute an interior vertex, since  $[X, Y]$  is a single edge of the triangle  $T$ . So any vertex in the relative interior of  $[X, Y]$  comes from the opposite side.

That side is covered by tiles abutting  $[X, Y]$ , each contributing a subsegment of one of its own edges, and consecutive contributions meet at points of  $[X, Y]$ ; a vertex in the relative interior is exactly such a meeting point. By hypothesis no covering edge overruns  $X$ : its continuation would leave  $ABC$ , or would enter the interior of a tile. Hence the first covering edge begins exactly at  $X$ , and the covering proceeds by whole tile edges laid end to end. The meeting points are therefore the partial sums of a sequence of tile side lengths, so each lies at distance from  $X$  in  $\langle a, b, c \rangle$ . (The last covering edge may overrun  $Y$ ; that affects no meeting point interior

to  $[X, Y]$ .) The two consequences are immediate, the second because  $\langle a, b, c \rangle$  has no element strictly between 0 and its smallest generator.  $\square$

**Corollary 113** (One-end-blocked chord dichotomy). [PROVED] *Let  $[X, Y]$  be a tile edge with the extension beyond  $X$  blocked. Then the far side of  $[X, Y]$  is covered by whole tile edges laid end to end from  $X$ , and exactly one of the following holds:*

- (i) *they partition  $[X, Y]$  exactly, so  $|XY|$  is a sum of tile sides and the arithmetic of Lemma 51 applies as it stands;*
- (ii) *the last one overruns  $Y$ , and those before it sum to some  $s \in \langle a, b, c \rangle \cap (0, |XY|)$ , with  $s = 0$  permitted.*

*In particular, if  $|XY| = \min(a, b, c)$  then that intersection is empty, so  $s = 0$  and the far side of  $[X, Y]$  is a single tile edge, either matching  $[X, Y]$  exactly or overrunning  $Y$ .*

*Proof.* Immediate from Lemma 112: the covering begins at  $X$  and proceeds by whole edges, so the partial sums before the final edge lie in  $\langle a, b, c \rangle$ , and the final edge either ends at  $Y$  or passes it.  $\square$

This is the common form of three arguments that were being made separately. Lemma 51 is the case where *both* ends are blocked, which deletes branch (ii) outright. The chord at  $V_k$  in Remark 106 is the case  $|XY| = b$  with the base end blocked, where  $\langle a, b, c \rangle \cap (0, b)$  consists of the multiples of  $a$  below  $b$  — which is exactly the “ $j$   $a$ -edges, then an overrun” description obtained there by hand. And the filler dichotomy of §5.7 is the same statement at a side junction. Branch (ii) is the crossing that Remark 111 shows cannot be removed by anchoring alone; the corollary does not close it, it states it once instead of three times.

**Remark 114** (Branch (ii) occurs, so no general lemma will delete it). It is worth being explicit that branch (ii) is not a formal possibility awaiting exclusion: it is realised in a kernel-checked tiling. In the 99-tiling, whose tile scaled by 16 is  $(a, b, c) = (32, 48, 64)$ , the  $b$ -edge from  $(88, 24)$  — a boundary point — to the interior point  $(136, 24)$  carries one interior vertex, at distance 32 from the blocked end. The remaining tail is  $48 - 32 = 16$ , and  $16 \notin \langle 32, 48, 64 \rangle$ , so the covering edge that meets  $Y$  must overrun it. Four  $b$ -edges of that shape occur in that tiling alone.

Together with Remark 111 this settles the shape of the obstruction. Three successive generalisations have now been refuted by explicit certified witnesses: that an edge with one end on the boundary is uncrossed (false,  $19 + 8 + 6 + 5$  counterexamples); that anchoring plus a length condition suffices (false, Remark 59); and that branch (ii) can be deleted (false, above). What survives is Corollary 113 itself, which is sharp, and the conclusion that the crossing at a given chord can only be excluded using the tiles the cascade has already forced there. No lemma quantified over chords will do it. That is a constraint on how the remaining work must be organised, and it is the reason the open steps in this note are stated as facts about particular configurations rather than as general pinning principles.

**Remark 115** (The quantization is exactly tight). The lemma is sharp in the strongest sense: not only does every crossing sit at a semigroup position, but *every* semigroup position below the edge length is realised. Across the eight kernel-checked tilings, all 52 crossings on edges with exactly one anchored end lie at a distance in  $\langle a, b, c \rangle$  from that end, with no exception; and edges of minimal length are crossed 0 times out of 74. Comparing permitted with observed

positions, by edge length:

|                      |                                              |                           |
|----------------------|----------------------------------------------|---------------------------|
| 44-, 77-, 99-tilings | $(a, b, c) = (16, 24, 32)$ or $(32, 48, 64)$ |                           |
| length $a$           | permitted $\emptyset$                        | observed $\emptyset$      |
| length $b$           | permitted $\{a\}$                            | observed $\{a\}$          |
| length $c$           | permitted $\{a, b\}$                         | observed $\{a, b\}$       |
| parallelogram 52     | $(a, b, c) = (18, 48, 54)$                   |                           |
| length $a$           | permitted $\emptyset$                        | observed $\emptyset$      |
| length $b$           | permitted $\{18, 36\}$                       | observed $\{18, 36\}$     |
| length $c$           | permitted $\{18, 36, 48\}$                   | observed $\{18, 36, 48\}$ |

So no strengthening of the conclusion is possible: the permitted set is attained.

Which side is minimal depends on the member:  $a < b$  exactly when  $f > \varphi e$  with  $\varphi$  the golden ratio, since  $a < b$  reads  $f^2 - ef - e^2 > 0$ . At the members carrying the certified tilings —  $(1, 2)$  and  $(1, 3)$  — one has  $a = m$ , which is why the observation appeared as a statement about  $a$ -edges.

The lemma is elementary and geometric, and its arithmetic content is only that a tile side is at least the least tile side, so we do not formalise it; it carries the label PROVED. It does not touch the chord at  $V_k$ , whose length is  $b$  and which is crossed in practice (Remark 111). It is recorded because it is the first general statement we have that excludes crossings outright, and because it fixes the exact radius within which the boundary still controls the interior.

**Corollary 116** (The walls form at  $e = 1$ ,  $f \geq 3$ ). [PROVED] *Let  $e = 1$  and  $f \geq 3$ . Then the base walk is the walls form  $(f, 1, 1)$ .*

*Proof.* Corollary 102 applies, since  $f \geq 3 > 2 = 2e$ , and leaves  $(0, 1, 2)$  and  $(f, 1, 1)$ ; the first is excluded by the three-letter argument of Remark 105.  $\square$

**5.7. The side parameter, and where its range is bounded.** Once the base walk is pinned, what remains is the side. The side question has a single integer parameter, and on the family of Corollary 116 that parameter takes only two open values, uniformly in the member.

The parametrisation itself is already in hand and is not new here. By Lemma 95 an equal side carries no  $b$ -edge at any member and its  $a$ -count is a multiple of  $f$ ; writing that count as  $pf$ , the side equation  $pf \cdot ef + n_c f^2 = f^3$  gives

$$n_c + pe = f,$$

and the  $\gamma$ -trap's  $n_c \geq 1$  gives  $0 \leq p \leq (f - 1)/e$ . Hypothesis 29 is read throughout this development as the assertion  $p = 0$ ; Remark 30 records that this reading and the block form of the hypothesis are not evidently equivalent. Lemma 99 already records this profile in the form “ $a$ -count  $fj$  with  $je \leq f - 1$ ”, and the Lean statements are `SideNoB.side_no_b_uncond`, `SideNoB.side_quantized` and `SideNoB.c_corner_forces_side_a`. What follows is the one consequence we did not find recorded: on the close pairs with  $f > 2e$  the range of  $p$  is not merely bounded but equal to  $\{0, 1, 2\}$ , uniformly in the member.

**Corollary 117** (A uniform two-case side problem). [CONJECTURE] *In the setting above, suppose in addition that  $2e < f$  and that  $(e, f)$  is a close pair,  $f^2 \leq 2ef + e^2$ . Then  $\lfloor (f - 1)/e \rfloor = 2$ . Hence on that family Hypothesis 29 reduces to excluding  $p \in \{1, 2\}$  — the same two cases for every member. (`side_p_le_two`.)*

*Proof.* From  $f \geq 2e + 1$  we get  $(f - 1)/e \geq 2$ . Conversely  $p \geq 3$  would give  $3e \leq pe \leq f - 1$ , so  $f \geq 3e + 1$  and then  $f^2 - 2ef - e^2 \geq (3e + 1)(e + 1) - e^2 = 2e^2 + 4e + 1 > 0$ , contradicting the close-pair inequality.  $\square$

**Remark 118** (What this buys, and what it does not). The count kills  $p = 2$  on  $f = 2e + 1$  but not  $p = 1$ . Full discussion in [1].

**5.8. A parity of the straight figures.** Lemma 90 records three counting identities whose published consequence is  $v_1 = 1 + n_2 + v_3 + 2v_4$ . The  $\alpha$ -identity also carries a parity statement, which we record together with the reason it does *not* exclude any member.

**Lemma 119** (Straight-figure parity). [PROVED] *Write  $S = n_1 + n_2$  for the total number of straight ( $\pi$ ) figures. Then  $S \equiv N + 1 \pmod{2}$ .*

*Proof.* The  $\alpha$ -identity is  $3 + n_1 + 3n_2 + 2v_2 + 4v_3 + 6v_4 = N$ ; substituting  $n_1 = S - n_2$  gives  $S + 2(n_2 + v_2 + 2v_3 + 3v_4) = N - 3$ . (**census\_parity**).  $\square$

The three identities have rank two and this is the only parity they contain: substituting the  $v_1$  relation into the  $\beta$ -identity returns the  $\alpha$ -identity exactly.

**Remark 120** (Why this does not exclude a member, and the trap it sets). The parity constrains only the number of interior  $T$ -junctions, so it excludes no member. Full discussion in [1].

**5.9. Apex rigidity, and a bound on the side parameter.** The apex of a base- $\beta$  target is completely rigid, and the rigidity is an identity of the whole family rather than a coincidence at one member. It forces the covering of the first chord exactly, and from that it bounds  $p$ .

Throughout,  $A$  is the apex, the equal sides have length  $f^3$ , the apex angle is  $3\alpha$ , and  $\sin(\alpha/2) = e/(2f)$ , whence

$$\cos \frac{\alpha}{2} = \frac{\sqrt{4f^2 - e^2}}{2f}, \quad \cos \frac{3\alpha}{2} = \frac{(f^2 - e^2)\sqrt{4f^2 - e^2}}{2f^3}, \quad \sin \frac{3\alpha}{2} = \frac{e(3f^2 - e^2)}{2f^3}.$$

By Lemma 45 an equal side carries no  $b$ -edge, and the apex figure is  $\{3\alpha\}$ , so the tile at  $A$  presents  $\alpha$  and its two edges there are  $b$  and  $c$ ; the one on the side is therefore a  $c$ . Hence the first junction  $J$  of the side sits at distance  $c = f^2$  from  $A$ . Write  $C$  for the chord through  $J$  parallel to the base. It cuts off a triangle similar to the target with ratio  $f^2/f^3 = 1/f$ , so the area above  $C$  is  $N/f^2$  tile areas and  $|C| = e(3f^2 - e^2)/f$ .

**Lemma 121** (The apex identities).

[VERIFIED] *For every  $(e, f)$ ,*

$$b \cos \frac{\alpha}{2} = c \cos \frac{3\alpha}{2}, \quad c \sin \frac{3\alpha}{2} - b \sin \frac{\alpha}{2} = a.$$

*Proof.* Neither identity is new, and it is worth saying which classical facts they are. Since  $3\alpha + 2\beta = \pi$  and  $\gamma = 2\alpha + \beta$  one has  $\gamma = \frac{\pi + \alpha}{2}$  and  $\beta = \frac{\pi}{2} - \frac{3\alpha}{2}$ , whence

$$\cos \frac{\alpha}{2} = \sin \gamma, \quad \cos \frac{3\alpha}{2} = \sin \beta, \quad \sin \frac{3\alpha}{2} = \cos \beta, \quad \sin \frac{\alpha}{2} = -\cos \gamma.$$

The first identity therefore reads  $b \sin \gamma = c \sin \beta$ , which is the *law of sines*  $b/\sin \beta = c/\sin \gamma$ ; the second reads  $c \cos \beta + b \cos \gamma = a$ , which is the *projection formula* for the tile. Both hold for every triangle whatever. Verified symbolically and numerically at two precisions, and machine-checked in the closed forms used here (`lean/ApexRigidity.lean: apex_drop_eq, apex_edge_eq`).  $\square$

**Remark 122** (What is actually new here). Since Lemma 121 is classical, the content of Theorem 123 lies entirely in the *configuration*, not in the algebra. The point is this. A base- $\beta$  target has base angle  $\beta$ , so an equal side descends at angle  $\beta$  to the horizontal. A tile presenting  $\alpha$  at a point of that side, with its  $c$  along the side, therefore has its other edge at that vertex — its  $b$  — inclined at  $\alpha + \beta = \pi - \gamma$  to the horizontal. The two edges drop by  $c \sin \beta$  and  $b \sin \gamma$ , and the law of sines makes those equal.

So the rigidity is not a coincidence of the family and not a property of the apex: it holds wherever a tile presents  $\alpha$  on an equal side with its  $c$  along that side, which is exactly why it recurs at every junction (Proposition 134). What the family supplies is the hypothesis that

makes it bite — that the target's base angle is the tile's own  $\beta$ , so that the tile's angle sum does the work. The consequences drawn from it in this section, Theorems 126, 138 and 168 and Corollary 131, are new; the identity underneath them is not.

**Theorem 123** (The first chord is covered exactly). [PROVED] *Let  $T_1, T_2, T_3$  be the three tiles at  $A$ , with  $T_1, T_3$  the outer ones. Then:*

- (a)  $T_1$ 's remaining apex edge is its  $b$ , ending at a point  $P$  lying exactly on  $C$ ;
- (b)  $T_1$ 's third edge  $JP$  lies along  $C$  and has length exactly  $a$ ;
- (c)  $T_2$  has exactly the fraction  $b/c$  of its area above  $C$ , and its trace on  $C$  is the segment  $PP'$  joining  $P$  to the mirror point  $P'$  of  $T_3$ , of length  $e(f^2 - e^2)/f$ ;
- (d) consequently  $a + |PP'| + a = |C|$  and the area above  $C$  is  $2 + b/c = N/f^2$  — so  $T_1, T_3$  and that fraction of  $T_2$  fill the region above  $C$  exactly, and no other tile has any area above  $C$ .

*Proof.* (a) The drop of  $J$  below  $A$  is  $c \cos \frac{3\alpha}{2}$  and that of  $P$  is  $b \cos \frac{\alpha}{2}$ ; they agree by Lemma 121. (b) The horizontal offsets of  $J$  and  $P$  from the axis of symmetry are  $c \sin \frac{3\alpha}{2}$  and  $b \sin \frac{\alpha}{2}$ , differing by  $a$ .

(c) Note first that  $P$  and  $P'$  are defined by  $T_1$  and  $T_3$ , at distance  $b$  along the two inner rays from  $A$ , independently of  $T_2$ . Now  $T_2$  presents  $\alpha$  at  $A$ , so its two edges there are a  $b$  and a  $c$ , laid on those same two rays — and *both orientations occur*, so we treat them together. Its  $b$ -edge terminates at drop  $b \cos \frac{\alpha}{2}$ , which is the chord by Lemma 121, hence at whichever of  $P, P'$  lies on that ray. Its  $c$ -edge terminates at drop  $c \cos \frac{\alpha}{2}$ , strictly below  $C$ , and the chord meets it at distance  $b$  from  $A$ , since  $\cos \frac{3\alpha}{2} / \cos \frac{\alpha}{2} = b/c$  — that is, at the other of  $P, P'$ . So in either orientation  $T_2 \cap C = PP'$ , and  $T_2$  is cut by a line through one of its vertices, giving area ratio  $b/c$  (middle.fraction). The length of  $PP'$  is  $2b \sin \frac{\alpha}{2} = e(f^2 - e^2)/f$ .

(d)  $2ef + e(f^2 - e^2)/f = e(3f^2 - e^2)/f = |C|$ , and  $2 + b/c = (3f^2 - e^2)/f^2 = N/f^2$  (area.above.chord), which is the area above  $C$ . Since the three tiles already supply all of it, nothing else can.  $\square$

**Corollary 124** (The tile below  $T_1$ 's  $a$ -edge has a vertex at  $P$ ). [PROVED]  *$T_1$  presents  $\gamma$  at  $P$ , and either  $T_2$  presents  $\gamma$  at  $P$  as well, or  $P$  lies interior to an edge of  $T_2$ . In both cases no further tile has  $P$  interior to an edge; in particular the tile below  $T_1$ 's  $a$ -edge has a vertex at  $P$ , and the residual figure there is  $\{2\alpha, 2\beta\}$ ,  $\{\beta, \gamma\}$  or  $\{\alpha, \beta\}$ .*

*Proof.*  $T_1$  has sides  $c, b, a$  at  $A, P, J$  by Theorem 123, so its angle at  $P$  is opposite  $c$ , namely  $\gamma$ . By Theorem 123(c) the trace of  $T_2$  on  $C$  is  $PP'$ , so  $P$  lies on  $\partial T_2$ : it is either a vertex of  $T_2$  or interior to one of its edges.

Suppose it is a vertex. The edge of  $T_2$  at  $P$  along the ray towards  $A$  terminates at  $P$ , at distance  $b$  from  $A$ ;  $T_2$ 's flanks at  $A$  are  $b$  and  $c$ , and a  $c$  there would terminate at distance  $c \neq b$ , so that edge is  $T_2$ 's  $b$ . Hence  $T_2$ 's angle at  $P$  has  $b$  as a flank and is  $\alpha$  or  $\gamma$ . If it were  $\alpha$ , the other flank at  $P$  would be  $c$  and  $T_2$  would be the triangle  $APR$  with  $AP = b$ ,  $PR = c$ ,  $AR = a$ , whose angle at  $A$  is opposite  $c$ , that is  $\gamma$  — contradicting that  $T_2$  presents  $\alpha$  at  $A$ . So  $T_2$  presents  $\gamma$ .

For the count: two  $\gamma$ 's total  $\pi + \alpha$ , leaving  $\pi - \alpha < \pi$ ; a  $\gamma$  and a straight angle total  $\pi + \gamma$ , leaving  $\pi - \gamma < \pi$ . In neither case does a further straight angle fit. Finally the tile below  $T_1$ 's  $a$ -edge is neither  $T_1$  nor  $T_2$ , since at  $P$  those occupy the wedge towards  $A$  and the wedge towards  $P'$  respectively, so it has a vertex at  $P$ .  $\square$

**Remark 125** (The two equal sides differ). The dichotomy in Corollary 124 is not an artefact.  $T_2$  lays  $b$  on one of the two rays it shares with  $T_1$  and  $T_3$  and  $c$  on the other, so exactly one of  $P, P'$  is a genuine vertex point and the other carries a  $T$ -junction. This is the asymmetry between the two equal sides already visible in Remark 167. It does not affect what follows: Corollary 124 delivers the same conclusion on both sides.

**Theorem 126** (The second edge of an equal side is a  $c$ ). [PROVED] Assume  $e^2 + ef < f^2$ , equivalently  $a < b$ . Then on every equal side the tile edge immediately below the first junction  $J$  is a  $c$ . The hypothesis is removable: Proposition 127 reaches the same conclusion assuming only  $\gcd(e, f) = 1$ , and it is that form the corollaries below use.

*Proof.* Let  $Y$  be the tile below  $T_1$ 's  $a$ -edge, adjacent to it at  $P$ . By Corollary 124,  $Y$  has a vertex at  $P$  and an edge running from  $P$  along  $C$  towards  $J$ . Since  $J$  lies on the boundary at distance  $a$ , that edge has length at most  $a$ ; as  $a < b < c$  it is exactly  $a$ , so  $Y$ 's  $a$ -edge is  $JP$ , matching  $T_1$ 's. In particular  $Y$  does not present  $\alpha$  at  $P$ , whose flanks are  $b$  and  $c$ .

At  $J$  the figure is straight, hence  $\{\alpha, \beta, \gamma\}$  or  $\{3\alpha, 2\beta\}$ .  $T_1$  presents  $\beta$  there (its angle opposite  $b$ ). The angles at the two ends of  $Y$ 's  $a$ -edge are  $\beta$  and  $\gamma$ , so  $Y$  presents  $\gamma$  or  $\beta$  at  $J$ . In the first case the figure must be  $\{\alpha, \beta, \gamma\}$  and the third tile presents  $\alpha$ ; in the second it must be  $\{3\alpha, 2\beta\}$  and the remaining three tiles all present  $\alpha$ . Either way,  $T_1$  occupies the extreme position against the ascending side and  $Y$  the next, so the tile against the *descending* side presents  $\alpha$ . Its flanks are  $b$  and  $c$ , and the side edge below  $J$  is that tile's edge; by Lemma 45 it is not a  $b$ , hence it is a  $c$ .  $\square$

**Proposition 127** (The golden-ratio hypothesis is removable). [PROVED] Theorem 126 holds at every member with  $\gcd(e, f) = 1$ , with no assumption relating  $a$  and  $b$ .

*Proof.* Only one step used  $a < b$ : that  $Y$ 's edge from  $P$  towards  $J$ , of length at most  $|PJ| = a$ , equals  $a$ . A  $c$ -edge never fits, since  $c = f^2 > ef = a$ . Suppose then  $b < a$  and  $Y$  lays  $b$ . Its far end is interior to  $T_1$ 's  $a$ -edge, and the residual segment from  $J$  to that point, of length  $a - b$ , must be partitioned by whole tile edges: no edge may overrun  $J$ , which lies on the boundary, and none may overrun  $Y$ 's vertex, which would put two tiles below the chord on the same ground. So  $a - b \in \langle a, b, c \rangle$ .

It is not. Modulo  $e$  the three lengths are 0,  $f^2$ ,  $f^2$ , and  $a - b = e^2 + ef - f^2 \equiv -f^2$ . A representation  $a - b = xa + yb + zc$  gives  $(y + z + 1)f^2 \equiv 0 \pmod{e}$ , and  $\gcd(e, f) = 1$  forces  $e \mid y + z + 1$ , hence  $y + z \geq e - 1$  and  $a - b \geq (e - 1)b$ . But  $(e - 1)b > a - b$ , since that is  $eb > a$ , which is  $b > f$ ; and  $b > f$  holds always, because  $e \leq f - 1$  gives  $e^2 \leq (f - 1)^2 < f^2 - f$  (ApexRigidity.b\_gt\_f, .eb\_gt\_a). Finally  $b < a$  forces  $e \geq 2$ , so the modular step is available (.e\_ge\_two\_of\_b\_lt\_a). Hence no representation exists,  $Y$  cannot lay  $b$ , and it lays  $a$ .

Checked on all 17 coprime members with  $b < a$  and  $e \leq 9$ ,  $f \leq 17$ : in every case  $a - b$  is a gap of  $\langle a, b, c \rangle$ .  $\square$

**Remark 128** (What this does and does not extend). The hypothesis  $a < b$  was later removed entirely by Proposition 127. Full discussion in [1].

**Theorem 129** (The side parameter vanishes on the family  $e = f - 1$ ). [CONJECTURE] Let  $e = f - 1$  with  $f \geq 2$ . Then every equal side of the base- $\beta$  target at  $m = 1$  carries no  $a$ -edge:  $p = 0$ .

This establishes the west half of Hypothesis 29 at every such member, that half being the assertion  $p = 0$  on the west side (Remark 30). We state the theorem in terms of  $p$ , which is what the proof gives.

*Proof.*  $\gcd(f - 1, f) = 1$ , so Proposition 127 applies and Corollary 131 gives  $n_c \geq 2$ . Here  $n_c = f - pe = f - p(f - 1)$ . For  $p = 1$  this is 1, contradicting  $n_c \geq 2$ ; for  $p \geq 2$  it is at most  $2 - f \leq 0$ , contradicting the  $\gamma$ -trap's  $n_c \geq 1$ . Hence  $p = 0$ .

Concretely at  $p = 1$  the side word would carry  $f$   $a$ -edges and a single  $c$ , the  $c$  at the apex; and Theorem 126 requires a second  $c$  immediately below it, which is not available. The side does have at least three edges, so the edge below the first junction exists and the theorem applies.  $\square$

**Remark 130** (Scope, and the status of this theorem). The theorem is *proved*, not machine-checked: steps 2–3 of its chain are planar geometry. Full discussion in [1].

**Corollary 131** (A bound on the side parameter). [PROVED] *Let  $\gcd(e, f) = 1$ . Then  $n_c \geq 2$ , hence*

$$pe + 2 \leq f.$$

*In particular  $p = 2$  is impossible whenever  $2e + 2 \leq f$  fails, and on the tight subfamily  $f = 2e + 1$  the bound reads  $pe \leq 2e - 1$ , so  $p \leq 1$ .*

*Proof.* The edge at the apex and the edge below  $J$  are both  $c$  by Theorem 126, in the form freed of the hypothesis  $a < b$  by Proposition 127, which needs only  $\gcd(e, f) = 1$ ; so  $n_c \geq 2$ , with  $n_c + pe = f$  this is  $pe + 2 \leq f$ . For  $f = 2e + 1$ ,  $p \geq 2$  would give  $2e \leq pe \leq 2e - 1$  (`side_p_bound`, `p_le_one_of_tight`).  $\square$

**Remark 132** (What this settles, and exactly where it stops). The bound reaches  $p = 1$  only at  $(1, 2)$ ; at every other member it stops at  $p \geq 2$ . Full discussion in [1].

**Remark 133** (The seven open primes of the tight subfamily). On the tight subfamily  $f = 2e + 1$  the tile count is  $N = 11e^2 + 12e + 3$ , and the members whose  $N$  is a prime congruent to 11 modulo 12 and still open are these:

| $N$   | $e$ | $f$ | tile $(a, b, c)$   | target                   |
|-------|-----|-----|--------------------|--------------------------|
| 227   | 4   | 9   | (36, 65, 81)       | (729, 729, 908)          |
| 1223  | 10  | 21  | (210, 341, 441)    | (9261, 9261, 12230)      |
| 3011  | 16  | 33  | (528, 833, 1089)   | (35937, 35937, 48176)    |
| 4643  | 20  | 41  | (820, 1281, 1681)  | (68921, 68921, 92860)    |
| 5591  | 22  | 45  | (990, 1541, 2025)  | (91125, 91125, 123002)   |
| 8963  | 28  | 57  | (1596, 2465, 3249) | (185193, 185193, 250964) |
| 13127 | 34  | 69  | (2346, 3605, 4761) | (328509, 328509, 446318) |

Every one satisfies the standing hypothesis: on  $f = 2e + 1$  the inequality  $e^2 + ef < f^2$  reduces to  $e^2 + 3e + 1 > 0$ , true for all  $e$ . So the whole of §5.9 applies at each, and Corollary 131 reads  $pe \leq 2e - 1$ , giving  $p \leq 1$ .

Each member is therefore in *exactly* the state of  $(3, 7)$ : the base walk is settled,  $p \in \{0, 1\}$ , and the single open question is  $p = 1$ , which by Corollary 139 reduces to whether a  $T$ -junction occurs at one of the points  $W(J)$ . Whatever settles  $(3, 7)$  by theory settles all seven, since no step of §5.9 uses the member beyond  $e^2 + ef < f^2$  — and conversely an exhaustive search at  $(3, 7)$  settles that member alone, the next target having  $N = 227$  and a target triangle nine times larger in linear scale.

**Proposition 134** (The descent is self-similar). [PROVED] *Suppose a tile  $Z$  lays a  $c$ -edge of an equal side between consecutive junctions  $J'$  (nearer the apex) and  $J$ , and presents  $\alpha$  at  $J'$ . Let  $W$  be its third vertex. Then  $W$  lies exactly on the chord  $C$  through  $J$ ; the segment  $JW$  is  $Z$ 's  $a$ -edge, of length  $a$ , lying along  $C$ ; and  $Z$  presents  $\beta$  at  $J$  and  $\gamma$  at  $W$ .*

*Proof.* The two edges of  $Z$  at  $J'$  are its flanks at  $\alpha$ , namely  $b$  and  $c$ ; the  $c$  runs down the side to  $J$  and the  $b$  into the interior, at angle  $\alpha$  from the descending side, hence at  $\frac{3\alpha}{2} - \alpha = \frac{\alpha}{2}$  from the downward vertical — the same inclination as the apex tile's  $b$ -edge in Theorem 123. So the drop from  $J'$  to  $W$  is  $b \cos \frac{\alpha}{2}$  and that from  $J'$  to  $J$  is  $c \cos \frac{3\alpha}{2}$ , equal by Lemma 121; and the horizontal offsets differ by  $c \sin \frac{3\alpha}{2} - b \sin \frac{\alpha}{2} = a$ . The angles at  $J$  and  $W$  are those opposite  $b$  and  $c$ , namely  $\beta$  and  $\gamma$ .  $\square$

**Remark 135** (Why the descent does not close, and it is not for want of trying). Proposition 134 says the configuration at  $J$  is a verbatim copy of the configuration at the first junction, with

$(A, J_1, P)$  replaced by  $(J', J, W)$ . If the argument of Theorem 126 could be run at  $J$  as it was at  $J_1$ , it would force the side edge below  $J$  to be a  $c$  as well, and induction down the side would give  $n_a = 0$ , that is  $p = 0$  — exactly Hypothesis 29. It cannot, and the obstruction is sharp.

The argument at  $J_1$  turns on Corollary 124: *two tiles present  $\gamma$  at  $P$ , so the residual is  $\pi - \alpha < \pi$  and no tile can have  $P$  interior to an edge. The second  $\gamma$  comes from  $T_2$ , which exists only because the apex figure is  $\{3\alpha\}$ . At  $W$  only  $Z$ 's  $\gamma$  is forced, and no counting argument can supply the other:*

**Lemma 136** (A single  $\gamma$  never excludes a  $T$ -junction). [PROVED] *Let  $x$  be an interior point at which one tile presents  $\gamma$  and one tile has  $x$  interior to an edge. Then the remaining tiles at  $x$  present angles totalling exactly  $\alpha + \beta$ , realised by one  $\alpha$  and one  $\beta$ . The angle count is therefore always consistent.*

*Proof.*  $2\pi = \gamma + \pi + R$  gives  $R = \pi - \gamma = \alpha + \beta$ , since  $\alpha + \beta + \gamma = \pi$ .  $\square$

So the obstruction is not a missing computation but the tile's own angle sum: a lone  $\gamma$  beside a straight edge balances at every member and at every point. Exclusion by angle count needs two  $\gamma$ 's, and the apex is the only place that supplies two.

We also record a near miss, since it shows how narrow the gap is. At  $W$  the  $T$ -junction *can* be excluded when it would sit on the tile  $V$  angularly adjacent to  $Z$  at  $J'$ : if  $V$  lays its  $c$  along  $Z$ 's  $b$ -ray, the budget is  $\gamma + \pi$  with residual  $\pi - \gamma < \pi$ , so no *second* straight angle fits, and the tile below  $Z$ 's  $a$ -edge is forced to have a vertex at  $W$  and to lay exactly  $a$  — which is all Theorem 126 needs. What defeats this is that the  $T$ -junction need not sit on  $V$  at all: the tile *below*  $C$  may carry an edge running along  $C$  from  $J$  through  $W$ , of length  $b$  or  $c$ , with its vertex on the boundary at  $J$ . That is precisely the tile the argument was trying to pin, so it closes on itself. Settling whether that placement can occur is, as far as we can see, the whole of what remains at  $p = 1$ .

**Definition 137** (The points  $W(J)$ ). Let  $J$  be a junction of an equal side whose edge *above* it is a  $c$ , laid by a tile presenting  $\alpha$  at the junction above. By Proposition 134 that tile's third vertex lies on the chord through  $J$  at distance exactly  $a$  from  $J$ ; call it  $W(J)$ .

**Theorem 138** (An  $a$ -edge forces a  $T$ -junction). [PROVED] *Assume  $e^2 + ef < f^2$ . Let  $J$  be as in Definition 137 and suppose the side edge below  $J$  is an  $a$ . Then  $W(J)$  lies in the interior of an edge of some tile.*

*Proof.* Every tile meeting  $J$  has a vertex there. A tile with  $J$  interior to an edge would contribute  $\pi$  and so exhaust the straight figure by itself, forcing the two side edges at  $J$  to be collinear parts of a single tile edge, in which case  $J$  would not be a junction; and an edge through  $J$  transversal to the side would carry that tile outside the target.

Write  $Z$  for the  $c$ -tile above  $J$ ; by Proposition 134 it presents  $\beta$  at  $J$  and its  $a$ -edge is  $JW$  with  $W = W(J)$ . Let  $M$  be the tile below sharing the chord ray with  $Z$ , and let  $X$  be the last tile in angular order, standing against the descending side;  $X$  lays the next side edge on one of its flanks, and that flank is not a  $b$  by Lemma 45.

The figure at  $J$  is  $\{\alpha, \beta, \gamma\}$  or  $\{3\alpha, 2\beta\}$ , with  $Z$  first. Exhausting the orders with first entry  $\beta$ :

| figure                      | order                                    | $M$      | side edge below $J$ |
|-----------------------------|------------------------------------------|----------|---------------------|
| $\{\alpha, \beta, \gamma\}$ | $(\beta, \alpha, \gamma)$                | $\alpha$ | $a$                 |
| $\{\alpha, \beta, \gamma\}$ | $(\beta, \gamma, \alpha)$                | $\gamma$ | $c$                 |
| $\{3\alpha, 2\beta\}$       | $(\beta, \alpha, \alpha, \alpha, \beta)$ | $\alpha$ | $a$ or $c$          |
| $\{3\alpha, 2\beta\}$       | $(\beta, \alpha, \alpha, \beta, \alpha)$ | $\alpha$ | $c$                 |
| $\{3\alpha, 2\beta\}$       | $(\beta, \alpha, \beta, \alpha, \alpha)$ | $\alpha$ | $c$                 |
| $\{3\alpha, 2\beta\}$       | $(\beta, \beta, \alpha, \alpha, \alpha)$ | $\beta$  | $c$                 |

(the side edge is read off  $X$ 's flanks with  $b$  removed:  $\alpha \mapsto c$ ,  $\beta \mapsto a$  or  $c$ ,  $\gamma \mapsto a$ ). In both rows permitting an  $a$ ,  $M$  presents  $\alpha$ . Its flanks are then  $b$  and  $c$ , so the edge  $M$  lays along the chord ray has length  $b$  or  $c$ , and  $e^2 + ef < f^2$  gives  $a < b < c$ . Since  $|JW| = a$ , the point  $W$  lies strictly inside that edge.  $\square$

**Corollary 139** (A sufficient condition for Hypothesis 29). [CONJECTURE] *Assume  $e^2 + ef < f^2$ . If for every junction  $J$  as in Definition 137 the point  $W(J)$  is a vertex of every tile meeting it, then  $p = 0$  on every equal side, i.e. Hypothesis 29 holds at that member.*

*Proof.* Suppose some equal side carries an  $a$ -edge, and take the first one counted from the apex. Every edge above it is a  $c$ , so the junction  $J$  immediately above it satisfies Definition 137, and Theorem 138 puts  $W(J)$  interior to an edge — contrary to hypothesis. Hence  $n_a = 0$ , that is  $p = 0$ .  $\square$

**Remark 140** (The crux, in its sharpest form). Corollary 139 replaces “exclude the surviving configuration” by a single question about an explicitly located finite set of points. The exclusion succeeds at the first junction: there  $W(J_1) = P$ , and Corollary 124 rules out a  $T$ -junction because two tiles present  $\gamma$  at  $P$ . It is unavailable at every later junction, and not by accident: Lemma 136 shows a single  $\gamma$  beside a straight edge always balances, so no angle count can do it, and Proposition 134 shows the configuration one level down is a verbatim copy, so no new local information appears. What is needed is a second  $\gamma$  at  $W(J)$  from geometry rather than from counting.

We note the numerical consequence is weak and should not be oversold: only a  $c \rightarrow a$  transition forces a  $T$ -junction, and a side word such as  $c c c c a^7$  has exactly one. The theorem gives one forced  $T$ -junction per equal side carrying an  $a$ , not one per  $a$ -edge, and in particular it does not revive the counting argument retracted in Remark 120.

**5.10. The inflation reformulation.** The base- $\beta$  tile is a rep-tile. Remark 33 records that the complete west block is the tile scaled by  $f$ , containing  $f^2$  tiles, and that at  $m = 1$  its far side is the whole equal side. Read the other way, that says the equal side's edge word is the  $c$ -side word of an inflated tile, and the side question becomes a question about the inflation alone.

**Proposition 141** (The inflated boundary). [PROVED] *Let the tile be inflated by  $f$ , so its sides are  $fa$ ,  $fb$ ,  $fc$ . Then the  $\alpha$ - and  $\beta$ -corners of the inflated triangle are each met by a single tile, which lays along the two sides the flanks of that angle. No assumption is made at the  $\gamma$ -corner, and none is needed: the  $c$ -side joins the  $\alpha$ - and  $\beta$ -vertices, so both its end edges are  $c$ ; the  $a$ -side has its  $\beta$ -end forced to  $a$  and the  $b$ -side its  $\alpha$ -end forced to  $b$ , and in each case the residual composition is unique. At  $(1, 3)$  and  $(3, 7)$  the outcome is:*

| $(e, f)$ | $a$ -side     | $b$ -side     | $c$ -side           |
|----------|---------------|---------------|---------------------|
| $(1, 3)$ | $a^3$ (rigid) | $b^3$ (rigid) | $c^3$ or $ca^3c$    |
| $(3, 7)$ | $a^7$ (rigid) | $b^7$ (rigid) | $c^7$ or $ca^7c^2c$ |

where the second  $c$ -side word at  $(3, 7)$  has 7  $a$ -edges and 4  $c$ -edges.

*Proof.* The residual lengths are  $fa - 2a$ ,  $fb - 2b$ ,  $fc - 2c$ . At  $(1, 3)$ , tile  $(3, 8, 9)$ : the residuals are 3, 8, 9, whose compositions over  $\{3, 8, 9\}$  are  $\{3\}$ ,  $\{8\}$  and  $\{9\}$ ,  $\{3, 3, 3\}$  respectively (Inflation.middle\_three, .middle\_eight, .middle\_nine). At  $(3, 7)$ , tile  $(21, 40, 49)$ : the  $c$ -side residual is 245, with compositions  $5c$  and  $7a + 2c$  (.residual\_245). For the  $a$ - and  $b$ -sides only one end is known, so the residuals are  $fa - a = 126$  and  $fb - b = 240$ ; their compositions are again unique,  $6a$  and  $6b$  (.residual\_126, .residual\_240), and at  $(1, 3)$  the corresponding residuals 6 and 16 give  $2a$  and  $2b$  (.residual\_13\_sides).

Only the  $\alpha$ - and  $\beta$ -corners were used. Two tiles at the  $\alpha$ -corner already give  $2\alpha > \alpha$ ; and a  $\beta$ -corner splits only as  $\beta = k\alpha$ , which with  $3\alpha + 2\beta = \pi$  gives  $\alpha(3 + 2k) = \pi$ , making  $\alpha/\pi$

rational and contradicting its irrationality. So the  $\beta$ -corner is single at every member, with no hypothesis (`Inflation.beta_not_nat_multiple`, `.beta_corner_single`).  $\square$

**Corollary 142** (The crux, on  $f^2$  tiles).

[PROVED]

At  $m = 1$  the inflated  $c$ -side is the whole equal side, so the two  $c$ -side words of Proposition 141 are exactly  $p = 0$  and  $p = 1$ . Deciding which occurs — that is, deciding whether the inflation is rigid — decides the side parameter on that side.

**Remark 143** (What this buys, and what it does not). Recorded in [1]. Full discussion in [1].

**Theorem 144** (The inflation is rigid at every member searched).

[CONJECTURE]

The inflated tile of Proposition 141 admits no tiling by  $f^2$  copies whose  $c$ -side carries the  $p = 1$  word, at each of the ten members

$$(1, 3), (1, 4), (1, 5), (2, 5), (3, 5), (1, 6), (1, 7), (3, 7), (5, 7), (4, 9).$$

Its only boundary is therefore the standard one,  $a^f$ ,  $b^f$ ,  $c^f$ .

*Proof.* Exhaustive exact search over the two profiles left open by Proposition 141. At (1, 3), target (9, 24, 27) with  $N = 9$ : the  $c$ -side profile (3, 0, 2) gives EXHAUSTED\_NO\_TILING in 9 nodes, and (0, 0, 3) gives FOUND\_TILING in 30. At (3, 7), target (147, 280, 343) with  $N = 49$ : the profile (7, 0, 4) gives EXHAUSTED\_NO\_TILING in 57 824 nodes, and (0, 0, 7) gives FOUND\_TILING in 443 347, with boundary words  $a^7$ ,  $b^7$ ,  $c^7$  as predicted. At (2, 5), target (50, 105, 125) with  $N = 25$ : the profile (5, 0, 3) gives EXHAUSTED\_NO\_TILING in 297 nodes and (0, 0, 5) gives FOUND\_TILING in 2 212. At (4, 9),  $N = 81$ : the profile (9, 0, 5) gives EXHAUSTED\_NO\_TILING in 5 161 190 nodes. The remaining six members were generated by `code/engine/gen_inflation.py`, which reproduces the five hand-built instances byte for byte; their node counts are

| $(e, f)$ | $f^2$ | $k = f - e$ | #words | $p = 1$ nodes |
|----------|-------|-------------|--------|---------------|
| (1, 3)   | 9     | 2           | 1      | 9             |
| (1, 4)   | 16    | 3           | 5      | 25            |
| (1, 5)   | 25    | 4           | 21     | 137           |
| (2, 5)   | 25    | 3           | 6      | 297           |
| (3, 5)   | 25    | 2           | 1      | 317           |
| (1, 6)   | 36    | 5           | 84     | 793           |
| (1, 7)   | 49    | 6           | 330    | 5 097         |
| (5, 7)   | 49    | 2           | 1      | 3 219         |
| (3, 7)   | 49    | 4           | 36     | 57 824        |
| (4, 9)   | 81    | 5           | 220    | 5 161 190     |

with the word count of Remark 149 beside each. Every member carries a FOUND\_TILING control on its  $p = 0$  instance.

The arithmetic behind the two-word dichotomy is machine-checked at the first three members (`Inflation.middle_nine`, `.residual_75`, `.residual_245`), as is the uniqueness of the  $a$ - and  $b$ -side compositions.

The FOUND runs are negative controls: a search exhausting both profiles would indicate a malformed instance rather than a theorem. Their node counts are reported for orientation only and are *not* reproducible constants: a FOUND search halts at the first tiling it meets, so the count depends on how work was distributed across threads. At (2, 5) the same instance returns 2 212, 2 199, 2 223 and 2 414 nodes on different thread counts. The EXHAUSTED counts carry no such dependence — an exhaustive search visits the same tree however it is scheduled, and this was checked directly at (3, 7), where 1, 2 and 4 threads all return 57 824 nodes with all four prune counters equal. At (4, 9) that control is still running at the time of writing, so the

fourth member is stated on the strength of its EXHAUSTED verdict alone and should be read as one degree weaker than the first three until the control returns. Instances, certificates and node counts are in `code/engine/inflation/`. Search verdicts are declared computer assistance and are not formalized.  $\square$

**Corollary 145** (Remark 47’s step, proved). [CONJECTURE] *At each of the ten members of Theorem 144: if the west corner block is complete, then the equal side on that side carries no  $a$ -edge.*

*Proof.* A complete west block is the tile inflated by  $f$  (Remark 33), and at  $m = 1$  its far side is the whole equal side. By Theorem 144 that side is  $c^f$ , so  $n_a = 0$ .  $\square$

**Remark 146** (What this settles and what it does not). This does not decide  $p$  at  $(3, 7)$ : the inflated tile is the west corner block only if that block is complete, which is the west half of Hypothesis 29 itself. Full discussion in [1].

Theorem 144 is proved one member at a time by search, which is unsatisfying: the statement is uniform in  $(e, f)$  and ought to have a uniform proof. One is available for half the family, at the cost of an extra hypothesis.

**Proposition 147** (Edge parity kills the  $p = 1$  boundary when  $e$  and  $f$  differ in parity). [PROVED] *Assume the dissection of the inflated tile is edge-to-edge. If  $e \not\equiv f \pmod{2}$ , then the boundary word with  $p = 1$  is impossible.*

*Proof.* The  $f^2$  tiles contribute  $3f^2$  edge-slots. Edge-to-edge means every edge of the subdivision is an entire edge of each tile meeting it, so each interior edge consumes exactly two slots and each boundary edge exactly one; hence  $3f^2 - B \equiv 0 \pmod{2}$ , where  $B$  is the number of boundary edges. The standard boundary has  $B = 3f$ , and  $3f^2 - 3f = 3f(f - 1)$  is always even. The  $p = 1$  boundary has  $f$  edges on the  $a$ -side,  $f$  on the  $b$ -side, and  $f + (f - e)$  on the  $c$ -side, so  $B = 4f - e$  and

$$3f^2 - (4f - e) \equiv f^2 + e \equiv f + e \pmod{2}.$$

This is odd exactly when  $e \not\equiv f$ , and an odd value contradicts the count.  $\square$

**Corollary 148** (The  $p = 1$  word needs two  $c$ -edges, so  $e = f - 1$  is rigid). [VERIFIED] *A  $p = 1$  word exists only if  $e + 2 \leq f$ . Consequently the inflation is rigid at every member with  $e = f - 1$ , uniformly in  $f$  and with no search.*

*Proof.* Proposition 141 forces both end letters of the  $c$ -side to be  $c$ -edges: the  $c$ -side joins the  $\alpha$ - and  $\beta$ -vertices, each met by a single tile laying the flanks of its angle, and neither  $\alpha$  (flanks  $\{b, c\}$ ) nor  $\beta$  (flanks  $\{a, c\}$ ) can present an  $a$  there once the  $a$ -side is  $a^f$  and the  $b$ -side  $b^f$ . The two ends are distinct positions, because the  $p = 1$  word has  $f + (f - e) \geq 2$  letters. Hence the word carries at least two  $c$ -edges, and its  $c$ -count is  $f - e$ , giving  $f - e \geq 2$ . At  $e = f - 1$  the  $c$ -count is exactly 1, so no  $p = 1$  word exists and the standard boundary is the only one. (Machine-checked, `p1_c_count_ge_two, no_p1_word_at_e_pred`.)  $\square$

**Remark 149** (The word census, and why the search cost is what it is). Corollary 148 also explains the observed cost of the searches. Writing  $k = f - e$ , a  $p = 1$  word is fixed by placing the remaining  $k - 2$   $c$ -edges among the  $f$   $a$ -edges between the two forced end  $c$ ’s, so the number of admissible words is

$$\binom{f + k - 2}{k - 2}.$$

This is 1 when  $k = 2$  — the word is forced to be  $ca^f c$  — and grows quickly with  $k$ . The search’s top-level task count is exactly this number: the  $(4, 9)$  run, with  $k = 5$ , dispatched

$\binom{12}{3} = 220$  tasks. The table below shows the count beside the measured node counts, and the  $k = 2$  members are cheap for a structural reason rather than an accidental one.

**Proposition 150** (The orientation of a boundary  $a$ -run is monotone). [VERIFIED]

*Let a maximal run of consecutive  $a$ -edges lie on a straight side of the target, or on a side of the inflated tile, and let  $T_1, \dots, T_L$  be the tiles laying them. Each  $T_i$  presents  $\beta$  at one end of its  $a$ -edge and  $\gamma$  at the other; write its orientation as BG or GB according to which end carries  $\beta$ . Then no BG is followed by a GB: the sequence is  $GB^j BG^{L-j}$  for a single  $j$ . Consequently at most one junction of the run carries the figure  $\{3\alpha, 2\beta\}$ , and it is the transition.*

*Proof.* Side  $a$  is opposite  $\alpha$ , so the angles at its two ends are  $\beta$  and  $\gamma$ ; a tile laying  $a$  on the boundary therefore presents one of each, never  $\beta$  twice. A junction interior to the run receives the right end of one tile and the left end of the next, so the four orientation pairs give contributions  $\{\gamma, \beta\}$ ,  $\{\gamma, \gamma\}$ ,  $\{\beta, \beta\}$ ,  $\{\beta, \gamma\}$  respectively. By Proposition 23 the only straight figures are  $\{\alpha, \beta, \gamma\}$  and  $\{3\alpha, 2\beta\}$ , carrying one  $\gamma$  and none. So  $\{\gamma, \gamma\}$  — the pair BG, GB — admits no figure at all, and  $\{\beta, \beta\}$  — the pair GB, BG — forces  $\{3\alpha, 2\beta\}$ . The first statement forbids the descent  $BG \rightarrow GB$ , which is the monotonicity; the second identifies the  $\{3\alpha, 2\beta\}$  junctions with the GB, BG transitions, of which a monotone sequence has at most one. (Machine-checked: BG\_GB\_forbidden, orient\_monotone, AAB\_iff\_transition.)  $\square$

**Remark 151** (What this buys). The branching at the  $k = 2$  word falls from  $3^{f-1}$  to  $f + 1$ , and the proposition applies verbatim to an  $a$ -run on an equal side. Full discussion in [1].

**Remark 152** (Scope, and why this is not a proof of Theorem 144). The obstruction assumes the dissection is edge-to-edge, which the main paper does not, and it is silent when  $e$  and  $f$  are both odd — which includes  $(1, 3)$  and  $(3, 7)$ . Full discussion in [1].

**5.11. The forced row and the slot dichotomy: the induction  $W(k)$ .** Write  $W(k)$  for the statement: *every scale- $k$  inflation occurring inside a tiling of the  $m = 1$  target is subdivided standardly* ( $1 \leq k \leq f$ ; at  $k = f$  this is the inflation of Proposition 141 again, now as an occurrence rather than in isolation). The induction  $W(k-1) \Rightarrow W(k)$  proceeds by forcing the first row of the occurrence tile by tile; the forcing stalls at exactly one per-slot dichotomy, whose surviving branch is the *rogue* configuration that the slot-kill theorems (RogueContainment, RogueChord, RogueFan, RogueMirror) and the engine certificates are stated against. Those modules cite the row forcing as “the forced row (paper)”; this subsection is that reference. The arithmetic skeleton — the position calculus, the junction ledger, the figure forcing and the dichotomy — is machine-checked in `lean/ForcedRow.lean`, cited step by step below.

*Setup.* Let  $\Delta_k$  be a scale- $k$  occurrence: a sub-region of the hypothetical tiling, bounded by tile edges, congruent to the tile scaled by  $k$ , with the tiling respecting its boundary. Place its  $\alpha$ -corner at  $A$ , its  $\gamma$ -corner at  $C = A + kbw$  and its  $\beta$ -corner at  $B = A + kcu$ , where  $w, u, \hat{v}$  are the unit vectors along the base  $AC$ , the  $c$ -side  $AB$ , and the  $a$ -side direction  $C \rightarrow B$  respectively; the chart identity  $cu - a\hat{v} = bw$  (RogueMirror.generator\_x/\_y) relates them. Write

$$Y_i = A + (i+1)bw \quad (0 \leq i \leq k-2)$$

for the base junctions and  $X_i = A + ibw + cu$  for the row apexes; the jump identity  $Y_i + a\hat{v} = X_i$  is `ForcedRow.jump_x`.

**Lemma 153** (Boundary words at scale  $k$ ). [PROVED] *For  $k < f$  the three side words of  $\Delta_k$  are forced outright: the base reads  $b^k$  (Inflation.b\_side\_rigid), the  $a$ -side reads  $a^k$  (.a\_side\_rigid), and the  $c$ -side carries no  $b$ -edge (.c\_side\_no\_b). At the wall scale  $k = f$  the same three readings hold: the  $c$ -side by the  $\gamma$ -trap (SideNoB.side\_no\_b\_uncond), the base by the corner coupling (RogueMirror.wall\_base\_reading), and the  $a$ -side is  $a^f$  or  $c^e$  (Inflation.a\_side\_words; the  $c^e$  branch is the transverse configuration, treated separately).*

**Lemma 154** (The corner tile). [PROVED] *The tile  $P_0$  of the occurrence at  $A$  is unique and standard: it presents  $\alpha$  at  $A$  with its  $b$ -flank on the base and its  $c$ -flank on the  $c$ -side.*

*Proof.* The region's corner angle at  $A$  is  $\alpha$ , and a fill  $x\alpha + y\beta + z\gamma = \alpha$  has  $(x, y, z) = (1, 0, 0)$  (`CChord.fill_alpha`, via the multiplicity bridge `Geometry.vertex_multiplicities`): a single tile presenting  $\alpha$ , whose two edges at  $A$  are the flanks  $\{b, c\}$  of  $\alpha$ . The  $c$ -side carries no  $b$ -edge (Lemma 153), so the  $b$ -flank lies on the base.  $\square$

**Lemma 155** (The first partner, and the parallelogram). [PROVED]  *$P_0$ 's  $a$ -edge  $[bw, cu]$  is matched by a single tile  $Q_0$ , the half-turn image of  $P_0$  about the edge midpoint;  $P_0 \cup Q_0$  is the parallelogram  $(A, bw, bw + cu, cu)$ .*

*Proof.* Both endpoints of the  $a$ -edge lie on the region boundary, and its extensions leave the region (below the base line at  $bw$ ; across the  $c$ -side at  $cu$ , since  $\hat{v}$  points outward there). The far side of the edge is therefore covered by whole tile edges partitioning it exactly (`Geometry.Dissection.wall_two_s` with tile edges walls by `edge_point_not_interior` and the covering tiles contained in the region — the containment obligation), with lengths summing to  $a$ ; `Inflation.a_unsplittable` leaves a single  $a$ -edge, hence a single partner tile carrying  $\{\beta, \gamma\}$  at the two ends. The reflected assignment puts its  $\gamma$  at  $bw$ , where  $P_0$  already presents  $\gamma$ : a junction interior to the base carries at most one  $\gamma$  (`BaseBetaCorners.pi_vertex_gamma_le_one`; directly,  $2\gamma = \pi + \alpha > \pi$ ). So the direct partner is forced, and the half-turn closes the parallelogram (`ForcedRow.pgram_x`).  $\square$

**Lemma 156** (The row advance at  $Y_0$ ). [PROVED] *The tile completing the junction  $Y_0 = bw$  is  $P_1 = P_0 + bw$ .*

*Proof.* At  $Y_0$  the figure sums to  $\pi$  on the region side;  $P_0$ 's  $\gamma$  and  $Q_0$ 's  $\beta$  leave the residual  $\alpha$  (`ForcedRow.junction_ledger`), filled by a single  $\alpha$ -tile (`CChord.fill_alpha`) with flanks  $\{b, c\}$  on the rays  $w$  (the base, forward) and  $u$  ( $Q_0$ 's  $c$ -edge ray). Laying  $c$  on the base would contradict the base word  $b^k$  ( $b \neq c$ , `ForcedRow.base_letter_distinct`), so the  $b$ -flank lies on the base and the tile is the translate  $P_1$ .  $\square$

**Proposition 157** (The slot dichotomy). [PROVED] *Let  $1 \leq i \leq k - 2$  and suppose the row is standard up to  $P_i$ . At the junction  $Y_i$  the figure is  $\{\alpha, \beta, \gamma\}$ : the  $\gamma$  is  $P_i$ 's, the  $\alpha$  is the next row tile  $P_{i+1} = P_0 + (i + 1)bw$ , forced with its  $b$ -flank on the base, and the  $\beta$ -tile spans the residual between the rays  $u$  and  $\hat{v}$  at  $Y_i$  with flanks  $\{a, c\}$ , laid in one of exactly two ways:*

- (std)  *$a$  along  $\hat{v}$  — its  $a$ -edge is exactly  $P_i$ 's, the tile is the direct partner  $Q_i = Q_0 + i bw$ , and the parallelogram closes as before;*
- (rogue)  *$c$  along  $\hat{v}$  — the  $c$ -edge overruns the row apex  $X_i$  by  $\Delta = c - a = f(f - e)$  (`ForcedRow.overrun_x`; `RogueChord.delta_identities`), an overhang shorter than every tile edge, making  $X_i$  a  $T$ -vertex.*

*At  $i = 0$  the rogue branch does not exist, since both ends of the chord are blocked (Lemma 155); the slot range is exactly  $1 \leq i \leq k - 2$ , i.e.  $M := i + 1 \in [2, k - 1]$ .*

*Proof.* When the forcing reaches  $Y_i$  only  $P_i$ 's  $\gamma$  is present there. The junction figure solves  $x\alpha + y\beta + z\gamma = \pi$ , so  $(x, y, z)$  is  $(3, 2, 0)$  or  $(1, 1, 1)$  (`BaseBetaCorners.pi_vertex_figures`); the  $\gamma$  kills the first, leaving exactly one  $\alpha$  and one  $\beta$  besides it (`ForcedRow.slot_figure`). The  $\beta$ -tile cannot be base-adjacent: its flanks are  $\{a, c\}$  (`Interface.beta_has_no_b_flank`) and the base letter is  $b$ , with  $a \neq b \neq c$  (`ForcedRow.base_letter_distinct`). So the base-adjacent tile is the  $\alpha$ , laying  $b$  on the base — the translate  $P_{i+1}$ , as in Lemma 156 — and the  $\beta$ -tile occupies the residual  $\pi - \gamma - \alpha = \beta$  (`ForcedRow.slot_residual_beta`) between  $u$  and  $\hat{v}$ , laying  $\{a, c\}$  on those rays in one of the two orders (`ForcedRow.flank_product`, `.std_iff_chord_a`). In the std case the  $a$ -edge coincides with  $P_i$ 's  $a$ -edge  $[Y_i, X_i]$  (`ForcedRow.jump_x`) and the tile is the direct

partner; in the rogue case  $c > a$  produces the stated overrun (`ForcedRow.rogue_overrun_pos`). Unlike at  $i = 0$ , the chord through  $Y_i$  and  $X_i$  is blocked only at  $Y_i$  — beyond  $X_i$  it runs into untiled interior — so the far-side covering may overrun at  $X_i$  and `a_unsplittable` does not exclude the rogue branch.  $\square$

**Corollary 158** (The induction step). [PROVED] *If every slot of  $\Delta_k$  takes the std branch, the first row is the standard one —  $P_0, \dots, P_{k-1}$  and  $Q_0, \dots, Q_{k-2}$ ,  $2k-1$  tiles — and the closure of  $\Delta_k$  minus the row is the scale- $(k-1)$  occurrence anchored at  $A + cu$  (`ForcedRow.reminder_x/y`; the count  $(2k-1) + (k-1)^2 = k^2$  is `ForcedRow.row_count`). Hence  $W(k-1) \Rightarrow W(k)$  holds whenever every slot's rogue branch is refuted, and the whole tower  $\{W(k) : k \leq f\}$  closes when this happens at every scale.*

The rogue branches are refuted per slot by the kill theorems: containment for  $Me < f$  (`RogueContainment`, wired as `ForcedRow.slot_dies_by_containment`), the first slot, boundary exit and top slot (`RogueChord K1-K3`), the low slots by the fan (`RogueFan`), the top-2/top-3 slots by the corridor (`RogueMirror`), and the residual  $(M, k)$  cells by engine certificate. The wall-scale input — the base of  $\Delta_f$  reads  $b^f$  — is `RogueMirror.wall_base_reading`, so every scale  $k \leq f$  consumes Lemma 153 uniformly.

**Remark 159** (Standing hypotheses, and labels). The chain above is proved on paper over exactly the following named inputs, and nothing else. (i) *Angle sums*: the tile angles at a point of the region sum to  $2\pi$  (interior),  $\pi$  (interior to a side), or the corner angle — the `Dissection.HasAngleSums` class, now proved for the target (`Geometry.Dissection.hasAngleSums` in `lean/AngleSumDissection.lean`, with the vertex-sector crux `Tri.volume_inter_ball.vertex` in `lean/VertexSector.lean`); the passage from these real equations to the integer multiplicities consumed by the fills is proved (`Geometry.vertex_multiplicities`, with  $\alpha/\pi \notin \mathbb{Q}$  proved at every member). The region-local instances (the figure at a point of a sub-region such as  $\Delta_k$  summing to the region's angle) are the same computation read inside the region and remain the named form in which the row forcing consumes them. (ii) *Coverings*: each side of a wall segment is partitioned exactly once by whole tile edges — proved (`Geometry.Dissection.side_partition`, `wall_two_sided`); what remains of that obligation is the ordered-run extraction (`ChordInterface.FarSide's` order), which is bookkeeping on the proved partition. (iii) *Containment*: a tile with interior points in a tile-bounded region lies in it. (iv) *The boundary words* of Lemma 153, themselves standing on (i)–(iii). The tower statement is about occurrences: it does not by itself decide Hypothesis 29, which asks whether the west corner block occurs at all (Remark 146).

**5.12. The tight subfamily  $f = 2e + 1$ .** Corollary 117 leaves the two cases  $p \in \{1, 2\}$ . The case  $p = 2$  is not uniformly hard: on one infinite subfamily it is completely rigid, and the rigidity is visible in a counting budget.

**Lemma 160** (The  $\gamma$ -injection budget at  $p = 2$ ). [PROVED]  
*A side of parameter  $p$  carries  $n_a = pf$ ,  $n_b = 0$ ,  $n_c = f - pe$ , hence  $k = pf + (f - pe)$  edges and  $k - 1$  junctions, so the  $\gamma$ -injection bound  $\#a + \#b \leq k - 1$  has slack  $f - pe - 1$ . At  $p = 1$  the slack is  $f - e - 1 > 0$  for every member. At  $p = 2$  it is  $f - 2e - 1$ , which vanishes exactly when  $f = 2e + 1$ . (*gamma\_slack, p\_two\_tight\_iff.*)*

The subfamily  $f = 2e + 1$  is infinite and automatically coprime, with  $N = 3f^2 - e^2 = 11e^2 + 12e + 3$ : the members (1, 3) at  $N = 26$ , (2, 5) at 71, (3, 7) at 138, (4, 9) at 227, (5, 11) at 338. The first is closed by Theorem 53 and the second by search, so (3, 7) is the smallest tight member not already settled.

**Proposition 161** (The  $p = 2$  side is forced). [PROVED] *Let  $f = 2e + 1$  and suppose an equal side has  $p = 2$ . Then  $n_c = f - 2e = 1$ , and the side word is  $a^{2f}c$ , read*

from the base corner to the apex. Every one of its  $2f$  junctions carries exactly one  $\gamma$  and is the figure  $\{\gamma, \alpha, \beta\}$ ; each  $a$ -tile presents  $\beta$  at its left junction and  $\gamma$  at its right; the corner tile lays  $a$  on the side and  $c$  on the base.

*Proof.*  $n_c = f - 2e = 1$  and  $2f \cdot ef + f^2 = f^2(2e + 1) = f^3$ , so the profile is consistent (`p_two_single_c`). Tightness makes the  $\gamma$ -injection a bijection, so every junction carries exactly one  $\gamma$ ; a  $\pi$ -vertex carrying a  $\gamma$  is  $\{\gamma, \alpha, \beta\}$  (`pi_vertex_with_gamma`). The apex figure is  $3\alpha$  and the sides carry no  $b$ -edge (Lemma 45), so the apex-end edge is a  $c$ ; with only one  $c$  available it sits there, fixing the word. The corner figure is a single  $\beta$  with flanks  $\{a, c\}$ , and the base-corner end being an  $a$ -edge puts the  $c$  on the base. The alternation follows: the corner tile presents  $\gamma$  at  $J_1$ , and each junction having a unique  $\gamma$  forces the next  $a$ -tile to present  $\beta$  there.  $\square$

**Remark 162** (The last junction, and what is still missing). Recorded in [1]. Full discussion in [1].

**Lemma 163** (The chord at the last junction). [PROVED] *Let  $f = 2e + 1$  and let a side have  $p = 2$ . The apex  $c$ -tile presents  $\beta$  at  $J_{2f}$  with its  $a$ -ray pointing along the horizontal, and  $J_{2f}$  lies at height fraction  $2e/f$ , so the horizontal chord from  $J_{2f}$  to the opposite side has length exactly  $e(3f^2 - e^2)(f - 2e)/f = eN/f$ . At  $(3, 7)$  this is  $414/7$ .*

*Moreover the tiles meeting that chord from either side meet it in whole edges: a tile whose interior misses the chord line lies in a closed half-plane of it, so its contact is a face (`tile_contact_face`), and at a non-vertex point that face is an edge (`contact_is_edge`). Disjointness of tile interiors then makes the covering a chain starting at the end of the  $c$ -tile's  $a$ -edge, and convexity of the target keeps every endpoint within the chord.*

**Theorem 164** ( $p = 2$  is excluded on the tight subfamily). [PROVED] *Let  $f = 2e + 1$ . Then no equal side of the base- $\beta$  target at  $m = 1$  has  $p = 2$ .*

*Proof.* By Proposition 161 the side word is  $a^{2f}c$  and the last junction carries  $\{\gamma, \alpha, \beta\}$ , so by Remark 162 a tile must sit above the chord of Lemma 163 with an edge beginning at chord position  $a$ . Enumerate those placements: the edge on the chord is  $a$ ,  $b$  or  $c$ , each with two chiralities.

If the edge is  $c$  it leaves the target, since  $(a + c)f - eN = f^3 - 2ef^2 + e^3 = (2e + 1)^2 + e^3 > 0$ .

If the edge is  $a$ , the height of the tile over its  $a$ -edge is  $2\text{Area}/a$ , and the apex stands  $H(f - 2e)/f$  above the chord where  $H$  is the target height. Writing  $A$  for the target area, these are  $2A/(Nef)$  and  $2A(f - 2e)/(eNf)$ , equal precisely when  $f - 2e = 1$ . So on this subfamily the third vertex lies at exactly apex height, and the target meets that height in one point: the apex. The two chiralities then demand  $N(f - 2e - 1) = 2f^2$  and  $N(f - 2e + 1) = 4f^2$ , i.e.  $0 = 2f^2$  and  $N = 2f^2$ , the latter giving  $e = f$ . Both are impossible. (At  $(3, 7)$  the tile would be  $(21, 40, 41)$ .)

If the edge is  $b$ , one chirality overruns the chord when  $-e^3 + 3e^2 + 4e + 1 > 0$ , i.e. for  $e \leq 4$ , and for  $e \geq 5$  its third vertex falls outside on the left, since  $(f^2 - e^2)(2f - e) \geq e(3f^2 - e^2)$  reduces to  $e^3 \leq 3e^2 + 4e + 1$ , false there; the ranges are complementary. The other chirality places the third vertex at chord position  $a + (2f^2 - e^2)/2$ , exceeding the chord by  $(7e^2 + 8e + 2)/(2f) > 0$  for every  $e$ .

No placement survives.  $\square$

*Second proof, by the angle count at  $P$ .* By Proposition 161 the last junction carries  $\{\gamma, \alpha, \beta\}$  with the  $\alpha$ -filler adjacent to the apex  $c$ -tile, so the filler lays a  $b$  or a  $c$  along that tile's  $a$ -ray. Both exceed  $a = |JP|$ , so  $P$  lies interior to the filler's edge and it contributes  $\pi$  there. But  $T_1$  presents  $\gamma$  at  $P$  and  $T_2$ 's chord trace ends at  $P$ , contributing at least  $\gamma$ ; the total is at least  $2\gamma + \pi = 2\pi + \alpha > 2\pi$  (`SecondEdge.filler_excluded`). No chord estimate and no

height identity is used, and the argument needs only  $e^2 + ef < f^2$  rather than  $f = 2e + 1$  — see Corollary 131, which draws the same conclusion from the side count.  $\square$

**Remark 165** (The audit of Theorem 164 against the apex rigidity). Recorded in [1]. Full discussion in [1].

**Theorem 166** (The last-junction dichotomy). [PROVED] *Let  $(e, f)$  be any member with  $f = 2e + 1$ , and consider the last junction  $J$  of an equal side — the junction where the final  $c$ -edge begins, at distance  $f^3 - f^2$  from the base corner, hence at height fraction  $(f - 1)/f$  for every value of  $p$ . Let  $X$  be the tile angularly adjacent to the apex  $c$ -tile  $C$  on the interior side of  $C$ 's  $a$ -ray. Then  $X$  lays an  $a$ -edge along that ray, coinciding with  $C$ 's. Equivalently,  $X$  does not present  $\alpha$  at  $J$ .*

*Proof.*  $X$  lays one of its own flanks along  $C$ 's  $a$ -ray. If it lays  $c$ , the run  $c - a$  must be covered inside the chord of Lemma 163 and that window is empty. If it lays  $b$ , the window pins the covering to a single  $a$ -edge, and the height identity of Theorem 164 then places that tile's third vertex at the apex, where its remaining sides fail to be tile sides. So  $X$  lays  $a$ . The flanks of  $\alpha$  are  $b$  and  $c$ , so  $X$  is not an  $\alpha$ -vertex. (`alpha_cannot_lay_a`, `adjacent_angle_not_alpha`.)  $\square$

**Remark 167** (What the dichotomy settles, and what it leaves). Theorem 164 is the case  $p = 2$ : there the tightness of Proposition 161 forces  $X$  to be the  $\alpha$ -filler, which Theorem 166 forbids. The dichotomy is strictly more general — it is stated for every  $p$  and every vertex figure — and at  $p = 1$  it forces structure without closing the case:

in the figure  $\{\gamma, \alpha, \beta\}$  it forces the angular order  $C(\beta), \gamma, \alpha$ , so the side word cannot end  $a c$  with the  $\alpha$  adjacent to  $C$ ;

in the figure  $\{3\alpha, 2\beta\}$  it forces  $X$  to be the second  $\beta$ . That  $\beta$  shares  $C$ 's  $a$ -edge and is the reflection of  $C$  across it, so both present  $\gamma$  at the far end  $Q$  of that edge. Since  $2\gamma = \pi + \alpha < 2\pi$ , the point  $Q$  is interior. Its figure is in fact *exactly*  $\{\beta, 3\gamma\}$ : by Theorem 123 the middle apex tile  $T_2$  also meets  $Q$ , its chord trace ending there, and a straight angle from  $T_2$  would give  $2\gamma + \pi > 2\pi$ ; so  $T_2$  contributes a third  $\gamma$ , leaving  $2\pi - 3\gamma = \beta$ . The alternative  $\{2\alpha, 2\beta, 2\gamma\}$ , which carries only two  $\gamma$ 's, is excluded.

At (3, 7) the reflected tile is  $(J, Q, R)$  with  $R = (207, 100\sqrt{187}/7)$ , a genuine (21, 49, 40) placement lying inside the target, and both figures at  $Q$  are consistent. So the two surviving configurations at  $p = 1$  — the mirror partner and the  $\gamma$  — are a single phenomenon, and neither is excluded by the chord or the height identity. Closing  $p = 1$ , and with it Hypothesis 29 on the whole subfamily  $f = 2e + 1$ , requires a further mechanism — and [1] rules out the two most natural candidates, so that mechanism must be metric.

**Theorem 168** (The side carrying  $T_2$ 's  $c$ -edge is not mirrored). [PROVED] *Assume  $e^2 + ef < f^2$ . The middle apex tile  $T_2$  lays its  $b$  on one of the two inner rays at  $A$  and its  $c$  on the other. Then the equal side on the ray carrying the  $c$  has figure  $\{\gamma, \alpha, \beta\}$  at its last junction, not  $\{3\alpha, 2\beta\}$ .*

*In particular the two equal sides are not both mirrored, and at least one carries  $\{\gamma, \alpha, \beta\}$ .*

*Proof.* Suppose both do. By Remark 167 the tile  $X$  adjacent to the apex  $c$ -tile on each side is that tile's reflection across their common  $a$ -edge, and by Theorem 123 that edge lies along the chord  $C$ . So each  $X$  is the reflection of the corresponding apex tile across  $C$ , and in particular each has the reflection  $R$  of the apex  $A$  across  $C$  as a vertex. Thus *both* mirror partners meet at the single point  $R$ , which lies on the axis of symmetry, at height  $H - 2H/f$  — that is, on the chord through the second junction.

Now consider  $T_2$ . It presents  $\alpha$  at  $A$ , laying its  $b$  on one of the two inner rays and its  $c$  on the other; let  $V$  be the far endpoint of the  $c$ , at distance  $c$  from  $A$ . Its horizontal offset from the

axis is  $c \sin \frac{\alpha}{2} = ef/2 = a/2$ , exceeding the offset  $b \sin \frac{\alpha}{2} = e(f^2 - e^2)/(2f)$  of  $P'$  by  $e^3/(2f) > 0$ . Testing  $V$  against the three edges of the mirror partner on that side gives cross products

$$\frac{e^3 \sqrt{4f^2 - e^2}}{2}, \quad \frac{e^3 \sqrt{4f^2 - e^2} (f - e)(f + e)}{2f^2}, \quad \frac{e \sqrt{4f^2 - e^2} (e^2 - ef - f^2)(e^2 + ef - f^2)}{2f^2}.$$

The first two are positive since  $e < f$ . In the third,  $e^2 - ef - f^2 < 0$  always, and  $e^2 + ef - f^2 < 0$  is exactly the hypothesis, so the product is positive. All three agree in sign, so  $V$  lies *strictly inside* that mirror partner. But  $V$  is a vertex of  $T_2$ , so the interiors of  $T_2$  and of the mirror partner meet — contradicting the dissection. Only the mirror partner on the ray carrying the  $c$  enters the test, so one side suffices; the other ray is handled by reflection.  $\square$

**Corollary 169** (The figure at  $P$  on that side is fully determined). [PROVED] *On the equal side carrying  $T_2$ 's  $c$ -edge, the point  $P$  lies at distance  $b < c$  from  $A$  along that edge, hence interior to it, so  $T_2$  is the straight tile at  $P$ . The figure at  $P$  is then forced:*

$$\underbrace{\gamma}_{T_1} + \underbrace{\pi}_{T_2} + \underbrace{\beta}_Y + \underbrace{\alpha}_U = 2\pi,$$

with  $Y$  the tile below  $T_1$ 's  $a$ -edge and  $U$  a single further tile presenting  $\alpha$ . No other distribution is possible.

*Proof.*  $T_1$  presents  $\gamma$  at  $P$  and  $T_2$  contributes  $\pi$ . By Theorem 168 the figure at the last junction of that side is  $\{\gamma, \alpha, \beta\}$ , in which  $Y$  presents  $\gamma$  at  $J$  and therefore  $\beta$  at  $P$ , those being the two angles at the ends of its  $a$ -edge. That leaves  $2\pi - \gamma - \pi - \beta = \pi - \gamma - \beta = \alpha$ , so exactly one further tile meets  $P$  and it presents  $\alpha$ .  $\square$

**Proposition 170** (The two placements of  $U$ ). [PROVED] *On the side carrying  $T_2$ 's  $c$ -edge, the tile  $U$  of Corollary 169 has  $\alpha$  at  $P$ , so its flanks there are  $b$  and  $c$ , one on the ray shared with  $Y$  and one on the ray shared with  $T_2$ . Exactly two placements arise.*

- (i)  $U$  lays  $c$  towards  $Y$  and  $b$  towards  $V$ . Then its  $c$  matches  $Y$ 's exactly, so the two share the vertex  $W$ ; and its third side is an  $a$ -edge lying along the chord through the second junction, immediately to the right of the  $a$ -edge that the  $c$ -tile below  $J$  lays there. At  $(3, 7)$  the two endpoints are  $(\frac{1182}{7}, \frac{100\sqrt{187}}{7})$  and  $(\frac{1329}{7}, \frac{100\sqrt{187}}{7})$ , at distance exactly 21.
- (ii)  $U$  lays  $b$  towards  $Y$  and  $c$  towards  $V$ . Then its vertex falls strictly inside  $Y$ 's  $c$ -edge, at distance  $b$  from  $P$ , and  $U$  presents  $\gamma$  there against  $Y$ 's straight angle.

In both placements the vertex  $V$  of  $T_2$  lies interior to  $U$ 's edge, since  $|PV| = c - b < b < c$ .

**Remark 171** (What the cascade gives). Placement (i) continues the pattern of §5.9 one level down: two consecutive  $a$ -edges lie flush along the chord through the second junction, laid by the  $c$ -tile below  $J$  and by  $U$ . That the two vertices land at the same height is the law of sines for the tile,  $c \sin \beta = b \sin \gamma$ , not a coincidence of the member.

The two placements are cleanly distinguished by that chord: in (i)  $U$  lies *flush* with it, its third side lying in the chord; in (ii)  $U$  *straddles* it, with vertices at heights  $+180\sqrt{187}/343$  and  $-9\sqrt{187}/14$  relative to it at  $(3, 7)$ . Placement (ii) puts a  $\gamma$  against a straight angle at a point interior to  $Y$ 's  $c$ -edge, which by Lemma 136 no angle count can exclude; and we have not excluded it geometrically either. Neither placement is contradictory as far as we can determine, so  $U$  does not close the case.

The two placements are separated by a further fact, obtained by the semigroup argument of Proposition 127. In placement (ii) the vertex  $X_1$  sits at distance  $b$  along  $Y$ 's  $c$ -edge  $[P, W]$ , leaving a residual  $[X_1, W]$  of length  $c - b = e^2$ . At  $(3, 7)$  that is 9, below every tile side, hence a gap of  $\langle a, b, c \rangle$ ; and the covering on  $U$ 's side of that edge cannot overrun  $X_1$ , which would place

two tiles under  $U$ . It must therefore overrun  $W$ , so a  $T$ -junction at  $W$  is *forced*, and the figure there is determined:

$$\underbrace{\gamma}_Z + \underbrace{\alpha}_Y + \pi + \beta = 2\pi.$$

Placement (i), by contrast, shares  $W$  with  $Y$  edge-to-edge and forces no such  $T$ -junction. So the two placements are distinguished by exactly the phenomenon Corollary 139 is about. This is consistent, not contradictory:  $p = 1$  requires a  $T$ -junction somewhere, and placement (ii) supplies one. The argument for  $e^2$  being a gap needs  $e^2 < \min(a, b)$ , which holds at  $(3, 7)$  but fails at several members with  $f$  close to  $e$ , so it is not family-wide.

One structural fact about the boundary of the forced region is worth recording, since it says where an argument has to engage. Every tile forced so far uses one of only *two* edge directions: the side direction, at  $\beta$  below the horizontal, and the direction at  $\alpha + \beta$  below it. Consequently every forced vertex sits at a *rational* multiple of the chord spacing  $\delta = H/f$ ; at  $(3, 7)$  the heights are  $0, 1, c/b, 2, (b+c)/c, (b+c)/b$ , with denominators dividing  $\text{lcm}(b, c)$ . This cannot persist, since the vertical projection of the  $\mathbb{Z}$ -module generated by all admissible edge vectors is dense ([1]). Of the four ways to continue past  $X_1$  in placement (ii), exactly two stay inside the two-direction system and two introduce a third direction. The rationality breaks precisely at those two, and that is the boundary of the region the present arguments control. What is gained is that the configuration on the unmirrored side is now determined down to this single binary choice, with every vertex located exactly.

**Proposition 172** (The figure at  $P'$  on the mirrored side). [PROVED] *On the equal side carrying  $T_2$ 's  $b$ -edge, the figure at  $P'$  is exactly  $\{\beta, 3\gamma\}$ , whichever straight figure that side carries at its last junction.*

*Proof.* There  $T_2$  has its  $b$ -endpoint at  $P'$ , so by Corollary 124 it presents  $\gamma$ ; the apex tile  $T_3$  presents  $\gamma$  as well. If the side carries  $\{3\alpha, 2\beta\}$ , its adjacent tile is the mirror partner and presents  $\gamma$  at  $P'$ , giving  $3\gamma$  and a residual  $2\pi - 3\gamma = \beta$ . If it carries  $\{\gamma, \alpha, \beta\}$ , the adjacent tile presents  $\beta$  at  $P'$ , giving  $2\gamma + \beta$  and a residual  $2\pi - 2\gamma - \beta = \gamma$ . Either way the multiset is  $\{\beta, 3\gamma\}$ .  $\square$

**Remark 173** (Both apex vertices are now determined). Corollary 169 and Proposition 172 together determine the vertex figures at  $P$  and  $P'$  completely: on the side carrying  $T_2$ 's  $c$  the figure is  $\gamma + \pi + \beta + \alpha$  with  $T_2$  straight, and on the side carrying its  $b$  it is  $\{\beta, 3\gamma\}$ . The Case A / Case B alternative of Remark 125 is therefore not a branching of the argument but a *labelling* of the two equal sides, and the labelling is decided by which ray  $T_2$  puts its  $c$  on.

**Remark 174** (The first coupling of the two sides). Every previous constraint in this branch has been local to one equal side; Theorem 168 is the first that forbids a combination across both. Its mechanism is that the two mirror partners are forced to share the vertex  $R$  on the axis, which leaves the middle apex tile nowhere to put its far vertex.

The sign condition governing the third cross product is not a second, independent appearance of  $e^2 + ef < f^2$ , and it would be misleading to present it as one. The quadratic factor is a difference of squares,

$$(e^2 - ef - f^2)(e^2 + ef - f^2) = (f^2 - e^2)^2 - (ef)^2 = b^2 - a^2,$$

so the third cross product is  $e\sqrt{4f^2 - e^2}(b^2 - a^2)/(2f^2)$  and is positive precisely when  $b > a$ . The containment test *is* the edge-length comparison, in disguise. For the same reason the first two cross products stand in the ratio  $b : c$ .

**Remark 175** (Scope, and where  $e \geq 2$  is easier). Recorded in [1]. Full discussion in [1].

**5.13. Close pairs: the extra base columns.** The close-pair regime  $f^2 \leq 2ef + e^2$  admits base decompositions the separated members do not. This subsection puts them in closed form and gives the sharp criterion, which is Hall's condition rather than the counting bound.

**Remark 176** (Close pairs: what the unconditional filters leave). At a close pair ( $f^2 \leq 2ef + e^2$ ) Lemma 96 no longer applies and the base admits extra columns, all with  $y = e + kf$  for some  $k \geq 1$ . Three unconditional filters apply: the  $\gamma$ -trap removes  $z = 0$ ; Theorem 100 removes  $x = 0$ ; and the corner parallelogram, which keeps  $b$ -edges out of the first two and last two positions, removes any column with  $x + z \leq 3$  in a word of at least four edges (`cornerpara_filter`). Over the range  $e \leq 9$ ,  $f \leq 15$  these leave 54 of the 62 extra columns standing, the first being  $(3, 7, 2)$  at  $(3, 4)$ ; every survivor has  $x < f$ . Closing them requires an argument bounding the number of base  $b$ -edges, and no such bound follows from size: writing  $y = e + kf$ , the size constraint  $yb \leq e(3f^2 - e^2)$  reads  $k \leq 2ef/(f^2 - e^2)$ , whose vanishing is precisely the separation condition  $f^2 > 2ef + e^2$ . The extra columns at close pairs are permitted by size by construction, so a bound on  $y$  there must be geometric.

Close pairs are not a finite exception: the condition  $f \leq (1 + \sqrt{2})e$  admits infinitely many coprime pairs, and among the smallest are  $(5, 6)$  and  $(4, 7)$ , that is  $N = 83$  and  $N = 131$ . Member by member certification therefore cannot settle this family.

**Remark 177** (The extra base columns, in closed form). The extra columns at a close pair all have  $y = e + kf$  with  $k \geq 1$  (Remark 176), and the obstruction is that no bound on  $k$  follows from size. Substituting  $y = e + kf$  into the base equation collapses it: the difference is  $f$  times a relation in  $x$  and  $z$  alone,

$$x(ef) + (e + kf)(f^2 - e^2) + zf^2 - e(3f^2 - e^2) = f(xe + zf - (2ef - kb)), \quad b = f^2 - e^2,$$

an identity (`base_column_reduction`). So for each  $k$  the surviving columns are the solutions of the two-variable linear Diophantine equation

$$xe + zf = 2ef - kb,$$

not of a three-variable search. Verified against brute enumeration on every close pair with  $e \leq 25$ ,  $f \leq 59$ : 655 surviving columns, no exception.

Two consequences. At  $k = 1$  the column is pinned,  $(x, y, z) = (e, e + f, 2e - f)$ , so it occurs exactly when  $f \leq 2e - 1$  (`base_column_k_one`). And the equation bounds  $k$ . Using only  $x, z \geq 1$  gives  $kb \leq 2ef - e - f$  (`base_column_k_bound`). The corner parallelogram also forces  $x + z \geq 4$ , and since  $e < f$  the minimum of  $xe + zf$  subject to  $x, z \geq 1$ ,  $x + z \geq 4$  sits at  $(x, z) = (3, 1)$ , so

$$kb \leq 2ef - 3e - f \quad (\text{k\_bound\_sharp}),$$

better by  $2e$ . Checked against brute enumeration on every close pair with  $e \leq 25$ ,  $f \leq 59$ : no violation, and the bound is *attained* — equal to the largest surviving  $k$  — at 186 of the 198 members carrying survivors. There is exactly one surviving column for each  $k$  in range. For fixed  $k$  the solutions form a chain  $x = x_0 + tf$ ,  $z = z_0 - te$ , so there are  $\lfloor (z_0 - 1)/e \rfloor + 1$  with  $z \geq 1$ , and exactly one when  $z_0 \leq e$ ; writing  $ke = qf + r$  this is  $e(1 + q) \leq kf$ , and since  $e < f$  we have  $ke/f < k$ , so  $q \leq k - 1$  and  $e(1 + q) \leq ek < fk$  (`one_column_per_k`; neither coprimality nor the close-pair hypothesis is used). Hence the surviving set at a member is described completely by

$$k = 1, \dots, \left\lfloor \frac{2ef - 3e - f}{b} \right\rfloor.$$

The description can be made exact. Reducing  $xe + zf = 2ef - kb$  modulo  $f$  gives  $e(x - ke) = f(2e - kf - z)$ , so  $f \mid e(x - ke)$  and, by coprimality,  $f \mid x - ke$  (`column_x_congruence`). For each  $k$  the admissible  $x$  therefore form a single residue class modulo  $f$ , and the surviving columns of that  $k$  are the members of that class with  $z = (2ef - kb - xe)/f \geq 1$  and  $x + z \geq 4$ . Over every

close pair with  $e \leq 25$ ,  $f \leq 59$  this criterion reproduces the surviving  $k$ -set at all 198 members carrying survivors, with no exception. The 12 members at which the bound  $kb \leq 2ef - 3e - f$  is not attained are exactly those where  $xe + zf = 2ef - kb$  has no solution with  $x, z \geq 1$  — a Frobenius condition on  $\langle e, f \rangle$ , not a further geometric filter; at  $(e, f) = (15, 23)$  and  $k = 2$ , for instance, the residue 44 is a gap of  $\langle 15, 23 \rangle$ .

All of this is a description of the surviving set, not the required  $k = 0$ . What it buys is that the residual question is one linear congruence per  $k$ , with  $k$  in an explicit finite range, in place of an unstructured three-variable family.

**Proposition 178** (The extra base columns at a close pair). [PROVED] *Let  $\gcd(e, f) = 1$  and  $1 \leq e < f$ , let  $b = f^2 - e^2$ , and let  $(x, y, z)$  be a base column with  $y = e + kf$ ,  $k \geq 1$ , satisfying  $x, z \geq 1$  and  $x + z \geq 4$ . Then*

- (i) *the base equation is equivalent to  $xe + zf = 2ef - kb$ ;*
- (ii)  *$f \mid x - ke$ , so  $x$  lies in a single residue class modulo  $f$ ;*
- (iii)  *$kb \leq 2ef - 3e - f$ ;*
- (iv) *for each  $k$  the pair  $(x, z)$  is unique.*

*Consequently the extra columns at a close pair are exactly one for each  $k = 1, \dots, \lfloor (2ef - 3e - f)/b \rfloor$  that admits a solution, with  $x$  determined modulo  $f$  and  $z$  determined by  $x$ . (`close_pair_column`, `close_pair_column_unique`, `one_column_per_k`.)*

**Proposition 179** (Sharp criterion for an extra base column). [VERIFIED] *Let  $\gcd(e, f) = 1$ ,  $1 \leq e < f$ ,  $b = f^2 - e^2$ . For  $k \geq 1$  let  $x_k$  be the least positive integer with  $x_k \equiv ke \pmod{f}$ , and write  $x_k = ke - m_k f$ , so that  $1 \leq x_k \leq f$ . Put*

$$g(k) = (m_k + 2)e - 1 - kf, \quad z_k = \frac{2ef - kb - x_k e}{f}.$$

*Then:*

- (i) *an extra base column with parameter  $k$  exists if and only if  $g(k) \geq 0$  and  $x_k + z_k \geq 4$ , and it is then  $(x_k, e + kf, z_k)$ ;*
- (ii)  *$g(k + 1) \leq g(k) - (f - e)$ , so  $g$  is strictly decreasing;*
- (iii)  *$g(1) = 2e - 1 - f$ . Hence no member with  $f \geq 2e$  carries an extra column at any  $k$ , and in general the admissible  $k$  form an initial segment  $1, \dots, K$  with  $K \leq 1 + \lfloor (2e - 1 - f)/(f - e) \rfloor$ .*

Machine-checked in `lean/Frontier.lean` as `column_criterion_identity`, `column_criterion`, `k_one_forces_f_le_two_e_sub_one`, `residue_step` and `g_strict_drop`, axiom-free beyond `propept`, `Classical.choice` and `Quot.sound`.

**Remark 180** (What the criterion settles, and what it leaves). The whole  $k \geq 1$  question is now confined to  $e < f < 2e$ , a strict sub-range of the close-pair range  $f \leq (1 + \sqrt{2})e$ . Of the 17296 close pairs with  $e \leq 200$ , 5065 have  $f \geq 2e$  and so carry no extra column at all: for those the base column is forced to  $k = 0$  outright. At  $e = 1$  the condition  $f \leq 2e - 1 = 1$  is incompatible with  $f > e$ , so no  $(1, f)$  member has an extra column, which is the arithmetic content of Theorem 20 recovered independently.

The criterion is exact, not merely necessary: it was checked against brute enumeration of the base equation on all 17296 close pairs with  $e \leq 200$ , covering 67218 columns, with no disagreement, and the initial-segment shape and the bound on  $K$  hold in every case. The corner-parallelogram condition  $x_k + z_k \geq 4$  is not redundant — it rejects an otherwise admissible  $k$  in 119 of the pairs with  $e \leq 120$  — so it is carried explicitly in (i).

What this does *not* do is exclude the surviving columns: 21837 of them exist as solutions of the base equation, and no arithmetic condition can remove them, since they are genuine

solutions. The arithmetic layer of the close-pair problem is now exactly described; the exclusion must come from the interior, as Remark 176 already recorded.

**5.14. Assembly.** Fix a hypothetical tiling. Its base walk is one of the three; each funnels as above to a configuration killed by the  $T_{\text{mid}}$  corner, the pentagon lemma, or the walk arithmetic (A1-type shielding, the  $a$ -prefix forcing with P6-type end constraints, the  $\beta$ -wedge). No case survives; no tiling exists.  $\square$

**5.15. Scope, certificates, verification.** Every arithmetic component is kernel-checked (`lean/BAAdjacency`, `Rigidity`, `W2Core`, `MidTriangle`, `SurplusLattice`, `GeneralPillars`, `MasterLemmas`, `Pentagon`; core toolchain, axiom-clean). For Lemma 41 the file `Pentagon.lean` proves, general in  $(e, f)$  and axiom-free, both facts the wedge analysis consumes: that  $(0, \min(a, b))$  is a gap of the numerical semigroup  $\langle a, b, c \rangle$ , and that the stub  $e^2 \bmod b$  lies in that gap *immediately* — splitting on  $e^2 < b$ : below, the stub is  $e^2$ , smaller than  $a = ef$  since  $e < f$  and smaller than  $b$  by hypothesis; above,  $b \leq e^2 \leq ef$  bounds the remainder by both. The second initial stub  $|a - b|$  can exceed  $\min(a, b)$  (e.g.  $5 > 3$  at  $(1, 3)$ ), so it escapes that gap; it is settled instead by Lemma 42 below, which removes the iteration from the argument entirely. Both stubs are therefore machine-checked, general in  $(e, f)$  and axiom-free. Certified seeded refutations:  $(1, 3)$ ,  $(1, 4)$  at  $j = 1, 2$ ,  $(2, 3)$  side-break,  $(2, 3)$  pentagon. Engine exhaustions independently confirm  $N = 11, 23, 26, 47, 59, 66, 71, 107$ . The exposition above is the assembled argument; an extended version with expanded strip/recursion diagrams is in preparation and does not affect the claims.

## 6. REALIZABILITY: THE COLLAR INDUCTION AND THE FIRST COMPLETE MEMBER

Theorem 31 is the exclusion side of the branch: no member has an instance at  $m = 1$ . This section supplies the construction side. For the member  $(e, f) = (1, 2)$  — tile  $(2, 3, 4)$ , unit target  $\Delta_1$  with base  $Y_1 = 11$ , sides  $X_1 = 8$ ,  $N_1 = 11$  — it settles *every* remaining instance, making  $(1, 2)$  the first member of any incommensurable-angles branch whose instance spectrum is completely determined.

Write  $\Delta_m$  for the instance target at scale  $m$  (base  $mY_1$ , sides  $mX_1$ , count  $N = m^2 N_1$ ), and subdivide  $\Delta_m$  by the standard  $m$ -fold edge subdivision into  $m^2$  cells, each a translate of  $\Delta_1$  (“up-cells”) or of its half-turn (“down-cells”). A cell is *not* individually tileable — that is exactly the  $m = 1$  exclusion — so all constructions below use two larger blocks whose interfaces are full straight segments.

**Lemma 181** (Unit parallelogram). [PROVED] *The parallelogram  $P_1$  with vertices  $(0, 0)$ ,  $(Y_1, 0)$ ,  $(Y_1 + X_1 \cos \beta, X_1 \sin \beta)$ ,  $(X_1 \cos \beta, X_1 \sin \beta)$  — one up-cell and one down-cell, glued along a full cell edge — admits a tiling by  $2N_1 = 22$  congruent copies of the tile. The tiling was produced by the exhaustive engine (125 nodes) and is kernel-verified in `lean/PgramTiling22.lean`: exact  $\mathbb{Z}[\sqrt{15}]$  coordinates scaled by 4, with congruence of all 22 tiles, containment in the closed parallelogram (4 half-planes), an explicit separating edge-line for each of the 231 tile pairs, and the exact area identity  $\sum 2A_i = 528\sqrt{15}$  — no axioms, no imports.*

**Lemma 182** (Collar decomposition). [VERIFIED] *For  $m \geq 3$ , let  $\Delta_{m-2}^{\text{apex}} \subset \Delta_m$  be the scale- $(m-2)$  copy anchored at the apex. The closure of  $\Delta_m \setminus \Delta_{m-2}^{\text{apex}}$  — the collar, the bottom two cell rows,  $4(m-1)$  cells — decomposes into*

$$(m-2) \text{ parallelogram columns} + \text{one scale-2 corner triangle},$$

where the  $j$ -th column ( $0 \leq j \leq m-3$ ) is the sheared strip with base  $[jY_1, (j+1)Y_1]$  and top  $[(j+1)Y_1, (j+2)Y_1]$  at height  $2X_1 \sin \beta$  — the union of two translates of  $P_1$  stacked flush — and the corner triangle is the scale-2 up-triangle on the base segment  $[(m-2)Y_1, mY_1]$ . Every

interface (column-to-column, column-to-corner, collar-to- $\Delta_{m-2}^{\text{apex}}$ ) is a full straight segment, so the pieces tile independently. In tiles:  $44(m-2)+44 = 44(m-1)$ , and  $N_1 m^2 = N_1(m-2)^2 + 44(m-1)$  exactly.

*Proof.* The subdivision arithmetic is immediate (and is kernel-checked, shape-free, as `collar_cells` in `lean/Collar.lean`); the only geometric content is flushness. The  $j$ -th column's lateral sides run parallel to the left leg (direction  $(X_1 \cos \beta, X_1 \sin \beta)$ , slope  $\tan \beta$ ); its right side, from  $((j+1)Y_1, 0)$  to  $((j+2)Y_1, 2X_1 \sin \beta)$ , coincides with the left side of column  $j+1$ , and for  $j = m-3$  with the left edge of the corner triangle. The corner triangle's right edge lies on the right leg of  $\Delta_m$ ; the collar's top edge, from  $(Y_1, 2X_1 \sin \beta)$  to  $((m-1)Y_1, 2X_1 \sin \beta)$ , is exactly the base of  $\Delta_{m-2}^{\text{apex}}$  ( $X_1 \cos \beta = Y_1/2$  by the isosceles altitude). Each column splits into two translates of  $P_1$  along the full segment at height  $X_1 \sin \beta$ .  $\square$

**Theorem 183** (Realizability, member  $(1, 2)$ ). [CONJECTURE]  $N = 11m^2$  is realizable for every  $m \geq 2$ . With Theorem 31 the instance spectrum of the member  $(1, 2)$  is completely determined:

$$\Delta_m \text{ is tileable} \iff m \neq 1.$$

*Proof.* Bases:  $m = 2$  and  $m = 3$  are the kernel-verified certificates `Tiling44.lean` and `Tiling99.lean`. Step: if  $\Delta_m$  is tileable ( $m \geq 2$ ) then so is  $\Delta_{m+2}$ , by Lemma 182 applied at scale  $m+2$ : tile  $\Delta_m^{\text{apex}}$  by hypothesis, each of the  $m$  columns by two copies of Lemma 181, the corner by the  $m = 2$  certificate. The two-step induction from bases 2, 3 reaching all  $m \geq 2$  is `two_step` in `lean/Collar.lean` (kernel-checked); the exclusion at  $m = 1$  is Theorem 31.  $\square$

**Remark 184** (Relation to Zhang's constructions). Zhang's construction paper [7] builds tiling families for the six sporadic targets whose tile has a  $2\pi/3$  angle (with a  $\pi/3$  variant), using ideal trapezoids as macro-tiles; the families there carry lower bounds and residue classes in  $m$  (e.g.  $m \geq 9$ , or  $m \in 5\mathbb{N}_0 + 7\mathbb{N}_0$ ), leaving the small members undecided. The branch treated here,  $3\alpha + 2\beta = \pi$ , is disjoint from those six cases ( $\gamma = 2\pi/3$  forces  $\beta = 0$  in this branch), and no construction family for it appears in the literature known to us; the 44-, 99- and 22-tilings above were found by our engine. The collar plays the role of Zhang's trapezoids — a macro-tile calculus — but the induction it drives has no lower bound and no residue class: every admissible  $m$  is realized. Member  $(1, 2)$  is thus, to our knowledge, the first incommensurable-angles (target, tile) pair with no undecided instance.

**Theorem 185** (The family is not uniform). [CONJECTURE] For the member  $(e, f) = (1, 3)$  — tile  $(3, 8, 9)$ ,  $N_1 = 26$  — the scale-2 target admits no tiling:  $N = 104$  is not realized. (Exhaustive search, 43 541 916 nodes.) Hence the conclusion of Theorem 183 does not extend along the family:  $\Delta_2$  exists for  $(1, 2)$  and does not exist for  $(1, 3)$ .

This is the reason every attempt to build the seeds at  $f = 3$  fails, and it is worth listing them because each is now explained rather than merely unsuccessful: the cevian piece  $C_2$  (68 tiles), the symmetric middle  $M_2$  (32 tiles), the maximal interior of the  $a \times c$  lattice (34 tiles left to place) and of the  $b$ -glued  $Q_c$  lattice (34 likewise) all return NO TILING, and a seeding by unit parallelograms is impossible by counting, since  $\Delta_M$  has  $M(M+1)/2$  up-cells against  $M(M-1)/2$  down-cells and each parallelogram consumes one of each, leaving  $M$  untileable cells. There was nothing to construct.

Two consequences. First, Remark 35 acquires content:  $(1, 2)$  is the unique member of its family with  $\beta < \pi/3$ , and it is also the only one known to realize its scale-2 target. Second, the collar of Theorem 183 has no base at  $m = 2$  for  $(1, 3)$ ; its bases must come from  $m \geq 3$ . Since  $\Delta_4$  would require  $\Delta_1 + \Delta_3$  and  $\Delta_5$  would require  $\Delta_2 + \Delta_3$ , both excluded, the set reachable from  $\Delta_3$  alone is exactly the multiples of 3. So if  $\Delta_3$  tiles, the spectrum of  $(1, 3)$  is plausibly  $N = 26m^2$  with  $3 \mid m$ , a shape quite unlike  $\{m \geq 2\}$ ; and if  $\Delta_3$  does not tile, the member

contributes nothing. We state this as a question rather than a conjecture, and the search is running.

The failure of the seeds to travel is not an accident of the search; it is visible in the certified tilings themselves.

**Proposition 186** (The relation lattice, and the interface floor). [PROVED] *For the tile  $(a, b, c) = (ef, f^2 - e^2, f^2)$  with  $\gcd(e, f) = 1$ , the lattice  $\Lambda(e, f) = \{n \in \mathbb{Z}^3 : n_a a + n_b b + n_c c = 0\}$  is free of rank two, generated by*

$$v_1 = (f, 0, -e) \quad (f \text{ a's} = e \text{ c's}), \quad v_2 = (e, f, -f) \quad (e \text{ a's} + f \text{ b's} = f \text{ c's}).$$

*Every  $n \in \Lambda$  has  $f \mid n_b$ , and every  $n$  with  $n_b = 0$  lies in  $\mathbb{Z}v_1$ . Consequently a nonzero relation has one-sided length at least  $f \min(a, b)$ , and every maximal straight interface of any tiling that is shorter than  $f \min(a, b)$  carries the same edge multiset on both sides. (`rel_v1`, `rel_v2`, `rel_b_mult`, `rel_param` in `lean/Primitives.lean`.)*

*Proof.* A maximal straight segment of the tiling is covered on each side by whole tile edges, and the two words have equal total length, so their multiset difference lies in  $\Lambda$ . Reducing the defining equation mod  $f$  gives  $f \mid n_b e^2$ , hence  $f \mid n_b$ ; writing  $n_b = fy$  and cancelling  $f$ , then reducing mod  $e$ , gives  $n_c = -fy + et$  and back-substitution  $n_a = ey - ft$ , i.e.  $n = yv_2 - tv_1$ . For the floor: if  $n_b \neq 0$  one side carries at least  $f$   $b$ -edges (length  $\geq fb$ ); if  $n_b = 0$  and  $n = sv_1 \neq 0$  one side carries at least  $f$   $a$ -edges (length  $\geq fa$ ).  $\square$

**Theorem 187** (The certified seeds are  $(1, 2)$ -degenerate). [CONJECTURE] *Sweep, exactly over  $\mathbb{Q}(\sqrt{15})$ , all eight kernel-certified tilings of the  $(1, 2)$  family — `Tiling44`, `N44B`, `N44C`, `Tiling99`, `PgramTiling22`, `CevianTiling28`, `CevianTiling63`, `Tiling28`, `Tiling77` (`code/interface_census.py`). The two-sided words of every maximal straight interface either agree as multisets or differ by one of the relations*

$$c = 2a, \quad 2b = 3a, \quad 2b = a + c, \quad 3c = 4b, \quad a + 2b = 2c,$$

*and each of these, evaluated at  $(a, b, c) = (ef, f^2 - e^2, f^2)$ , factors as a unit times  $(f - 2e)$  times a quantity positive on  $0 < e < f$ . Every nontrivial interface equation used by every certified seed therefore vanishes exactly on  $f = 2e$ : the seed complexes close on the member  $(1, 2)$  and on no other member, so no relabelling of any certified seed transfers along the family — in structural agreement with Theorem 185. What a construction at a general member must use instead is the member's own lattice  $\Lambda(e, f)$ , whose shortest nonzero relations have length  $ef^2$  and  $f^3$ ; the two-lattice architecture of Theorem 192 is the one known construction that manages with the universal relations alone, and it is exactly the one that generalises.*

**Proposition 188** (Cevian reduction: two tiles and an atom). [PROVED] *In  $\Delta_m$  let  $P_e$  be the base point at distance  $ecm$  from the east corner. The cevian from the apex to  $P_e$  has length  $fbm$  and cuts  $\Delta_m$  into the triangle  $(ef^2m, fbm, f^3m) = fm \cdot (a, b, c)$  — the tile at scale  $fm$ , tileable by  $(fm)^2$  copies for every  $m$  — and the west piece  $W_m$  with sides  $(e(2f^2 - e^2)m, f^3m, fbm)$  and angles  $(\beta, 2\alpha, \alpha + \beta)$ . The wall from  $(fam, 0)$  to the apex cuts  $W_m$  in turn into the tile at scale  $fm$  and the atom  $A_m$ : the isosceles triangle with legs  $fbm$ , base  $ebm$ , apex angle  $\alpha$ , carrying  $bm^2$  tiles. Hence*

$$\Delta_m = fm \cdot T \sqcup fm \cdot T \sqcup A_m, \quad m^2 N_1 = 2(fm)^2 + bm^2$$

*(`two_tiles_plus_atom` in `lean/Primitives.lean`), and realizability at multiplier  $m$  follows from a tiling of  $A_m$  alone. The certified instances of  $W_m$  at  $(1, 2)$  are `CevianTiling28` ( $W_2$ ) and `CevianTiling63` ( $W_3$ ), and the 28-tile triangle  $(12, 14, 16)$  of `Tiling28` is the same shape as  $W_2$ . No certified tiling of any  $A_m$  is known: both cevian certificates cross the internal wall (3 resp. 7 straddlers, exact sweep), the unit atom  $A_1$  admits no tiling at any  $e = 1$  member — its*

base word is forced to the single letter  $b^e$  by the residue census (`atom_base_forced`), and at  $e = 1$  the single  $b$ -edge's  $\gamma$ -end overflows the base corner opening  $\alpha + \beta$  (`corner_fill_unique`) — and  $A_1$  admits none at  $(2, 3)$  by the forced overlap of the retraction commit. The atom spectrum  $\{m : A_m \text{ tiles}\}$  is open at every member, including  $(1, 2)$ ; a member-uniform tiling of some  $A_{m_0}$  would make every family's realizable set nonempty with primitives bounded by  $m_0$ , and is the sharpest open construction question of the branch.

The cevian piece, unlike the atom, has seeds — and at  $e = 1$  its own addition law, which closes its spectrum at  $(1, 2)$  completely.

**Lemma 189** (The  $W$ -parallelogram,  $e = 1$ ). [PROVED] *Let  $e = 1$ . The parallelogram spanned by  $fb$  (cevian direction) and  $2f^2 - 1$  (base direction), included angle  $\alpha + \beta$ , admits a tiling by  $2(2f^2 - 1)$  tiles. Explicitly: the  $c$ -glued two-tile brick is the parallelogram with sides  $a, b$  and angle  $\alpha + \beta$ ; the base side is  $Y = 2f^2 - 1 = f \cdot a + b$  (recall  $a = f$ ,  $b = f^2 - 1$ ), so  $f$  slabs of height  $a$  (each  $fb/b = f$  bricks laid  $b$ -along-the-cevian-direction) and one slab of height  $b$  ( $fb/a = b$  bricks laid  $a$ -along) fill it with  $f^2 + b = 2f^2 - 1$  bricks (`w_slab_counts` in `lean/Primitives.lean`). Every interface is a full straight segment.*

**Theorem 190** ( $W$ -addition, and the second complete pair). [CONJECTURE]  
*Let  $e = 1$  and write  $S_W = \{m : W_m \text{ admits a tiling}\}$ . Then  $S_W$  is closed under addition:  $W_{m_1+m_2}$  decomposes as  $W_{m_1}$  anchored at the  $\beta$ -corner, the parallelogram of Lemma 189 at scale  $m_1 \times m_2$ , and an upright translate of  $W_{m_2}$ , all interfaces full straight segments. At  $(e, f) = (1, 2)$  the kernel-certified seeds `CevianTiling28` ( $W_2$ ) and `CevianTiling63` ( $W_3$ ) therefore give  $W_m$  for every  $m \geq 2$ ; and  $W_1$  admits no tiling unconditionally, since a 7-tiling of  $W_1$  together with the east  $f$ -scaled tile would tile  $\Delta_1 = (8, 8, 11)$ , refuted by the certified  $N = 11$  exhaustion. Hence the instance spectrum of the pair (scalene  $(2\alpha, \beta, \alpha + \beta)$  target, tile  $(2, 3, 4)$ ) is completely determined:*

$W_m$  is tileable  $\iff m \neq 1$ , realizing  $N = 7m^2 \in \{28, 63, 112, 175, 252, \dots\}$  for every  $m \geq 2$ .

*This is the second (target, tile) pair with no undecided instance, over the same tile as Theorem 183, and it contributes the count  $175 = 7 \cdot 5^2$ , reachable neither from 28 nor 63 by the subdivision ladder.*

*Proof.* The decomposition is the barycentric cut of the band below the apex-anchored copy, exactly as in Theorem 197, transported to the scalene shape: in the affine frame  $u =$  (west side),  $v =$  (base) at the  $\beta$ -corner,  $W_s = \{xu + yv : x, y \geq 0, x + y \leq s\}$ , and the cut  $x = m_1$  splits the band  $m_1 \leq x + y \leq s$  into the parallelogram spanned by  $m_1(u - v)$  and  $m_2v$  — sides  $m_1 fb$  and  $m_2(2f^2 - 1)$ , angle  $\alpha + \beta$ , tiled by Lemma 189 — and the upright translate  $\{x \geq m_1\}$  of  $W_{m_2}$ . Counts:  $(m_1 + m_2)^2(2f^2 - 1) = m_1^2(2f^2 - 1) + 2m_1m_2(2f^2 - 1) + m_2^2(2f^2 - 1)$  (`add_count`); the anchors are `w_counts`. Sums of 2s and 3s reach every  $m \geq 2$  (`two_step`, `lean/Collar.lean`).  $\square$

**Remark 191** (The program for the other members). Lemma 182 is stated for  $(1, 2)$  but its geometry is member-free: the subdivision, the column shear, and the flushness argument use only the isosceles altitude identity, valid for every  $(e, f)$ . For any member, realizability of *all*  $m \geq 2$  therefore reduces to three finite seeds: a tiling of  $\Delta_2$ , of  $\Delta_3$ , and of the unit parallelogram  $P_1$ . The  $m$ -direction is closed once and for all; what remains per member is the three seed searches (or a uniform seed construction in  $(e, f)$ , which we have not yet found). The seeds grow quickly — for  $(1, 3)$  they are 104, 234 and 52 tiles — so this is engine territory; the collar lemma converts each trio of successes into a complete member. The parallelogram seed of  $(1, 3)$  is already in hand: the tile  $(3, 8, 9)$  tiles its unit parallelogram into 52 copies, found by the

engine and kernel-verified in `lean/PgramTiling52.lean` (exact  $\mathbb{Z}[\sqrt{35}]$ , 1326 separating lines, zero axioms).

For the family  $e = 1$  the seed is not needed at all: the parallelogram can be written down.

**Theorem 192** (The unit parallelogram,  $e = 1$ ). [PROVED] *Let  $e = 1$  and  $f \geq 2$ , so  $(a, b, c) = (f, f^2 - 1, f^2)$  and  $N_1 = 3f^2 - 1$ . Then  $P_1$  admits an explicit tiling by  $2N_1$  copies of the tile.*

*Proof.* Work in the affine basis  $u = (1, 0)$ ,  $w = (\cos \beta, \sin \beta)$ , in which  $P_1$  is the box  $[0, Y_1] \times [0, X_1]$  with  $Y_1 = 3f^2 - 1$  and  $X_1 = f^3 = fc$ ; recall  $X_1 \cos \beta = Y_1/2$ . The  $a \times c$  cell of this basis is the parallelogram obtained by gluing two tiles along their  $b$ -edge (sides  $a, c$ , angle  $\beta$ ), so every whole cell carries two tiles and a cell cut along its own  $b$ -diagonal carries one. Split  $P_1$  by the two parallel lines through  $(fa, 0)$  and  $(fa + b, 0)$  with direction  $X_1w - fau$ :

- $\Lambda$  the triangle  $(0, 0), (fa, 0), (0, X_1)$ . Its hypotenuse drops exactly  $a$  per row of height  $c$ , hence passes through cell corners; cells with  $i + j \leq f - 2$  are whole and those with  $i + j = f - 1$  are cut along their  $b$ -diagonal, giving  $2\binom{f}{2} + f = f^2$  tiles.
- $\Sigma$  the strip between the lines: the parallelogram spanned by  $bu$  and  $X_1w - fau$ . Put  $\sigma = (X_1w - fau)/b$ . Then

$$|\sigma|^2 = \frac{b^2 + 4f^6 - N_1^2}{4b^2} = f^2, \quad \cos \angle(\sigma, u) = \frac{1}{2f} = \cos(\alpha + \beta),$$

so  $\sigma$  has length  $a$  and makes angle  $\alpha + \beta$  with  $u$ . Consequently the parallelogram spanned by  $bu$  and  $\sigma$  has sides  $b, a$  and angles  $\alpha + \beta, \gamma$ : it is the parallelogram obtained by gluing two tiles along their  $c$ -edge, and its long diagonal has length  $c$  because  $c^2 = a^2 + b^2 + 2ab \cos(\alpha + \beta)$ . Stacking  $b$  of them along  $\sigma$  closes exactly on  $(0, X_1)$  and gives  $2b$  tiles.

- $P$  the remainder, which is the  $\Lambda$ -staircase in the lattice translated by  $b$ , run to  $u = Y_1$ ; this is legitimate since  $Y_1 - b = 2f^2 = 2fa$ . It gives  $2(f^2 + \binom{f}{2}) + f = 3f^2$  tiles.

The three pieces are disjoint with union  $P_1$ , and  $f^2 + 2b + 3f^2 = 4f^2 + 2(f^2 - 1) = 2N_1$ .  $\square$

The construction has been generated and verified exactly in  $\mathbb{Q}(\sqrt{D})$  — congruence, containment, pairwise separation, area — for  $f = 2, \dots, 12$ , i.e. 22, 52, 94, 148, 214, 292, 382, 484, 598, 724 and 862 tiles; only  $f = 2, 3$  had previously been obtained by search.

**Remark 193** (The obstruction at  $e \geq 2$ ). The mechanism is specific to  $e = 1$ . For general  $(e, f)$  the strip is always the parallelogram spanned by  $ebu$  and  $X_1w - fau$ , whose slant has length exactly  $fb$  and whose angle satisfies  $\cos \theta = e/(2f) = \sin(\alpha/2)$ , i.e.  $\theta = \alpha + \beta$ ; but in the sheared basis it is combinatorially an  $s \times fb$  rectangle to be filled by  $a \times b$  bricks, and the de Bruijn–Klarner criterion requires  $a \mid s$  or  $a \mid fb$ . The latter says  $e \mid f^2$ , i.e.  $e = 1$ ; the former contradicts the compatibility condition  $s \equiv -e^3 \pmod{ef}$  unless  $f \mid e^2$ . So no admissible width admits a brick interior when  $e \geq 2$ , and exhaustive search confirms that no interior of any kind exists in the cases tested: on  $(2, 3)$  the widths  $s = 4, 10, 16, 22$  all return NO TILING (1, 30, 181, 855 nodes), and on  $(3, 4)$  the width  $s = 21$  returns NO TILING (72 nodes), while the  $e = 1$  control is found in 10 nodes.

**Theorem 194** (The dichotomy). [PROVED] *The unit parallelogram of the member  $(e, f) = (2, 3)$  — sides  $Y_1 = 46$  and  $X_1 = 27$  with angle  $\beta$ , tile  $(6, 5, 9)$ , 46 tiles — admits no tiling. (Exhaustive search, 4334789 nodes.) Together with Theorem 192 this separates the branch: at  $e = 1$  the unit parallelogram exists for every  $f$ , and at  $e = 2$  it does not exist at all.*

The obstruction of Remark 193 is therefore not a limitation of the two-lattice architecture: at  $e \geq 2$  there is nothing for any architecture to find. At  $(2, 3)$  both the unit cell (Theorem 31) and the unit parallelogram are untileable.

**Remark 195** (What this does to the induction). Recorded in [1]. Full discussion in [1].

**Proposition 196** (Wide parallelograms, all  $(e, f)$ ). [PROVED] *A parallelogram with sides  $pY_1$  (base direction) and  $qX_1$  (leg direction), angle  $\beta$ , is a pure  $a \times c$  lattice — a  $(pN_1/f) \times (qf)$  grid of two-tile cells, hence tileable by  $2pqN_1$  tiles — if and only if  $f \mid p$  (the side divisibilities and count are `pgram_cols`, `pgram_cell_count` in `lean/Primitives.lean`).*

**Theorem 197** (Addition law). [PROVED] *Fix a member  $(e, f)$  and  $m_1, m_2 \geq 1$  with  $\Delta_{m_1}$  and  $\Delta_{m_2}$  tileable. If  $f \mid m_1$ , or if  $e = 1$ , then  $\Delta_{m_1+m_2}$  is tileable.*

*Proof.* Anchor  $\Delta_{m_1}$  at the apex of  $\Delta_{m_1+m_2}$ . The band below it is the trapezoid with parallel horizontal sides  $m_1Y_1$  (top) and  $(m_1+m_2)Y_1$  (base) and legs  $m_2X_1$ . Cut it from the top-west vertex parallel to the east leg: by the isosceles altitude identity  $X_1 \cos \beta = Y_1/2$  (as in Lemma 182) the cut has length  $m_2X_1$  and lands at  $(m_2Y_1, 0)$ , splitting the band into the upright scale- $m_2$  triangle on  $[0, m_2Y_1]$  and the parallelogram  $P$  spanned by  $m_1Y_1$  (base direction) and  $m_2X_1$  (east-leg direction). Tile the triangle by hypothesis. For  $P$ : if  $f \mid m_1$ , Proposition 196 applies with  $(p, q) = (m_1, m_2)$ ; if  $e = 1$ ,  $P$  is an  $m_1 \times m_2$  grid of translates of the unit parallelogram  $P_1$  of Theorem 192. All three interfaces are full straight segments. The count bookkeeping is `add_count` in `lean/Primitives.lean`.  $\square$

**Remark 198** (Correction, and the closure structure). An earlier version of Proposition 196 asserted the single-seed step “ $\Delta_M$  is tileable whenever  $\Delta_t$  is tileable and  $f \mid M - t$ ”. That step is withdrawn: slicing the band below  $\Delta_t^{\text{apex}}$  by lines parallel to a leg always leaves an upright corner triangle of scale  $M - t$ , which the hypothesis does not tile; the two-seed form of Theorem 197 is what the geometry gives. The closure structure of  $S(e, f) = \{m : \Delta_m \text{ tileable}\}$  is therefore: closed under  $m \mapsto km$  (subdivision), closed under addition of an  $f$ -divisible element, and closed under addition outright at  $e = 1$ . With seeds  $\{2, 3\}$  at  $(1, 2)$  this recovers Theorem 183; at  $(1, 3)$ , where  $m = 1$  is exhausted and  $m = 2$  excluded (Theorem 185), the set generated by a  $\Delta_3$  seed under all three operations is exactly  $3\mathbb{Z}_{\geq 1}$  (`reachable_13` in `lean/Primitives.lean`).

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